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## MATHS - IIA

# DEMOIVRE'S THEOREM

1) (i) If n is an integer then show that $(1+i)^{2n}+(1-i)^{2n}=2^{n+1}\cos\frac{n\pi}{2}$

**Sol**:- Now, $(1+i)^{2n}$

Multiply and divide with $\sqrt{a^2+b^2}=\sqrt{1+1}=\sqrt{2}$

$$(1+i)^{2n}=\left[\sqrt{2}\left[\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right]\right]^{2n}$$

$$=(\sqrt{2})^{2n}\left[\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right]^{2n}$$

$$=2^n\left[\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right]^{2n} \qquad \boxed{(\cos\theta+i\sin\theta)^n=(\cos n\theta+i\sin n\theta)}$$

$$(1+i)^{2n}=2^n\left[\cos\frac{\cancel{2}n\pi}{\cancel{4}_2}+i\sin\frac{\cancel{2}n\pi}{\cancel{4}_2}\right]$$

$$(1+i)^{2n}=2^n\left[\cos\frac{n\pi}{2}+i\sin\frac{n\pi}{2}\right]\longrightarrow (1)$$

Similarly, $(1-i)^{2n}=2^n\left[\cos\frac{n\pi}{2}-i\sin\frac{n\pi}{2}\right]\longrightarrow (2)$

Adding (1) & (2)

$$(1+i)^{2n}+(1-i)^{2n}=2^n\left[\cos\frac{n\pi}{2}+\cancel{i\sin\frac{n\pi}{2}}+\cos\frac{n\pi}{2}-\cancel{i\sin\frac{n\pi}{2}}\right]$$

$$=2^n\left[2^1\cos\frac{n\pi}{2}\right]$$

$$=2^{n+1}\cos\frac{n\pi}{2}$$

(ii) If 'n' is a positive integer then show that $(1+i)^n+(1-i)^n=2^{\frac{n+2}{2}}\cos\left[\frac{n\pi}{4}\right]$.

**Sol**:- Now $(1+i)^n$

Multiply and divide with $\sqrt{a^2+b^2}=\sqrt{1+1}=\sqrt{2}$

$$(1+i)^n=\left[\sqrt{2}\left[\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right]\right]^n$$

$$=(\sqrt{2})^n\left[\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right]^n$$

$$=2^{n/2}\left[\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right]^n$$

$$(1+i)^n=2^{n/2}\left[\cos\frac{n\pi}{4}+i\sin\frac{n\pi}{4}\right]\longrightarrow (1)$$

Similarly, $(1-i)^n=2^{n/2}\left[\cos\frac{n\pi}{4}-i\sin\frac{n\pi}{4}\right]\longrightarrow (2)$

Adding (1) & (2)

$$(1+i)^n+(1-i)^n=2^{n/2}\left[\cos\frac{n\pi}{4}+i\sin\frac{n\pi}{4}\right]+2^{n/2}\left[\cos\frac{n\pi}{4}-i\sin\frac{n\pi}{4}\right]$$

$$= 2^{n/2}\left[\cos\frac{n\pi}{4} + i\cancel{\sin\frac{n\pi}{4}} + \cos\frac{n\pi}{4} - i\cancel{\sin\frac{n\pi}{4}}\right]$$

$$= 2^{n/2}\left[2\cos\frac{n\pi}{4}\right]$$

$$= 2^{\frac{n+1}{2}}\cos\frac{n\pi}{4}$$

② If $\alpha,\beta$ are the roots of the equation $x^2-2x+4=0$ then for any positive integer 'n'. Show that $\alpha^n+\beta^n = 2^{n+1}\cos\left[\frac{n\pi}{3}\right]$.

Sol:- Given $\alpha,\beta$ are the roots of $x^3-2x+4=0$

$$\boxed{x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}}$$

$$(\alpha,\beta) = \frac{2\pm\sqrt{4-4(1)(4)}}{2}$$

$$\alpha,\beta = \frac{2\pm\sqrt{4-16}}{2}$$

$$= \frac{2\pm\sqrt{-12}}{2}$$

$$\alpha,\beta = \frac{2\pm i\sqrt{12}}{2} \Rightarrow \frac{2\pm i2\sqrt{3}}{2}$$

$$\alpha,\beta = 1\pm i\sqrt{3}$$

$$(\alpha,\beta) = (1+i\sqrt{3}),(1-i\sqrt{3})$$

$$\alpha^n+\beta^n = (1+i\sqrt{3})^n + (1-i\sqrt{3})^n$$

Multiply and divide with $\sqrt{a^2+b^2} = \sqrt{1+3} = \sqrt{4} = 2$

$$\alpha^n+\beta^n = \left[2\left[\frac{1}{2}+i\frac{\sqrt{3}}{2}\right]\right]^n + \left[2\left[\frac{1}{2}-i\frac{\sqrt{3}}{2}\right]\right]^n$$

$$= 2^n\left[\frac{1}{2}+i\frac{\sqrt{3}}{2}\right]^n + 2^n\left[\frac{1}{2}-i\frac{\sqrt{3}}{2}\right]^n$$

$$= 2^n\left[\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right]^n + 2^n\left[\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}\right]^n$$

$$= 2^n\left[\cos\frac{n\pi}{3}+i\sin\frac{n\pi}{3}\right] + 2^n\left[\cos\frac{n\pi}{3}-i\sin\frac{n\pi}{3}\right]$$

$$= 2^n\left[\cos\frac{n\pi}{3}+i\cancel{\sin\frac{n\pi}{3}} + \cos\frac{n\pi}{3} - i\cancel{\sin\frac{n\pi}{3}}\right]$$

$$= 2^n\left[2\cos\frac{n\pi}{3}\right]$$

$$\alpha^n+\beta^n = 2^{n+1}\cos\frac{n\pi}{3}$$

③ If 'n' is an integer then show that $(1+\cos\theta+i\sin\theta)^n + (1+\cos\theta-i\sin\theta)^n = 2^{n+1}\cos^n\left[\frac{\theta}{2}\right]\cos\left[\frac{n\theta}{2}\right]$

$$\boxed{\begin{array}{c}1+\cos\theta = 2\cos^2\theta/2 \\ \sin\theta = 2\sin\theta/2\cos\theta/2\end{array}}$$

Sol:- L.H.S $= (1+\cos\theta+i\sin\theta)^n + (1+\cos\theta-i\sin\theta)^n$

$$= \left(2\cos^2\frac{\theta}{2} + i2\sin\theta/2\cos\theta/2\right]^n + \left[2\cos^2\theta/2 - i2\sin\theta/2\cos\theta/2\right]^n$$

$$= 2^n\cos^n\theta/2(\cos\theta/2+i\sin\theta/2)^n + 2^n\cos^n\theta/2[\cos\theta/2 - i\sin\theta/2]^n$$

$$= 2^n\cos^n\theta/2(\cos n\theta/2+i\sin n\theta/2) + 2^n\cos^n\theta/2(\cos n\theta/2 - i\sin n\theta/2)$$

$$= 2^n(\cos^n\theta/2(\cos n\theta/2 + i\cancel{\sin n\theta/2} + \cos n\theta/2 - i\cancel{\sin n\theta/2})$$

$$= 2^n\cos^n\theta/2\,(2\cos n\theta/2)$$

$$= 2^{n+1}\cos^n\theta/2\,\cos n\theta/2$$

(4) Show that one value of $\left[\dfrac{1+\sin\pi/8+i\cos\pi/8}{1+\sin\pi/8-i\cos\pi/8}\right]^{8/3}=1$

Sol:- $L.H.S = \left[\dfrac{1+\sin\pi/8+i\cos\pi/8}{1+\sin\pi/8-i\cos\pi/8}\right]^{8/3}$

$$=\left[\dfrac{1+\cos[\pi/2-\pi/8]+i\sin[\pi/2-\pi/8]}{1+\cos[\pi/2-\pi/8]-i\sin[\pi/2-\pi/8]}\right]^{8/3}$$

$$=\left[\dfrac{1+\cos 3\pi/8+i\sin 3\pi/8}{1+\cos 3\pi/8-i\sin 3\pi/8}\right]^{8/3}$$

$$=\left[\dfrac{2\cos^2 3\pi/16+i2\sin 3\pi/16\cos 3\pi/16}{2\cos^2 3\pi/16-i2\sin 3\pi/16\cos 3\pi/16}\right]^{8/3}$$

$$=\left[\dfrac{\cancel{2\cos} 3\pi/16\,[\cos 3\pi/16+i\sin 3\pi/16]}{\cancel{2\cos} 3\pi/16\,[\cos 3\pi/16-i\sin 3\pi/16]}\right]^{8/3}$$

$$=\left[\dfrac{\cos 8/3\times 3\pi/16+i\sin 8/3\times 3\pi/16}{\cos 8/3\times 3\pi/16-i\sin 8/3\times 3\pi/16}\right]$$

$$=\left[\dfrac{\cos\pi/2+i\sin\pi/2}{\cos\pi/2-i\sin\pi/2}\right]$$

$$=\left[\dfrac{0+i}{0-i}\right]$$

$$=\dfrac{i}{-i}$$

$$=-1$$

(5) If $\cos\alpha+\cos\beta+\cos\delta=0=\sin\alpha+\sin\beta+\sin\delta$ Then prove that $\cos^2\alpha+\cos^2\beta+\cos^2\delta=3/2=\sin^2\alpha+\sin^2\beta+\sin^2\gamma$

Sol:- $x=\text{Cis}\,\alpha$ $\quad y=\text{Cis}\,\beta$ $\quad z=\text{Cis}\,\gamma$

$x=\cos\alpha+i\sin\alpha$ $\quad y=\cos\beta+i\sin\beta$ $\quad z=\cos\gamma+i\sin\gamma$

$$x+y+z=\cos\alpha+i\sin\alpha+\cos\beta+i\sin\beta+\cos\gamma+i\sin\gamma$$

$$=(\cos\alpha+\cos\beta+\cos\gamma)+i(\sin\alpha+\sin\beta+\sin\gamma)$$

$$=0+i(0)$$

$$x+y+z=0$$

$$(x+y+z)^2=0$$

$$x^2+y^2+z^2+2(xy+yz+zx)=0$$

$$x^2+y^2+z^2=-2(xy+yz+zx)$$

$$=-2xyz\left[\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right]$$

$$=-2xyz\left[\frac{1}{\cos\alpha+i\sin\alpha}+\frac{1}{\cos\beta+i\sin\beta}+\frac{1}{\cos\gamma+i\sin\gamma}\right]$$

$$=-2xyz\left[\cos\alpha-i\sin\alpha+\cos\beta-i\sin\beta+\cos\gamma-i\sin\gamma\right]$$

$$=-2xyz\left[\cos\alpha+\cos\beta+\cos\gamma\right]-i\left[\sin\alpha+\sin\beta+\sin\gamma\right]$$

$$=-2xyz\left[0-i(0)\right]$$

$x^2+y^2+z^2=0$

$(\cos\alpha+i\sin\alpha)^2+(\cos\beta+i\sin\beta)^2+(\cos\gamma+i\sin\gamma)^2=0$

$(\cos 2\alpha+i\sin 2\alpha)+(\cos 2\beta+i\sin 2\beta)+(\cos 2\gamma+i\sin 2\gamma)=0$

$(\cos 2\alpha+\cos 2\beta+\cos 2\gamma)+i(\sin 2\alpha+\sin 2\beta+\sin 2\gamma)=0$

comparing real part on both sides

$\Rightarrow \cos 2\alpha+\cos 2\beta+\cos 2\gamma=0$

$\Rightarrow 2\cos^2\alpha-1+2\cos^2\beta-1+2\cos^2\gamma-1=0$

$\Rightarrow 2\cos^2\alpha+2\cos^2\beta+2\cos^2\gamma-3=0$

$\Rightarrow 2(\cos^2\alpha+\cos^2\beta+\cos^2\gamma)=3$

$\Rightarrow \cos^2\alpha+\cos^2\beta+\cos^2\gamma=3/2 \longrightarrow (1)$

$\Rightarrow 1-\sin^2\alpha+1-\sin^2\beta+1-\sin^2\gamma=3/2$

$\Rightarrow 3-(\sin^2\alpha+\sin^2\beta+\sin^2\gamma)=3/2$

$\Rightarrow 3-3/2=\sin^2\alpha+\sin^2\beta+\sin^2\gamma$

$\Rightarrow \frac{6-3}{2}=\sin^2\alpha+\sin^2\beta+\sin^2\gamma$

$\therefore \cos^2\alpha+\cos^2\beta+\cos^2\gamma=\frac{3}{2}=\sin^2\alpha+\sin^2\beta+\sin^2\gamma$

(6) If $\cos\alpha+\cos\beta+\cos\gamma=0=\sin\alpha+\sin\beta+\sin\gamma$

(i) $\cos 3\alpha+\cos 3\beta+\cos 3\gamma=3\cos(\alpha+\beta+\gamma)$

$x=\text{Cis}\,\alpha$ $\quad y=\text{Cis}\,\beta$ $\quad z=\text{Cis}\,\gamma$

$x=\cos\alpha+i\sin\alpha$ $\quad y=\cos\beta+i\sin\beta$ $\quad z=\cos\gamma+i\sin\gamma$

Now $x+y+z$

$=\cos\alpha+i\sin\alpha+\cos\beta+i\sin\beta+\cos\gamma+i\sin\gamma$

$=\cos\alpha+\cos\beta+\cos\gamma+i(\sin\alpha+\sin\beta+\sin\gamma)$

$=0+i(0)$

$=0$

$x+y+z=0$

c.o.B.S

$(x+y+z)^3=0$

$x^3+y^3+z^3=3xyz$

~~$\cos\alpha+\cos\beta$~~

$\Rightarrow(\cos\alpha+i\sin\alpha)^3+(\cos\beta+i\sin\beta)^3+(\cos\gamma+i\sin\gamma)^3=3\,\text{cis}\,\alpha\cdot\text{Cis}\,\beta\cdot\text{Cis}\,\gamma$

$\Rightarrow \cos 3\alpha+i\sin 3\alpha+\cos 3\beta+i\sin 3\beta+\cos 3\gamma+i\sin 3\gamma=3\,\text{Cis}(\alpha+\beta+\gamma)$

$\Rightarrow \cos 3\alpha+\cos 3\beta+\cos 3\gamma+i(\sin 3\alpha+\sin 3\beta+\sin 3\gamma)=3\cos(\alpha+\beta+\gamma)+i\,3\sin(\alpha+\beta+\gamma)$

(i) Comparing real parts on b.s

$\cos 3\alpha+\cos 3\beta+\cos 3\gamma=3\cos(\alpha+\beta+\gamma)$

(ii) Comparing imaginary parts on b.s

$\sin 3\alpha+\sin 3\beta+\sin 3\gamma=3\sin(\alpha+\beta+\gamma)$

(iii) $xy+yz+zx = xyz\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$

$\Rightarrow xyz\left[\frac{1}{\cos\alpha+i\sin\alpha}+\frac{1}{\cos\beta+i\sin\beta}+\frac{1}{\cos\gamma+i\sin\gamma}\right]$

$\Rightarrow xyz\,(\cos\alpha-i\sin\alpha+\cos\beta-i\sin\beta+\cos\gamma-i\sin\gamma)$

$\Rightarrow xyz\,(\cos\alpha+\cos\beta+\cos\gamma)-i(\sin\alpha+\sin\beta+\sin\gamma)$

$\Rightarrow xyz\,(0-i(0))$

$xy+yz+zx=0$

$\Rightarrow \text{cis}\,\alpha.\text{cis}\,\beta+\text{cis}\,\beta.\text{cis}\,\gamma+\text{cis}\,\gamma.\text{cis}\,\alpha=0$

$\Rightarrow \text{cis}(\alpha+\beta)+\text{cis}(\beta+\gamma)+\text{cis}(\gamma+\alpha)=0$

$\Rightarrow \cos(\alpha+\beta)+i\sin(\alpha+\beta)+\cos(\beta+\gamma)+i\sin(\beta+\gamma)+\cos(\gamma+\alpha)+i\sin(\gamma+\alpha)=0$

$\Rightarrow [\cos(\alpha+\beta)+\cos(\beta+\gamma)+\cos(\gamma+\alpha)]+i[\sin(\alpha+\beta)+\sin(\beta+\gamma)+\sin(\gamma+\alpha)]=0$

Comparing real part on b.s

$\cos(\alpha+\beta)+\cos(\beta+\gamma)+\cos(\gamma+\alpha)=0$

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7. If 'n' is a positive integer, show that $(P+iQ)^{1/n}+(P-iQ)^{1/n}=2(P^2+Q^2)^{1/2n}\cos\left[\frac{1}{n}\tan^{-1}\frac{Q}{P}\right]$

Sol:- Now $(P+iQ)^{1/n}$

Multiply and divide with $\sqrt{a^2+b^2}=\sqrt{P^2+Q^2}$

$(P+iQ)^{1/n}=\left[\sqrt{P^2+Q^2}\left[\frac{P}{\sqrt{P^2+Q^2}}+i\frac{Q}{\sqrt{P^2+Q^2}}\right]\right]^{1/n}$

$=(\sqrt{P^2+Q^2})^{1/n}\left[\frac{P}{\sqrt{P^2+Q^2}}+i\frac{Q}{\sqrt{P^2+Q^2}}\right]^{1/n}$

let $r=\sqrt{P^2+Q^2}$, $\cos\theta=\frac{P}{\sqrt{P^2+Q^2}}$, $\sin\theta=\frac{Q}{\sqrt{P^2+Q^2}}$

$\tan\theta=\frac{\sin\theta}{\cos\theta}=\frac{Q}{P}$

$\theta=\tan^{-1}\left(\frac{Q}{P}\right)$

$(P+iQ)^{1/n}=r^{1/n}(\cos\theta+i\sin\theta)^{1/n}$

$(P+iQ)^{1/n}=r^{1/n}(\cos\theta/n+i\sin\theta/n)\rightarrow 1$

Similarly $(P-iQ)^{1/n}=r^{1/n}(\cos\theta/n-i\sin\theta/n)\rightarrow 2$

Adding 1 & 2

$(P+iQ)^{1/n}+(P-iQ)^{1/n}=r^{1/n}[\cos\theta/n+\cancel{i\sin\theta/n}+\cos\theta/n-\cancel{i\sin\theta/n}]$

$=r^{1/n}.\,2\cos\theta/n$

$=2r^{1/n}\cos\theta/n$

$=2(\sqrt{P^2+Q^2})^{1/n}.\cos\left(\frac{1}{n}\theta\right)$

$=2(P^2+Q^2)^{1/2n}\left[\cos\left(\frac{1}{n}\right)\tan^{-1}\left(\frac{Q}{P}\right)\right]$

(8) Find all the roots of equation $x^{11}-x^7+x^4-1=0$

Sol:- $x^7(x^4-1)+1(x^4-1)=0$

$(x^4-1)(x^7+1)=0$

$x^4-1=0 \quad | \quad x^7+1=0$

$x^4=1 \quad | \quad x^7=-1$

Case (i) :- $x^4=1$

$x=(1)^{1/4}$

$x=(1+i(0))^{1/4}$

$x=(\cos 0+i\sin 0)^{1/4}$

$x=(\text{cis}\,0)^{1/4}$

$x=\text{cis}\left(\frac{2k\pi+0}{\cancel{4}\,2}\right)$ where $k=0,1,2,3$

$x=\text{cis}\ \frac{k\pi}{2}$

$x=\text{cis}\,0, \text{cis}\,\pi/2, \text{cis}\,\pi, \text{cis}\,3\pi/2$

$x=\pm 1, \pm i$

case (ii) $x^7=-1$

$x=(-1)^{1/7}$

$x=(-1+i(0))^{1/7}$

$x=(\cos\pi+i\sin\pi)^{1/7}$

$x=(\text{cis}\,\pi)^{1/7}$

$x=\text{cis}\left(\frac{2k\pi+\pi}{7}\right)$ $k=0,1,2,3,4,5,6$

$x=\cos\pi/7, \text{cis}\,3\pi/7, \text{cis}\,5\pi/7,$
$\text{cis}\,7\pi/7, \text{cis}\,9\pi/7, \text{cis}\,11\pi/7,$
$\text{cis}\,13\pi/7$

$\therefore$ The roots are $\pm 1, \pm i,$
$\text{cis}\,\pi/7, \text{cis}\,3\pi/7 ---- \text{cis}\,13\pi/7$

(9) Solve $x^9-x^5+x^4-1=0$

Sol:- $x^5(x^4-1)+1(x^4-1)=0$

$(x^4-1)(x^5+1)=0$

$x^4-1=0 \qquad x^5+1=0$

$x^4=1 \qquad x^5=-1$

Case (i) $x^4=1$

$x=(1+i(0))^{1/4}$

$=(\cos 0+i\sin 0)^{1/4}$

$x=(\text{cis}\,0)^{1/4}$

$x=\text{cis}\left(\frac{2k\pi+0}{4}\right)$ where $k=0,1,2,3$

$x=\text{cis}\left(\frac{k\pi}{2}\right)$ where $k=0,1,2,3$

$x=\text{cis}\,0, \text{cis}\,\pi/2, \text{cis}\,\pi, \text{cis}\,3\pi/2$

$x=\pm 1, \pm i$

Case (ii) $x^5=-1$

$x=(-1+i(0))^{1/5}$

$x=(\cos\pi+i\sin\pi)^{1/5}$

$x=(\text{cis}\,\pi)^{1/5}$

$x=\text{cis}\left(\frac{2k\pi+\pi}{5}\right)$ where $k=0,1,2,3,4$

$x=\text{cis}\,\pi/5, \text{cis}\,3\pi/5, \text{cis}\,5\pi/5, \text{cis}\,9\pi/5$

$\therefore$ The roots are $\pm 1, \pm i, \text{cis}\,\pi/5, ---- \text{cis}\,9\pi/5$

(10) If 'n' is an integer and $z = cis\theta\left(\theta+(2n+1)\pi/2\right)$, then show that $\frac{z^{2n}-1}{z^{2n}+1} = i\tan(n\theta)$

Sol:- Given $z = cis\theta$

$z = \cos\theta + i\sin\theta$

$z^n = (\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

$\frac{1}{z^n} = \frac{1}{\cos\theta + i\sin n\theta} = \cos n\theta - i\sin n\theta$

$$L.H.S = \frac{z^{2n}-1}{z^{2n}+1}$$

$$= \frac{\cancel{z^n}\left(z^n - \frac{1}{z^n}\right)}{\cancel{z^n}\left(z^n + \frac{1}{z^n}\right)}$$

$$= \frac{\cancel{\cos n\theta} + i\sin n\theta - \cancel{\cos n\theta} + i\sin n\theta}{\cos n\theta + \cancel{i\sin n\theta} + \cos n\theta - \cancel{i\sin n\theta}}$$

$$= \frac{\cancel{2}i\sin n\theta}{\cancel{2}\cos n\theta}$$

$$= i\tan n\theta$$

R.H.S

# THEORY OF EQUATIONS

## LAQ's (7M)

1) Solve the equation $x^4-10x^3+26x^2-10x+1=0$

Sol: Given Equation is even degree reciprocal of class - I

Divide by $x^2$ on Both sides of given equation

$$\frac{x^4}{x^2}-\frac{10x^3}{x^2}+\frac{26x^2}{x^2}-\frac{10x}{x^2}+\frac{1}{x^2}=0$$

$$x^2-10x+26-\frac{10}{x}+\frac{1}{x^2}=0$$

$$\left(x^2-\frac{1}{x^2}\right)-10\left(x+\frac{1}{x}\right)+26=0$$

let $x+\frac{1}{x}=t$

squaring on both sides

$$x^2+\frac{1}{x^2}=t^2-2$$

$$t^2-2-10t+26=0$$

$$t^2-10t+24=0$$

$$t^2-6t-4t+24=0$$

$$t(t-6)-4(t-6)=0$$

$$(t-6)(t-4)=0$$

$t-6=0$, $t=6$ | $t-4=0$, $t=4$

**case (i)** $t=6$

$$x+\frac{1}{x}=6$$

$$\frac{x^2+1}{x}=6$$

$$x^2-6x+1=0$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

**case (ii)** $t=4$

$$x+\frac{1}{x}=4$$

$$\frac{x^2+1}{x}=4$$

$$x^2-4x+1=0$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(1)}}{2(1)}$$

$$x = \frac{6 \pm \sqrt{36-4}}{2}$$

$$x = \frac{6 \pm \sqrt{32}}{2}$$

$$x = \frac{6 \pm 4\sqrt{2}}{2}$$

$$\boxed{x = 3 \pm 2\sqrt{2}}$$

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)}$$

$$x = \frac{4 \pm \sqrt{16-4}}{2}$$

$$x = \frac{4 \pm \sqrt{12}}{2}$$

$$x = \frac{4 \pm 2\sqrt{3}}{2}$$

$$\boxed{x = 2 \pm \sqrt{3}}$$

$\therefore$ The roots are $3 \pm 2\sqrt{2}$, $2 \pm \sqrt{3}$

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2.) Solve $6x^6 - 25x^5 + 31x^4 - 31x^2 + 25x - 6 = 0$

Sol:- Given Equation is even degree reciprocal Equation of class-II

1, -1 are roots of given equation

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 6 | -25 | 31 | 0 | -31 | 25 | -6 |
| | 0 | 6 | -19 | 12 | 12 | -11 | 6 |
| -1 | 6 | -19 | 12 | 12 | -19 | 6 | 0 |
| | 0 | -6 | 25 | -37 | 25 | -6 | |
| | 6 | -25 | 37 | -25 | 6 | 0 | |

The equation is $6x^4 - 25x^3 + 37x^2 - 25x + 6 = 0$

divide $x^2$ on Both sides

$$\frac{6x^4}{x^2} - \frac{25x^3}{x^2} + \frac{37x^2}{x^2} - \frac{25x}{x^2} + \frac{6}{x^2} = 0$$

$$6x^2 - 25x + 37 - \frac{25}{x} + \frac{6}{x^2} = 0$$

$$6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x + \frac{1}{x}\right) + 37 = 0$$

Let $x^2 + \frac{1}{x^2} = t^2 - 2$

$$6(t^2 - 2) - 25t + 37 = 0$$

$$6t^2 - 12 - 25t + 37 = 0$$

$$6t^2 - 25t + 25 = 0$$

$$6t^2 - 10t - 15t + 25 = 0$$

$$2t(3t-5) - 5(3t-5) = 0$$

$$(3t-5)(2t-5) = 0$$

| | |
|---|---|
| $3t - 5 = 0$ | $2t - 5 = 0$ |
| $3t = 5$ | $2t = 5$ |
| $\boxed{t = \frac{5}{3}}$ | $\boxed{t = \frac{5}{2}}$ |

**case (i)** $t = \frac{5}{3}$

$$x + \frac{1}{x} = \frac{5}{3}$$

$$\frac{x^2+1}{x} = \frac{5}{3}$$

$$3x^2 + 3 = 5x$$

$$3x^2 - 5x + 3 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(3)(3)}}{2(3)}$$

$$x = \frac{5 \pm \sqrt{25-36}}{6}$$

$$x = \frac{5 \pm \sqrt{-11}}{6}$$

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

**case (ii)** $t = \frac{5}{2}$

$$x + \frac{1}{x} = \frac{5}{2}$$

$$\frac{x^2+1}{x} = \frac{5}{2}$$

$$2x^2 + 2 = 5x$$

$$2x^2 - 5x + 2 = 0$$

$$2x^2 - 4x - x + 2 = 0$$

$$2x(x-2) - 1(x-2) = 0$$

$$(2x-1)(x-2) = 0$$

| | |
|---|---|
| $2x - 1 = 0$ | $x - 2 = 0$ |
| $2x = 1$ | $x = 2$ |
| $x = \frac{1}{2}$ | |

$\therefore$ The roots are $\frac{5 \pm i\sqrt{11}}{6}$, $\frac{1}{2}$, 2, 1, −1

3) Solve $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$

Sol:- Given Equation is even degree reciprocal equation of class-I divide with $x^2$ on Both sides

$$\frac{6x^4}{x^2} - \frac{35x^3}{x^2} + \frac{62x^2}{x^2} - \frac{35x}{x^2} + \frac{6}{x^2} = 0$$

$$6x^2 - 35x + 62 - \frac{35}{x} + \frac{6}{x^2} = 0$$

$$6\left(x^2 + \frac{1}{x^2}\right) - 35\left(x + \frac{1}{x}\right) + 62 = 0$$

let $x + \frac{1}{x} = t$

squaring on Both sides

$$x^2 + \frac{1}{x^2} = t^2 - 2$$

$$6(t^2 - 2) - 35t + 62 = 0$$

$$6t^2 - 12 - 35t + 62 = 0$$

$$6t^2 - 35t + 50 = 0$$

$$6t^2 - 15t - 20t + 50 = 0$$

$$3t(2t - 5) - 10(2t - 5) = 0$$

$$(3t - 10)(2t - 5) = 0$$

$3t - 10 = 0$      $2t - 5 = 0$

$3t = 10$      $2t = 5$

$\boxed{t = \frac{10}{3}}$      $\boxed{t = \frac{5}{2}}$

case (i) $t = \frac{5}{2}$

$$x + \frac{1}{x} = \frac{5}{2}$$

$$\frac{x^2 + 1}{x} = \frac{5}{2}$$

$$2x^2 + 2 = 5x$$

$$2x^2 - 5x + 2 = 0$$

$$2x^2 - x - 4x + 2 = 0$$

$$x(2x - 1) - 2(2x - 1) = 0$$

$$(2x - 1)(x - 2) = 0$$

$2x - 1 = 0$      $x - 2 = 0$

$2x = 1$      $\boxed{x = 2}$

$\boxed{x = \frac{1}{2}}$

case (ii) $t = \frac{10}{3}$

$$x + \frac{1}{x} = \frac{10}{3}$$

$$\frac{x^2 + 1}{x} = \frac{10}{3}$$

$$3x^2 + 3 = 10x$$

$$3x^2 - 10x + 3 = 0$$

$$3x^2 - x - 9x + 3 = 0$$

$$x(3x - 1) - 3(3x - 1) = 0$$

$$(3x - 1)(x - 3) = 0$$

$3x - 1 = 0$      $x - 3 = 0$

$3x = 1$      $\boxed{x = 3}$

$\boxed{x = \frac{1}{3}}$

$\therefore$ The roots are $\frac{1}{2}, 2, \frac{1}{3}, 3$.

4) Solve the Equation $x^5-5x^4+9x^3-9x^2+5x-1=0$

Sol:- Given Equation is odd degree reciprocal equation of class-II

'1' is root of given equation

By using synthetic division

$$\begin{array}{c|cccccc} 1 & 1 & -5 & 9 & -9 & 5 & -1 \\ & 0 & 1 & -4 & 5 & -4 & 1 \\ \hline & 1 & -4 & 5 & -4 & 1 & \boxed{0} \end{array}$$

$x^4-4x^3+5x^2-4x+1=0$

divide with $x^2$ on Both sides

$$\frac{x^4}{x^2}-\frac{4x^3}{x^2}+\frac{5x^2}{x^2}-\frac{4x}{x^2}+\frac{1}{x^2}=0$$

$$x^2-4x+5-\frac{4}{x}+\frac{1}{x^2}=0$$

$$\left(x^2+\frac{1}{x^2}\right)-4\left(x+\frac{1}{x}\right)+5=0$$

let $x+\frac{1}{x}=t$

Squaring on Both sides

$x^2+\frac{1}{x^2}=t^2-2$

$t^2-2-4t+5=0$

$t^2-4t+3=0$

$t^2-t-3t+3=0$

$t(t-1)-3(t-1)=0$

$(t-1)(t-3)=0$

$t-1=0 \qquad t-3=0$

$\boxed{t=1} \qquad \boxed{t=3}$

case (i) $t=1$

$x+\frac{1}{x}=1$

$\frac{x^2+1}{x}=1$

$x^2+1=x$

$x^2-x+1=0$

case-(ii) $t=3$

$x+\frac{1}{x}=3$

$\frac{x^2+1}{x}=3$

$x^2+1=3x$

$x^2-3x+1=0$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2-4(1)(1)}}{2(1)}$$

$$x = \frac{1 \pm \sqrt{1-4}}{2}$$

$$x = \frac{1 \pm \sqrt{-3}}{2}$$

$$x = \frac{1 \pm i\sqrt{3}}{2}$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$x = \frac{-(-3) \pm \sqrt{9-4(1)(1)}}{2(1)}$$

$$x = \frac{3 \pm \sqrt{9-4}}{2}$$

$$x = \frac{3 \pm \sqrt{5}}{2}$$

$\therefore$ The roots are $1$, $\frac{1 \pm i\sqrt{3}}{2}$, $\frac{3 \pm \sqrt{5}}{2}$

5.) Solve the equation $2x^5 + x^4 - 12x^3 - 12x^2 + x + 2 = 0$

Sol:- Given Equation is odd degree reciprocal Equation of class-I

"$-1$" is root of given equation

By using synthetic division

$$\begin{array}{r|rrrrrr} -1 & 2 & 1 & -12 & -12 & 1 & 2 \\ & 0 & -2 & 1 & 11 & 1 & -2 \\ \hline & 2 & -1 & -11 & -1 & 2 & \boxed{0} \end{array}$$

$2x^4 - x^3 - 11x^2 - x + 2 = 0$

divide with $x^2$

$$\frac{2x^4}{x^2} - \frac{x^3}{x^2} - \frac{11x^2}{x^2} - \frac{x}{x^2} + \frac{2}{x^2} = 0$$

$$2x^2 - x - 11 - \frac{1}{x} + \frac{2}{x^2} = 0$$

$$2\left(x^2 + \frac{1}{x^2}\right) - 1\left(x + \frac{1}{x}\right) - 11 = 0$$

put $x + \frac{1}{x} = t$

Squaring on Both sides

$$x^2 + \frac{1}{x^2} = t^2 - 2$$

$$2(t^2 - 2) - t - 11 = 0$$

$$2t^2 - 4 - t - 11 = 0$$

$$2t^2 - t - 15 = 0$$

$$2t^2 - 6t + 5t - 15 = 0$$

$$2t(t-3) + 5(t-3) = 0$$

$$(t-3)(2t+5) = 0$$

$t - 3 = 0$ $\quad\quad$ $2t = -5$

$t = 3$ $\quad\quad$ $t = -\frac{5}{2}$

<u>case (i)</u> $t = 3$

$$x + \frac{1}{x} = 3$$

$$\frac{x^2+1}{x} = 3$$

$$x^2 + 1 = 3x$$

$$x^2 - 3x + 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(1)}}{2(1)}$$

$$x = \frac{3 \pm \sqrt{9-4}}{2}$$

$$x = \frac{3 \pm \sqrt{5}}{2}$$

<u>case (ii)</u> $t = -\frac{5}{2}$

$$x + \frac{1}{x} = -\frac{5}{2}$$

$$\frac{x^2+1}{x} = -\frac{5}{2}$$

$$2x^2 + 2 = -5x$$

$$2x^2 + 5x + 2 = 0$$

$$2x^2 + 4x + x + 2 = 0$$

$$2x(x+2) + 1(x+2) = 0$$

$$(x+2)(2x+1) = 0$$

$x + 2 = 0$ $\quad\quad$ $2x + 1 = 0$

$x = -2$ $\quad\quad$ $2x = -1$

$x = -\frac{1}{2}$

$\therefore$ The roots are $-1, \frac{3 \pm \sqrt{5}}{2}, -2, -\frac{1}{2}$

6.) Given that the roots of $x^3 + 3px^2 + 3qx + r = 0$ are in

(i.) A.P show that $2p^3 - 3pq + r = 0$

(ii.) G.P show that $p^3 r = q^3$

(iii.) H.P show that $2q^3 = r(3pq - r)$

Given that the roots of $x^3 + 3px^2 + 3qx + r = 0$ —— (1)

<u>sol</u>:- i, A.P show that $2p^3 - 3pq + r = 0$

let $a+d, a, a-d$ roots are in A.P

$$S_1 \Rightarrow a + \cancel{d} + a + a - \cancel{d} = -3P$$

$$\cancel{3}a = -\cancel{3}p$$

$$\boxed{a = -p}$$

$a = -p$ one of the root of equation — (1)

$$(-p)^3 + 3p(-p)^2 + 3q(-p) + r = 0$$

$$-p^3 + 3p^3 - 3pq + r = 0$$

$$\boxed{2p^3 - 3pq + r = 0}$$

Hence proved

(ii.) G.P show that $p^3 r = q^3$

let $\frac{a}{r}, a, ar$ are the roots are in G.P

$$S_3 = \left(\frac{a}{r}\right)(a)(ar) = -r$$

$$a^3 = -r \Rightarrow a = (-r)^{\frac{1}{3}}$$

$a = (-r)^{1/3}$ is one of root of equation (1)

$$\left((-r)^{\frac{1}{3}}\right)^{\cancel{3}} + 3p\left((-r)^{\frac{1}{3}}\right)^2 + 3q(-r)^{\frac{1}{3}} + r = 0$$

$$-\cancel{r} + 3pr^{\frac{2}{3}} - 3qr^{\frac{1}{3}} + \cancel{r} = 0$$

$$\cancel{3}pr^{\frac{2}{3}} = \cancel{3}qr^{\frac{1}{3}}$$

cubing on Both sides

$$p^3 r^{\cancel{2}} = q^3 \cancel{r}$$

$$\boxed{p^3 r = q^3}$$ hence proved

(iii.) H.P show that $2q^3 = r(3pq - r)$

$x^3 + 3px^2 + 3qx + r = 0$ are in H.P

$\frac{1}{x^3} + \frac{3p}{x^2} + \frac{3q}{x} + r = 0$ are in A.P

$$\frac{1 + 3px + 3qx^2 + rx^3}{x^3} = 0$$

$rx^3 + 3qx^2 + 3px + 1 = 0$ are in A.P — (2)

let $a+d, a, a-d$ roots are in A.P

$$S_1 = a+\cancel{d}+a+a-\cancel{d} = -\frac{3q}{r}$$

$$\cancel{3}a = -\frac{\cancel{3}q}{r}$$

$$\boxed{a = -\frac{q}{r}}$$

$a = -\frac{q}{r}$ is one of root of equation (2)

$$r\left(-\frac{q}{r}\right)^3 + 3q\left(-\frac{q}{r}\right)^2 + 3p\left(-\frac{q}{r}\right) + 1 = 0$$

$$\frac{-rq^3}{r^3} + \frac{3q^3}{r^2} - \frac{3pq}{r} + 1 = 0$$

$$\frac{-\cancel{r}q^3}{\cancel{r}r^2} + \frac{3q^3}{r^2} - \frac{3pq}{r} + 1 = 0$$

$$\frac{-q^3 + 3q^3 - 3pqr + r^2}{r^2} = 0$$

$$2q^3 - 3pqr + r^2 = 0$$

$$2q^3 = 3pqr - r^2$$

$$\boxed{2q^3 = r(3pq - r)}$$ Hence proved

7.) Solve $18x^3 + 81x^2 + 121x + 60 = 0$, given that one root is equal to half the sum of remaining roots

Sol:- let $\alpha, \beta, \gamma$ are roots of equation

given that one root is equal to half of remaining roots

$$\alpha = \frac{\beta+\gamma}{2}$$

$$2\alpha = \beta + \gamma$$

$$S_1 = \alpha + \beta + \gamma = -\frac{81}{18} = -\frac{9}{2}$$

$$\alpha + 2\alpha = -\frac{9}{2}$$

$$\cancel{3}\alpha = -\frac{\cancel{9}^{3}}{2}$$ $$\boxed{\alpha = -\frac{3}{2}}$$

$\alpha = -\frac{3}{2}$ is one root of equation

$$\begin{array}{c|cccc} -\frac{3}{2} & 18 & 81 & 121 & 60 \\ & 0 & -27 & -81 & -60 \\ \hline & 18 & 54 & 40 & \boxed{0} \end{array}$$

$18x^2+54x+40=0$

$9x^2+27x+20=0$

$9x^2+15x+12x+20=0$

$3x(3x+5)+4(3x+5)=0$

$(3x+5)(3x+4)=0$

$3x+5=0 \qquad 3x+4=0$

$3x=-5 \qquad 3x=-4$

$\boxed{x=-\frac{5}{3}} \qquad \boxed{x=-\frac{4}{3}}$

$\therefore$ The roots are $-\frac{3}{2}, \; -\frac{5}{3}, \; -\frac{4}{3}$

8) Solve $x^4+x^3-16x^2-4x+48=0$, given that product of two roots is 6

Sol:- let $\alpha, \beta, \gamma, \delta$ are roots of equation

given that product of two roots is 6

$\boxed{\alpha\beta = 6}$

$S_1 = \alpha+\beta+\gamma+\delta = -1 \longrightarrow ①$

$S_2 = \alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\gamma\delta+\gamma\alpha = -16 \longrightarrow ②$

$S_3 = \alpha\beta\gamma+\beta\gamma\delta+\gamma\delta\alpha+\delta\alpha\beta = 4 \longrightarrow ③$

$S_4 = \alpha\beta\gamma\delta = 48 \longrightarrow ④$

from equation ④

$\alpha\beta\gamma\delta = 48$

$(\cancel{6})\gamma\delta = \cancel{48}^{8}$

$\boxed{\gamma\delta = 8}$

from equation ③ $\alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \delta\alpha\beta$

$$\alpha\beta(\gamma+\delta) + \gamma\delta(\alpha+\beta) = 4$$

$$6(-1-\alpha-\beta) + 8(\alpha+\beta) = 4$$

$$-6-6\alpha-6\beta+8\alpha+8\beta = 4$$

$$2\alpha+2\beta = 4+6$$

$$2(\alpha+\beta) = 10$$

$$\alpha+\beta = 5 \longrightarrow ⑤$$

from equation ①

$$\gamma+\delta = -1-(\alpha+\beta)$$

$$\gamma+\delta = -1-5$$

$$\boxed{\gamma+\delta = -6} \longrightarrow ⑥$$

case-i

$\alpha+\beta = 5$ , $\alpha\beta = 6$

$$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$$

$$(\alpha-\beta)^2 = 25-24$$

$$(\alpha-\beta)^2 = 1$$

$$\boxed{\alpha-\beta = 1} \longrightarrow ⑦$$

from equation ⑤ & ⑦

$$\alpha+\beta = 5$$
$$\alpha-\beta = 1$$
$$\overline{2\alpha = 6}$$

$$\boxed{\alpha = 3}$$

$\alpha = 3$ in equation ⑤

$$3+\beta = 5 \Rightarrow \boxed{\beta = 2}$$

case-ii

$\gamma+\delta = -6$ , $\gamma\delta = 8$

$$(\gamma-\delta)^2 = (\gamma+\delta)^2 - 4\gamma\delta$$

$$(\gamma-\delta)^2 = 36-32$$

$$(\gamma-\delta)^2 = 4 \Rightarrow \gamma-\delta = 2 \longrightarrow ⑧$$

from equation 6 and 8

$$r+\delta=-6$$
$$r-\delta=2$$

$$2r=-4$$
$$\boxed{r=-2}$$

$r=-2$ in equation 6

$$-2+\delta=-6$$
$$\boxed{\delta=-4}$$

$\therefore$ The roots are $3, 2, -2, -4$

9.) Solve $4x^3-24x^2+23x+18=0$ given that roots are A.P

Sol:- let $a+d, a, a-d$ are roots of equation

$$S_1=a+d+a+a-d=-\frac{(-24)}{4}$$
$$3a=6$$
$$\boxed{a=2}$$

$a=2$ is one of root of given equation

By using synthetic division

| | | | | |
|---|---|---|---|---|
| 2 | 4 | −24 | 23 | 18 |
| | 0 | 8 | −32 | −18 |
| | 4 | −16 | −9 | 0 |

$$4x^2-16x-9=0$$
$$4x^2-18x+2x-9=0$$
$$2x(2x-9)+1(2x-9)=0$$
$$(2x-9)(2x+1)=0$$

$2x-9=0$    $2x+1=0$

$2x=9$    $2x=-1$

$x=\frac{9}{2}$    $x=-\frac{1}{2}$

$\therefore$ The roots are $2, \frac{9}{2}, -\frac{1}{2}$

10.) Solve the equation $x^3 - 7x^2 + 14x - 8 = 0$, given that roots are in G.P

Sol:- let $\frac{a}{r}, a, ar$ are the roots of equation

$$S_3 = \left(\frac{a}{r}\right)(a)(ar) = -\frac{(-8)}{1}$$

$$a^3 = 8$$

$$a^3 = 2^3$$

$$\boxed{a = 2}$$

$a = 2$ is one of root of equation

By using synthetic division

$$\begin{array}{c|cccc} 2 & 1 & -7 & 14 & -8 \\ & 0 & 2 & -10 & 8 \\ \hline & 1 & -5 & 4 & \boxed{0} \end{array}$$

$$x^2 - 5x + 4 = 0$$

$$x^2 - x - 4x + 4 = 0$$

$$x(x-1) - 4(x-1) = 0$$

$$(x-1)(x-4) = 0$$

$x - 1 = 0$ $\quad$ $x - 4 = 0$

$x = 1$ $\quad$ $x = 4$

$\therefore$ The roots are $2, 1, 4$

11.) Solve the equation $x^4 - 6x^3 + 13x^2 - 24x + 36 = 0$, given that they have multiple roots

Sol:- $f(x) = x^4 - 6x^3 + 13x^2 - 24x + 36$

$$f'(x) = 4x^3 - 18x^2 + 26x - 24$$

$$f''(x) = 12x^2 - 36x + 26$$

$$f(3) = (3)^4 - 6(3)^3 + 13(3)^2 - 24(3) + 36$$

$$= 81 - 162 + 117 - 72 + 36$$

$$= 234 - 234$$

$$= 0$$

$f'(3) = 4(3)^3 - 18(3)^2 + 26(3) - 24$

$= 108 - 162 + 78 - 24$

$= 186 - 186$

$= 0$

$f''(3) = 12(3)^2 - 36(3) + 26$

$= 108 - 108 + 26$

$= 26 \neq 0$

$\therefore$ '3' is two times root of equation by using synthetic division

$$\begin{array}{c|ccccc} 3 & 1 & -6 & 13 & -24 & 36 \\ & 0 & 3 & -9 & 12 & -36 \\ \hline 3 & 1 & -3 & 4 & -12 & \boxed{0} \\ & 0 & 3 & 0 & 12 & \\ \hline & 1 & 0 & 4 & \boxed{0} & \end{array}$$

$x^2 + 4 = 0$

$x^2 = -4$

$x = \pm\sqrt{-4}$

$x = \pm 2i$

$\therefore$ The roots are $3, 3, \pm 2i$

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12.) Solve $x^4 + 4x^3 - 2x^2 - 12x + 9 = 0$ given that it has two pairs of equal roots

Sol:- $f(x) = x^4 + 4x^3 - 2x^2 - 12x + 9 = 0$

$f'(x) = 4x^3 + 12x^2 - 4x - 12$

$f(1) = 1 + 4 - 2 - 12 + 9$

$= 0$

$f'(1) = 4 + 12 - 4 - 12$

$= 0$

"1" is two times roots of equation

By using synthetic division

12

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 1 | 4 | -2 | -12 | 9 | |
| | 0 | 1 | 5 | 3 | -9 | |
| | 1 | 5 | 3 | -9 | 0 | |
| 1 | 0 | 1 | 6 | 9 | | |
| | 1 | 6 | 9 | 0 | | |

$x^2+6x+9=0$

$x^2+2\cdot3\cdot x+3^2=0$

$(x+3)^2=0$

$(x+3)(x+3)=0$

$x+3=0$ $\qquad$ $x+3=0$

$\boxed{x=-3}$ $\qquad$ $\boxed{x=-3}$

$\therefore$ The roots are $1, 1, -3, -3$

13.) Find the polynomial equation whose roots are the translate of equation $x^5-4x^4+3x^2-4x+6=0$ by $-3$

Sol:-

| | | | | | | |
|---|---|---|---|---|---|---|
| 3 | 1 | -4 | 0 | 3 | -4 | 6 |
| | 0 | 3 | -3 | -9 | -18 | -66 |
| 3 | 1 | -1 | -3 | -6 | -22 | -60 |
| | 0 | 3 | 6 | 9 | 9 | |
| | 1 | 2 | 3 | 3 | -13 | |
| 3 | 0 | 3 | 15 | 54 | | |
| | 1 | 5 | 18 | 57 | | |
| 3 | 0 | 3 | 24 | | | |
| | 1 | 8 | 42 | | | |
| 3 | 0 | 3 | | | | |
| | 1 | 11 | | | | |

$\therefore$ The required equation is $x^5+11x^4+42x^3+57x^2-13x-60=0$

# BINOMIAL THEOREM (LAQ's)

1. Find the sum of the series. $\frac{3\cdot5}{5\cdot10}+\frac{3\cdot5\cdot7}{5\cdot10\cdot15}+\frac{3\cdot5\cdot7\cdot9}{5\cdot10\cdot15\cdot20}+\cdots\infty$

Sol:- let $S=\frac{3\cdot5}{(1\cdot5)(2\cdot5)}+\frac{3\cdot5\cdot7}{1\cdot5\cdot2\cdot5\cdot3\cdot5}+\frac{3\cdot5\cdot7\cdot9}{1\cdot5\cdot2\cdot5\cdot3\cdot5\cdot2\cdot5\cdot2}+\cdots\infty$

Adding $1+\frac{3}{5}$ on Both sides

$$S+1+\frac{3}{5}=1+\frac{3}{5}+\frac{3\cdot5}{2!\,(5^2)}+\frac{3\cdot5\cdot7}{3!\,(5^3)}+\cdots\infty$$

$$S+1+\frac{3}{5}=1+\frac{3}{1!}\left(\frac{1}{5}\right)+\frac{3\cdot5}{2!}\left(\frac{1}{5}\right)^2+\frac{3\cdot5\cdot7}{3!}\left(\frac{1}{5}\right)^3+\cdots\infty$$

This in the form of

$$\boxed{1+\frac{p}{1!}\left(\frac{x}{q}\right)^1+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+\cdots\infty=(1-x)^{-p/q}}$$

By comparing

$\boxed{p=3}$ $\quad q=p-1$ $\quad \frac{x}{q}=\frac{1}{5}$

$q=3-1$

$\boxed{q=2}$ $\quad \boxed{x=\frac{2}{5}}$

Now, $S+1+\frac{3}{5}=\left[1-\frac{2}{5}\right]^{-3/2}=\left[\frac{5-2}{5}\right]^{-3/2}=\left[\frac{3}{5}\right]^{-3/2}$

$$S+\frac{5+3}{5}=\left[\frac{3}{5}\right]^{-3/2}$$

$$S+\frac{8}{5}=\left[\frac{5}{3}\right]^{3/2}$$

$$S=\left[\frac{5}{3}\right]^{3/2}-\frac{8}{5}=\frac{\sqrt{125}}{\sqrt{27}}-\frac{8}{5}=\frac{5\sqrt{5}}{3\sqrt{3}}-\frac{8}{5}$$

$$\boxed{\therefore S=\frac{5\sqrt{5}}{3\sqrt{3}}-\frac{8}{5}}$$

$$\frac{3\cdot5}{5\cdot10}+\frac{3\cdot5\cdot7}{5\cdot10\cdot15}+\cdots\infty=\frac{5\sqrt{5}}{3\sqrt{3}}-\frac{8}{5}.$$

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2. Find the sum of infinite series $\frac{3}{4}+\frac{3\cdot5}{4\cdot8}+\frac{3\cdot5\cdot7}{4\cdot8\cdot12}+\cdots\infty$

Sol:- let $S=\frac{3}{4}+\frac{3\cdot5}{4\cdot8}+\frac{3\cdot5\cdot7}{4\cdot8\cdot12}+\cdots\infty$

Add one on B.S

$$S+1 = 1+\frac{3}{4}+\frac{3\cdot5}{4\cdot8}+\frac{3\cdot5\cdot7}{4\cdot8\cdot12}+\cdots\infty$$

$$S+1 = 1+\frac{3}{1}\left(\frac{1}{4}\right)+\frac{3\cdot5}{1\cdot4\cdot2\cdot4}+\frac{3\cdot5\cdot7}{1\cdot4\cdot4\cdot2\cdot3\cdot2\cdot2}+\cdots\infty$$

$$S+1 = 1+\frac{3}{1!}\left(\frac{1}{4}\right)+\frac{3\cdot5}{2!}\left(\frac{1}{4}\right)^2+\frac{3\cdot5\cdot7}{3!}\left(\frac{1}{4}\right)^3+\cdots\infty$$

Comparing with

$$\boxed{(1-x)^{-p/q} = 1+\frac{p}{1!}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+\cdots\infty}$$

$\boxed{p=3}$ $\quad p+q=5 \quad \frac{x}{q}=\frac{1}{4} \quad \boxed{x=1/2}$

$q=5-3 \quad x=2/4=1/2$

$\boxed{q=2}$

$$S+1 = \left[1-\frac{1}{2}\right]^{-3/2} = \left[\frac{2-1}{2}\right]^{-3/2} = \left[\frac{1}{2}\right]^{-3/2} = (2)^{3/2}$$

$$S+1 = \sqrt{8} = 2\sqrt{2}$$

$$S = 2\sqrt{2}-1$$

$$\therefore \frac{3}{4}+\frac{3\cdot5}{4\cdot8}+\frac{3\cdot5\cdot7}{4\cdot8\cdot12}+\cdots\infty = 2\sqrt{2}-1$$

3. Find the sum of the series $1+\frac{1}{3}+\frac{1\cdot3}{3\cdot6}+\frac{1\cdot3\cdot5}{3\cdot6\cdot9}+\cdots\infty$.

Sol:- Let $S = 1+\frac{1}{3}+\frac{1\cdot3}{3\cdot6}+\cdots\infty$

$$S = 1+\frac{1}{3}+\frac{1\cdot3}{1\cdot3\cdot2\cdot3}+\frac{1\cdot3\cdot5}{1\cdot3\cdot3\cdot2\cdot3\cdot3}+\cdots\infty$$

$$S = 1+\frac{1}{1!}\left(\frac{1}{3}\right)+\frac{1\cdot3}{2!}\left(\frac{1}{3}\right)^2+\frac{1\cdot3\cdot5}{3!}\left(\frac{1}{3}\right)^2+\cdots\infty$$

Comparing with

$$\boxed{1+\frac{p}{1!}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+\frac{p(p+q)(p+2q)}{3!}\left(\frac{x}{q}\right)^3+\cdots\infty = (1-x)^{-p/q}}$$

$\boxed{p=1}$ $\quad p+q=3 \quad \frac{x}{q}=\frac{1}{3}$

$q=3-1 \quad \boxed{x=2/3}$

$\boxed{q=2}$

$$S = \left[1-\frac{2}{3}\right]^{-1/2} = \left[\frac{3-2}{3}\right]^{-1/2} = \left[\frac{1}{3}\right]^{-1/2}$$

$$S=(3)^{1/2}=\sqrt{3}$$

$$1+\frac{1}{3}+\frac{1\cdot3}{3\cdot6}+\frac{1\cdot3\cdot5}{3\cdot6\cdot9}+\cdots\infty=\sqrt{3}.$$

4. Find the sum of infinite series

$$1+\frac{2}{3}\cdot\frac{1}{2}+\frac{2\cdot5}{3\cdot6}\left(\frac{1}{2}\right)^2+\frac{2\cdot5\cdot8}{3\cdot6\cdot9}\left(\frac{1}{2}\right)^3+\cdots\infty$$

Sol:- let $S=1+\frac{2}{3}\left(\frac{1}{2}\right)+\frac{2\cdot5}{3\cdot6}\left(\frac{1}{2}\right)^2+\frac{2\cdot5\cdot8}{3\cdot6\cdot9}\left(\frac{1}{2}\right)^3+\cdots\infty$

$$S=1+\frac{2}{1!}\left(\frac{1}{6}\right)+\frac{2\cdot5}{1\cdot3\cdot2\cdot3}+\frac{2\cdot5\cdot8}{1\cdot3\cdot2\cdot3\cdot3\cdot3}+\cdots\infty$$

$$S=1+\frac{2}{1!}\left(\frac{1}{6}\right)+\frac{2\cdot5}{2!}\left(\frac{1}{6}\right)^2+\frac{2\cdot5\cdot8}{3!}\left(\frac{1}{6}\right)^3+\cdots\infty$$

Comparing with

$$\boxed{(1-x)^{-p/q}=1+\frac{p}{1!}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+\cdots\infty}$$

$\boxed{p=2}$ $\quad p+q=5$ $\quad \frac{x}{q}=\frac{1}{6}$ $\quad \boxed{x=1/2}$

$\boxed{q=3}$

$$\therefore S=\left(1-\frac{1}{2}\right)^{-2/3}$$

$$S=\left(\frac{1}{2}\right)^{-2/3}$$

$$S=(2)^{2/3}=\sqrt[3]{4}.$$

$$S=1+\frac{2}{3}\cdot\frac{1}{2}+\frac{2\cdot5}{3\cdot6}\left(\frac{1}{2}\right)^2+\cdots\infty=\sqrt[3]{4}.$$

5. If $x=\frac{1\cdot3}{3\cdot6}+\frac{1\cdot3\cdot5}{3\cdot6\cdot9}+\frac{1\cdot3\cdot5\cdot7}{3\cdot6\cdot9\cdot12}+\cdots$ then prove that $9x^2+24x=11$

Sol: Add $1+\frac{1}{3}$ on B.S

$$1+\frac{1}{3}+x=1+\frac{1}{3}+\frac{1\cdot3}{3\cdot6}+\frac{1\cdot3\cdot5}{3\cdot6\cdot9}+\cdots\infty$$

$$\frac{3+1+3x}{3}=1+\frac{1}{3}+\frac{1\cdot3}{1\cdot3\cdot2\cdot3}+\frac{1\cdot3\cdot5}{1\cdot3\cdot2\cdot3\cdot3\cdot3}+\cdots\infty$$

$$\frac{4+3x}{3}=1+\frac{1}{1!}\left(\frac{1}{3}\right)+\frac{1\cdot3}{2!}\left(\frac{1}{3}\right)^2+\frac{1\cdot3\cdot5}{3!}\left(\frac{1}{3}\right)^3+\cdots\infty$$

$$\frac{4}{3}+x = 1+\frac{1}{1!}\left(\frac{1}{3}\right)+\frac{1\cdot 3}{2!}\left(\frac{1}{3}\right)^2+ \cdots \infty$$

Comparing with

$$\boxed{(1-x)^{-p/q} = 1+\frac{p}{1!}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+ \cdots \infty}$$

$\boxed{p=1}$ $\quad p+q=3 \quad \frac{x}{q}=\frac{1}{3}$

$\boxed{q=2}$ $\quad \boxed{x=2/3}$

$$\frac{4}{3}+x = \left[1-\frac{2}{3}\right]^{-1/2}$$

$$\frac{4}{3}+x = \left[\frac{3-2}{3}\right]^{-1/2} = \left[\frac{1}{3}\right]^{-1/2}$$

$$\frac{4}{3}+x = \sqrt{3}$$

$$4+3x = 3\sqrt{3}$$

S.O.B.S

$$9x^2+16+24x = 27.$$

$$\boxed{9x^2+24x=11}$$

6. If $t = \frac{4}{5}+\frac{4\cdot 6}{5\cdot 10}+\frac{4\cdot 6\cdot 8}{5\cdot 10\cdot 15}+ \cdots \infty$ then prove that $9t=16$.

Sol:- Adding one on B.S

$$1+t = 1+\frac{4}{5}+\frac{4\cdot 6}{5\cdot 10}+\frac{4\cdot 6\cdot 8}{5\cdot 10\cdot 15}+ \cdots \infty$$

$$1+t = 1+\frac{4}{1!}\left(\frac{1}{5}\right)+\frac{4\cdot 6}{2!}\left(\frac{1}{5}\right)^2+\frac{4\cdot 6\cdot 8}{3!}\left(\frac{1}{5}\right)^3+ \cdots \infty$$

Comparing with

$$\boxed{(1-x)^{-p/q} = 1+\frac{p}{1!}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+ \cdots \infty}$$

$\boxed{p=4}$ $\quad p+q=6 \quad \frac{x}{q}=\frac{1}{5}$

$\boxed{q=2}$ $\quad \boxed{x=2/5}$

$$1+t = \left(1-\frac{2}{5}\right)^{-4/2}$$

$$1+t = \left(\frac{3}{5}\right)^{-4/2}$$

$$1+t=\left(\frac{5}{3}\right)^2$$

$$1+t=\frac{25}{9}$$

$$t=\frac{25}{9}-1 \quad =\frac{25-9}{9} \quad =\frac{16}{9}$$

$$\boxed{9t=16}$$

Hence proved.

7. If $x=\frac{5}{(2!)3}+\frac{5\cdot 7}{(3!)3^2}+\frac{5\cdot 7\cdot 9}{4!(3^3)}+\cdots$, then find the value of $x^2+4x$.

Sol:- multiply & divide by 3

$$x=\frac{3\cdot 5}{(2!)3^2}+\frac{3\cdot 5\cdot 7}{(3!)3^3}+\frac{3\cdot 5\cdot 7\cdot 9}{4!(3^4)}+\cdots$$

Adding $1+\frac{3}{3}$ on B.S

$$1+\frac{3}{3}+x=1+\frac{3}{3}+\frac{3\cdot 5}{2!}\left(\frac{1}{3^2}\right)+\frac{3\cdot 5\cdot 7}{3!}\left(\frac{1}{3^3}\right)+\cdots$$

$$2+x=1+\frac{3}{1!}\left(\frac{1}{3}\right)+\frac{3\cdot 5}{2!}\left(\frac{1}{3}\right)^2+\frac{3\cdot 5\cdot 7}{3!}\left(\frac{1}{3}\right)^3+\cdots$$

Comparing with

$$\boxed{1+\frac{p}{1!}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+\cdots\infty=(1-x)^{-p/q}}$$

$\boxed{p=3}$ $\quad p+q=3$ $\quad \frac{x}{q}=\frac{1}{3}$

$\boxed{q=2}$ $\quad \boxed{x=2/3}$

Now, $2+x=\left[1-\frac{2}{3}\right]^{-3/2}$

$$2+x=\left[\frac{3-2}{3}\right]^{-3/2}=\left[\frac{1}{3}\right]^{-3/2}$$

$$2+x=(3)^{3/2}$$

$$2+x=\sqrt{27}$$

S.O.B.S

$$4+x^2+4x=27.$$

$$\boxed{x^2+4x=23}.$$

Hence Proved.

8. If $x = \frac{1}{5} + \frac{1\cdot3}{5\cdot10} + \frac{1\cdot3\cdot5}{5\cdot10\cdot15} + \cdots \infty$, then find $3x^2+6x$

Sol:- Adding one on B·S

$$1+x = 1+\frac{1}{5}+\frac{1\cdot3}{5\cdot10}+\frac{1\cdot3\cdot5}{5\cdot10\cdot15}+\cdots\infty$$

$$1+x = 1+\frac{1}{5}+\frac{1\cdot3}{1\cdot5\cdot2\cdot5}+\frac{1\cdot3\cdot5}{1\cdot5\cdot2\cdot5\cdot3\cdot5}+\cdots\infty$$

$$1+x = 1+\frac{1}{1!}\left(\frac{1}{5}\right)+\frac{1\cdot3}{2!}\left(\frac{1}{5}\right)^2+\frac{1\cdot3\cdot5}{3!}+\cdots\infty$$

comparing

$$\boxed{1+\frac{p}{1!}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+\frac{p(p+q)(p+2q)}{2}\left(\frac{x}{q}\right)^3+\cdots(1-x)^{-p/q}}$$

$\boxed{p=1}$ $\quad p+q=3$ $\quad \frac{x}{q}=\frac{1}{5}$

$\boxed{q=2}$ $\quad \boxed{x=2/5}$

$$x+1=\left[1-\frac{2}{5}\right]^{-1/2}$$

$$=\left[\frac{5-2}{5}\right]^{-1/2}$$

$$=\left[\frac{3}{5}\right]^{-1/2}$$

$$x+1=\left(\frac{5}{3}\right)^{1/2}$$

$$x+1=\sqrt{\frac{5}{3}}$$

S·O·B·S

$$x^2+1+2x=\frac{5}{3}$$

$$3x^2+3+6x=5$$

$$\boxed{3x^2+6x=2}$$

Hence proved.

9. Find the sum of the infinite series.

$$\frac{7}{5}\left[1+\frac{1}{10^2}+\frac{1\cdot3}{1\cdot2}\left[\frac{1}{10^4}\right]+\frac{1\cdot3\cdot5}{1\cdot2\cdot3}\left[\frac{1}{10^6}\right]+\cdots\right]$$

Sol:- $\frac{7}{5}\left[1+\frac{1}{1!}\left(\frac{1}{10^2}\right)^1+\frac{1\cdot3}{2!}\left(\frac{1}{10^2}\right)^2+\frac{1\cdot3\cdot5}{3!}\left(\frac{1}{10^2}\right)^3+\cdots\right]$

$$= \frac{7}{5}\left[1+\frac{1}{1!}\left[\frac{1}{100}\right]+\frac{1\cdot3}{2!}\left[\frac{1}{100}\right]^2+\frac{1\cdot3\cdot5}{3!}\left[\frac{1}{100}\right]^3+\cdots\cdots\right]$$

$$\boxed{1+\frac{p}{1!}\left[\frac{x}{q}\right]+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2+\cdots\cdots=(1-x)^{-p/q}}$$

$\boxed{p=1}$ $\quad p+q=3 \quad \frac{x}{q}=\frac{1}{100}$

$\boxed{q=2}$ $\quad \boxed{x=\frac{1}{50}}$

$$= \frac{7}{5}\left[\left[\frac{50-1}{50}\right]^{-1/2}\right] = \frac{7}{5}\left[\frac{49}{50}\right]^{-1/2} = \frac{7}{5}\left[\frac{50}{49}\right]^{1/2}$$

$$= \frac{7}{5}\sqrt{\frac{50}{49}} = \frac{\cancel{7}}{5}\times\frac{\sqrt{50}}{\cancel{7}} = \frac{\sqrt{50}}{5} = \frac{5\sqrt{2}}{5} = \sqrt{2}$$

$$= \frac{7}{5}\left[1+\frac{1}{10^2}+\frac{1\cdot3}{1\cdot2}\left(\frac{1}{10^4}\right)+\cdots\cdots\right] = \sqrt{2}.$$

10. Find the sum of the infinite series.

$$S = \frac{3}{4\cdot8} - \frac{3\cdot5}{4\cdot8\cdot12} + \frac{3\cdot5\cdot7}{4\cdot8\cdot12\cdot16} - \cdots\cdots$$

Sol:- Multiply with one on both sides.

$$S = \frac{1\cdot3}{4\cdot8} - \frac{1\cdot3\cdot5}{4\cdot8\cdot12} + \cdots\cdots\infty$$

Add $1-\frac{1}{4}$ on B.S

$$S+1-\frac{1}{4} = 1-\frac{1}{4}+\frac{1\cdot3}{4\cdot8}-\frac{1\cdot3\cdot5}{4\cdot8\cdot12}+\cdots\cdots$$

$$S+\frac{3}{4} = 1-\frac{1}{4}+\frac{1\cdot3}{1\cdot4\cdot2\cdot4}-\frac{1\cdot3\cdot5}{1\cdot4\cdot2\cdot4\cdot3\cdot4}+\cdots\cdots$$

$$S+\frac{3}{4} = 1-\frac{1}{4}+\frac{1\cdot3}{2!}\left(\frac{1}{4}\right)^2-\frac{1\cdot3\cdot5}{3!}\left(\frac{1}{4}\right)^3+\cdots\cdots$$

$$S+\frac{3}{4} = 1-\frac{1}{1!}\left(\frac{1}{4}\right)+\frac{1\cdot3}{2!}\left(\frac{1}{4}\right)^2-\frac{1\cdot3\cdot5}{3!}\left(\frac{1}{4}\right)^3+\cdots\cdots$$

Comparing B.S

$$\boxed{1-\frac{p}{1!}\left(\frac{x}{q}\right)+\frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2-\cdots\cdots=(1+x)^{-p/q}}$$

$\boxed{p=1}$ $\quad p+q=3 \quad \frac{x}{q}=\frac{1}{4}$

$\boxed{q=2}$ $\quad \boxed{x=1/2}$

$x = 2/4 = 1/2$

$$S+\frac{3}{4}=\left(1+\frac{1}{2}\right)^{-1/2}$$

$$S+\frac{3}{4}=\left(\frac{3}{2}\right)^{-1/2}=\left(\frac{2}{3}\right)^{1/2}=\sqrt{\frac{2}{3}}$$

$$\boxed{S=\sqrt{\frac{2}{3}}-\frac{3}{4}}$$

11. Show that for any non-zero rational number $x$,

$$1+\frac{x}{2}+\frac{x(x-1)}{2\cdot 4}+\frac{x(x-1)(x-2)}{2\cdot 4\cdot 6}+\cdots=1+\frac{x}{3}+\frac{x(x+1)}{3\cdot 6}+\frac{x(x+1)(x+2)}{3\cdot 6\cdot 9}.$$

Sol:- $LHS=1+\frac{x}{2}+\frac{x(x-1)}{1\cdot 2\cdot 2\cdot 2}+\frac{x(x-1)(x-2)}{1\cdot 2\cdot 2\cdot 2\cdot 2\cdot 3}+\cdots$

$$=1+\frac{x}{2}+\frac{x(x-1)}{2!}\left(\frac{1}{2}\right)^2+\frac{x(x-1)(x-2)}{3!}\left(\frac{1}{2}\right)^3+\cdots$$

$$=1+\frac{x}{1!}\left(\frac{1}{2}\right)+\frac{x(x-1)}{2!}\left(\frac{1}{2}\right)^2+\frac{x(x-1)(x-2)}{3!}\left(\frac{1}{2}\right)^3+\cdots$$

$\boxed{P=x}$ $\quad P+q=x-1$ $\quad \frac{x}{q}=\frac{1}{2}$

$\cancel{x}+q=\cancel{x}-1$ $\quad \boxed{x=-1/2}$

$\boxed{q=-1}$

$$\left(1+\frac{1}{2}\right)^{-x/-1}=\left(1+\frac{1}{2}\right)^{x}=\left(\frac{3}{2}\right)^{x}$$

$RHS=1+\frac{x}{3}+\frac{x(x+1)}{1\cdot 3\cdot 3\cdot 2}+\frac{x(x+1)(x+2)}{3\cdot 6\cdot 9}+\cdots$

$$=1+\frac{x}{1!}+\frac{x(x+1)}{2!}\left(\frac{1}{3}\right)^2+\frac{x(x+1)(x+2)}{3!}\left(\frac{1}{3}\right)^3+\cdots$$

$\boxed{P=x}$ $\quad P+q=x+1$ $\quad \frac{x}{q}=\frac{1}{3}$

$\cancel{x}+q=\cancel{x}+1$ $\quad x=1/3$

$\boxed{q=1}$

$$\left[1-\frac{1}{3}\right]^{-x}=\left[\frac{3-1}{3}\right]^{-x}=\left[\frac{2}{3}\right]^{-x}$$

$$=\left[\frac{3}{2}\right]^{x}.$$

$\therefore LHS=RHS$

(17)

12. If 'n' is a positive integer & x is any non-zero real, ~~real~~ number, then prove that.

$$C_0+C_1\cdot\frac{x}{2}+C_2\cdot\frac{x^2}{3}+\cdots\cdots+C_n\frac{x^n}{n+1}=\frac{(1+x)^{n+1}-1}{(n+1)x}$$

Sol:- $LHS = C_0+C_1\cdot\frac{x}{2}+C_2\frac{x^2}{3}+\cdots\cdots+C_n\frac{x^n}{n+1}$

let $S={}^nC_0+{}^nC_1\frac{x}{2}+{}^nC_2\frac{x^2}{3}+\cdots\cdots+{}^nC_n\frac{x^n}{n+1}$

Multiply $x$ on B.S

$x\cdot S={}^nC_0\,x+{}^nC_1\frac{x^2}{2}+\cdots\cdots{}^nC_n\frac{x^{n+1}}{n+1}$

Multiply $n+1$ on B.S

$(n+1)x\cdot S=n+1\,{}^nC_0\,x^1+n+1\,{}^nC_1\frac{x^2}{2}+\cdots\,n+1\,{}^nC_n\frac{x^{n+1}}{n+1}$.

$(n+1)x\cdot S={}^{n+1}C_1\,x^1+{}^{n+1}C_2\,x^2+\cdots\cdots+{}^{n+1}C_{n+1}\,x^{n+1}$.

Adding & sub ${}^{n+1}C_0$ on B.S.

$(n+1)x\cdot S={}^{n+1}C_0+{}^{n+1}C_1\,x^1+{}^{n+1}C_2\,x^2+\cdots\cdots+{}^{n+1}C_{(n+1)}x^{n+1}-{}^{n+1}C_0$

$(n+1)x\cdot S=(1+x)^{n+1}-1$

$$S=\frac{(1+x)^{n+1}-1}{(n+1)x}$$

LHS = RHS.

13. For $r=0,1,2\cdots\cdots n$ prove that $C_0C_r+C_1C_{r+1}+C_2C_{r+2}+\cdots\cdots+C_{n-r}\cdot C_n={}^{2n}C_{n+r}$. Hence deduce that

i, $C_0^2+C_1^2+C_2^2+\cdots\cdots+C_n^2={}^{2n}C_n$

ii, $C_0\cdot C_1+C_1\cdot C_2+C_2\cdot C_3+\cdots\cdots+C_{n-1}\cdot C_n={}^{2n}C_{n+1}$

Sol:- $(1+x)^n=C_0+C_1x^1+C_2x^2+\cdots\cdots+C_rx^2+\cdots\cdots+C_nx^n$ ——— (1)

$(x+1)^n=C_0x^n+C_1x^{n-1}+C_2x^{n-2}+\cdots\cdots+C_n$ ——— (2)

from (1) & (2)

$(1+x)^n(x+1)^n=[C_0+C_1x^1+C_2x^2+\cdots\cdots+C_2x^r+C_nx^n]$

$=[C_0x^n+C_1x^{n-1}+\cdots C_rx^{n-r}+C_n]$

RHS = in the coefficient of $x^{n+r}$ is

$$C_0C_r + C_1\cdot C_{r+1} + C_2\cdot C_{r+2} + \cdots + C_{n-r}\cdot C_n \quad \text{——— (3)}$$

LHS $= (1+x)^n(x+1)^n = (1+x)^{2n}$.

The coefficient of $x^{n+r}$ in LHS ${}^{2n}C_{n+r}$ ——— (4).

from eq (3) & (4).

$$C_0C_r + C_1\cdot C_{r+1} + C_2\cdot C_{r+2} + \cdots + C_{n-r}\cdot C_n = {}^{2n}C_{(n+r)} \quad \text{——— (5)}$$

Case i, Put $r=0$ in eq (5).

$$= C_0\cdot C_0 + C_1C_1 + C_2\cdot C_2 = {}^{2n}C_n$$

$$C_0^2 + C_1^2 + C_2^2 + \cdots + C_n^2 = {}^{2n}C_n.$$

Case ii Put $r=1$ in eq (5).

$$C_0C_1 + C_1C_2 + C_2C_3 + C_{n-1}\cdot C_n = {}^{2n}C_{n+1}.$$

14. Prove that $(C_0+C_1)\cdot(C_1+C_2)\cdot(C_2+C_3)\cdots(C_{n-1}+C_n) = \frac{(n+1)^n}{n!}C_0\cdot C_1\cdot C_2\cdots C_n$.

Sol:- Given,

$$(C_0+C_1)(C_1+C_2)\cdot(C_2+C_3)\cdots(C_{n-1}+C_n) = \frac{(n+1)^n}{n!}C_0\cdot C_1\cdot C_2\cdots C_n$$

Now, take LHS.

$$\text{LHS} = (C_0+C_1)(C_1+C_2)(C_2+C_3)\cdots(C_{n-1}+C_n)$$

$$= C_0\left(1+\frac{C_1}{C_0}\right)\cdot C_1\left(1+\frac{C_2}{C_1}\right)\cdot C_2\left(1+\frac{C_3}{C_2}\right)\cdots C_{n-1}\left(1+\frac{C_n}{C_{n-1}}\right) C_0C_1C_2\cdots C_{n-1}$$

$$\boxed{\text{WKT } \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}}$$

$$= \left(1+\frac{n-1+1}{1}\right)\left(1+\frac{n-2+1}{2}\right)\left(1+\frac{n-3+1}{3}\right)\cdots\left(1+\frac{n-n+1}{n}\right)C_0\cdot C_1C_2\cdots C_{n-1}$$

$$= \left(\frac{1+n}{1}\right)\left(\frac{2+n-2+1}{2}\right)\left(\frac{3+n-3+1}{3}\right)\cdots\left(\frac{n+1}{n}\right)C_0C_1C_2\cdots C_n$$

$$\boxed{C_n = 1}$$

$$= \left(\frac{n+1}{1}\right)\left(\frac{n+1}{2}\right)\left(\frac{n+1}{3}\right)\cdots\left(\frac{n+1}{n}\right)C_0\cdot C_1\cdot C_2\cdots C_n.$$

$$= \frac{(n+1)^n}{n!} \quad C_0 \cdot C_1 \cdot C_2 \text{ - - - - } C_n$$

$= RHS$.

15. If the coefficients of 4 consecutive terms in the expansion of $(1+x)^n$ are $a_1, a_2, a_3, a_4$ respectively then show that

$$\frac{a_1}{a_1+a_2} + \frac{a_3}{a_3+a_4} = \frac{2a_2}{a_2+a_3}.$$

Sol:- Expansion.

$$(1+x)^n = {}^nC_0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \text{ - - - - - -}$$

$$a_1 = {}^nC_{r-1} \quad a_2 = {}^nC_r \quad a_3 = {}^nC_{r+1} \quad a_4 = {}^nC_{r+2}.$$

$$LHS = \frac{a_1}{a_1+a_2} + \frac{a_3}{a_3+a_4}$$

$$= \frac{{}^nC_{r-1}}{{}^nC_{r-1} + {}^nC_r} + \frac{{}^nC_{r+1}}{{}^nC_{r+1} + {}^nC_{r+2}}$$

$$\boxed{\text{WKT}, \ {}^nC_r + {}^nC_{r-1}}$$

$$= \frac{{}^nC_{r-1}}{{}^{n+1}C_r} + \frac{{}^nC_{r+1}}{{}^{n+1}C_{r+2}}$$

$$= \frac{\cancel{{}^nC_{r-1}}}{\frac{n+1}{r}\,\cancel{{}^nC_{r-1}}} + \frac{\cancel{{}^nC_{r+1}}}{\frac{n+1}{r+2}\,\cancel{{}^nC_{r+1}}}$$

$$= \frac{r}{n+1} + \frac{r+2}{n+1} = \frac{2r+2}{n+1}$$

$$LHS = \frac{2(r+1)}{n+1} \text{ ———————— } (1)$$

$$RHS = \frac{2a_2}{a_2+a_3}$$

$$= \frac{2\,{}^nC_r}{{}^nC_r + {}^nC_{r+1}}$$

$$\boxed{{}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r}$$

$$= \frac{2\,{}^nC_r}{{}^{n+1}C_{r+1}}$$

$$= \frac{2\,{}^n\cancel{C_r}}{\frac{n+1}{r+1}\,{}^n\cancel{C_r}}$$

$$\boxed{RHS = \frac{2(r+1)}{n+1}} \quad \text{———— (2)}$$

From eq (1) & (2).

$$\frac{a_1}{a_1+a_2} + \frac{a_3}{a_3+a_4} = \frac{2a_2}{a_3+a_2}$$

LHS = RHS

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16. If the $2^{nd}$, $3^{rd}$, $4^{th}$ terms in the expansion of $(a+x)^n$ are respectively 240, 720, 1080 & find $a, x, n$.

Sol:- Given,

$$(a+x)^n = {}^nC_0\, a^n + {}^nC_1\, a^{n-1} x + {}^nC_2\, a^{n-2} x^2 + \ldots$$

$$\boxed{\text{WKT} \quad T_{r+1} = {}^nC_r\, x^{n-r} a^r}$$

$2^{nd}$ term $= T_2 \quad = {}^nC_1\, a^{n-1} x^1 = 240$ ———— (1)

$3^{rd}$ term $= T_3 \quad = {}^nC_2\, a^{n-2} x^2 = 720$ ———— (2)

$4^{th}$ term $= T_4 \quad = {}^nC_3\, a^{n-3} x^3 = 1080$ ———— (3)

$$\frac{eq\,(2)}{eq\,(1)} = \frac{{}^nC_2\, a^{n-2} x^2}{{}^nC_1\, a^{n-1} x} = \frac{\cancel{720}\ 3}{\cancel{240}}$$

$$\boxed{\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}}$$

$$\left(\frac{n-1}{2}\right)\frac{x}{a} = 3$$

$$\boxed{\frac{x}{a} = \frac{6}{n-1}} \quad \text{———— (4)}$$

$$\frac{eq\,(3)}{eq\,(2)} = \frac{{}^nC_3\, a^{n-3} \cancel{x^3}}{{}^nC_2\, a^{n-2} \cancel{x^2}} = \frac{1080}{720} = \frac{9}{6} = \frac{3}{2}$$

$$\left(\frac{n-3+1}{3}\right)a^{\cancel{n}-3-\cancel{n}+2} x = \frac{3}{2}.$$

$$\left(\frac{n-2}{3}\right)a^{-1}x = \frac{3}{2}$$

$$\frac{x}{a} = \left(\frac{9}{2(n-2)}\right)$$

$$\boxed{\frac{x}{a} = \frac{9}{2n-4}} \quad \text{——— (5)}$$

from eq (4) & (5)

$$\frac{6}{n-1} = \frac{9}{2n-4}$$

$$12n-24 = 9n-9$$

$$3n = 15$$

$$\boxed{n=5}$$

Put $n=5$ in eq (4)

$$\frac{x}{a} = \frac{6}{5-1}$$

$$2x = 3a.$$

$$\boxed{x = \frac{3a}{2}} \quad \text{——— (6)}$$

$n, x$ values sub in eq (1)

$$^{n}c_1\, a^{n-1} x^1 = 240.$$

$$5c_1\, a^{5-1}\left(\frac{3a}{2}\right) = 240.$$

$$\cancel{5}\cdot a^4\left(\frac{\cancel{3}a}{2}\right) = \cancel{240}\ \cancel{48}\ 16$$

$$a^5 = 32$$

$$a^5 = 2^5$$

$$\boxed{a=2}$$

Put $x = \frac{6}{2} = 3$

$$\boxed{x=3}$$

$$\therefore \boxed{x=3,\ a=2,\ n=5}$$

17. If the coefficient of $r^{th}$, $(r+1)^{th}$ & $(r+2)^{nd}$ terms in the expansion of $(1+x)^n$ are in A.P, then show that

$n^2-(4r+1)n+4r^2-2=0$

Sol:- $(1+x)^n = {}^nC_0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \cdots$

$$\boxed{T_{r+1} = {}^nC_r\, x^{n-r} a^r}$$

$$T_r = T_{r-1+1} = {}^nC_{r-1}$$

$$T_{r+1} = T_{r+1} = {}^nC_r$$

$$T_{r+2} = T_{r+1+1} = {}^nC_{r+1}$$

${}^nC_{r-1}, {}^nC_r, {}^nC_{r+1}$ are in AP then

$$2 = \frac{{}^nC_{r-1}}{{}^nC_r} + \frac{{}^nC_{r+1}}{{}^nC_r} \qquad \boxed{\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}}$$

$$2 = \frac{r}{n-(r-1)} + \frac{n-r}{r+1}$$

$$2 = \frac{r}{n-r+1} + \frac{n-r}{r+1}$$

$$2 = \frac{r(r+1)+(n-r+1)(n-r)}{(n-r+1)(r+1)}$$

$$2(n-r+1)(r+1) = r^2+\cancel{r}+n^2-rn+n-rn+r^2-\cancel{r}$$

$$2nr-2r^2+2r+2n-2r+2 = -2rn+2r^2-n^2+n$$

$$2nr+2rn-2r^2-2r^2+2n-n-n^2+2=0$$

$$n^2-4rn+4r^2-n-2=0$$

$$\boxed{n^2-(4r+1)n+4r^2-2=0}$$

18. If the coefficients of $x^9, x^{10}, x^{11}$ terms in the expansion of $(1+x)^n$ are in AP then PT $n^2-41n+398=0$

Sol:- Expansion $(1+x)^n = {}^nC_0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \cdots$

The coefficient of $x^9 = {}^nC_9$

The coefficient of $x^{10} = {}^nC_{10}$

The coefficient of $x^{11} = {}^nC_{11}$.

${}^nC_9, {}^nC_{10}, {}^nC_{11}$ are in A.P.

$\boxed{a, b, c \text{ are in A.P then } 2b = a+c}$

$$2\,{}^nC_{10} = {}^nC_9 + {}^nC_{11}$$

$$\boxed{\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}}$$

$$2 = \frac{{}^nC_9}{{}^nC_{10}} + \frac{{}^nC_{11}}{{}^nC_{10}}$$

$$2 = \frac{10}{n-10+1} + \frac{n-11+1}{11}$$

$$2 = \frac{10}{n-9} + \frac{n-10}{11}$$

$$2 = \frac{110+(n-9)(n-10)}{(n-9)11}$$

$$22(n-9) = 110 + n^2 - 10n - 9n + 90.$$

$$22n - 198 = 110 + n^2 - 19n + 90.$$

$$\boxed{n^2 - 41n + 398 = 0}$$

---

19. If the P & Q are the sum of odd terms & the sum of even terms respectively in the expansion of $(x+a)^n$ then prove that i) $P^2 - Q^2 = (x^2-a^2)^n$.

ii) $4PQ = (x+a)^{2n} - (x-a)^{2n}$.

Sol:- Given, P & Q are the sum of odd terms & the sum of even terms respectively $(x+a)^n$.

$$\boxed{(x+a)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1}a + {}^nC_2 x^{n-2}a^2 + {}^nC_3 x^{n-3}a^3 + \cdots}$$

$$= ({}^nC_0 x^n + {}^nC_2 x^{n-2}a^2 + \cdots) + ({}^nC_1 \cancel{xa} x^{n-1} + {}^nC_3 x^{n-3}a^3 + \cdots)$$

= (sum of even terms) + (sum of odd terms).

$$(x-a)^n = P - Q \quad \text{———} \quad (1).$$

$$\boxed{(x-a)^n = {}^nC_0 x^n - {}^nC_1 x^{n-1}a^1 - {}^nC_2 x^{n-2}a^2 - {}^nC_3 x^{n-3}x^3 + \cdots}$$

$$\cancel{= ({}^nC_0 x^n + {}^nC_2 x^{n-2}a^2 + \cdots) - (\text{sum of odd terms})}$$

$$= ({}^nC_0 x^n + {}^nC_2 x^{n-2}a^2 + \cdots) - ({}^nC_1 x + {}^nC_3 x^{n-3}a^3 + \cdots).$$

= (sum of even terms) − (sum of odd terms)

$(x-a)^n = P - Q$ ——— (2).

i, $(P^2-Q^2) = (P+Q)(P-Q)$.

$= (x+a)^n (x-a)^n$.

$\boxed{(P^2-Q^2) = (x^2-a^2)^n}$

ii, $4PQ = (P+Q)^2 - (P-Q)^2$

$= \left((x+a)^n\right)^2 - \left((x-a)^n\right)^2$.

$\boxed{4PQ = (x+a)^{2n} - (x-a)^{2n}}$.

20. If the coefficient of $x^{10}$ in the expansion of $\left(ax^2+\frac{b}{x}\right)^{11}$ is equal to the coefficient of $x^{-10}$ in the expansion of $\left(ax-\frac{1}{bx^2}\right)^{11}$, then find the relation b/w a & b, where a & b are real numbers.

Sol:- The general term in expansion of $\left(ax^2+\frac{1}{bx^2}\right)$.

Step 1: $T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx}\right)^r$.

$= {}^{11}C_r \frac{a^{11-r}}{b^r} x^{22-2r} \quad x^{-r}$

$= {}^{11}C_r \frac{a^{11-r}}{b^r} x^{22-3r}$.

The coefficient of $x^{10}$ in the expansion is

$22 - 3r = 10$

$12 = 3r$

$\boxed{r=4}$

$T_{r+1} = {}^{11}C_4 \frac{a^{11-4}}{b^4}$

$= {}^{11}C_4 \frac{a^7}{b^4}$ ——— (1).

Step-2: The general term in the expansion $\left(ax-\frac{1}{bx^2}\right)$

$T_{r+1} = {}^{11}C_r (ax)^{11-r} \left(\frac{-1}{bx^2}\right)^r$.

$$= (-1)^r \, {}^{11}C_r \, \frac{a^{11-r}}{b^r} \, x^{11-r} \, x^{-2r}$$

$$= (-1)^r \, {}^{11}C_r \, \frac{a^{11-r}}{b^r} \, x^{11-3r}.$$

The coefficient of $x^{-10}$ in the expansion of.

$11-3r=-10$.

$21 = 3r$

$\boxed{r=7}$

$$T_{r+1} = (-1)^7 \, {}^{11}C_7 \, \frac{a^{11-7}}{b^7}.$$

$$= -{}^{11}C_7 \, \frac{a^4}{b^7} \quad \text{———} (2)$$

The coefficients 1 & 2 are equal.

$$ {}^{11}C_4 \frac{a^7}{b^4} = -{}^{11}C_7 \frac{a^4}{b^7}$$

$$ {}^{11}C_4 \frac{\cancel{a^4}.a^3}{\cancel{b^4}} = -{}^{11}C_4 \frac{\cancel{a^4}}{\cancel{b^4}.b^3}$$

$$a^3 = -\frac{1}{b^3}$$

$$a^3b^3 = -1$$

$$(ab)^3 = -1$$

$$\boxed{ab=1}$$

21. Suppose that 'n' is a natural number and I, F are integral part & fractional part of $(7+4\sqrt{3})^n$, then show that i, I is an odd Integer. ii, $(I+F)(I-F)=1$

Sol:- Given,

$I+f = (7+4\sqrt{3})^n$ if $0<f<1$ ———— (1)

det $G = (7-4\sqrt{3})^n$.

$6<4\sqrt{3}<7$.

$-7<-4\sqrt{3}<-6$

$7-7<(7-4\sqrt{3})<7-6$.

$0<7-4\sqrt{3}<1$.

$0<(7-4\sqrt{3})^n<1$.

$0<G<1$ ——— (2)

from (1) & (2).

$0<f<1$

$\underline{0<G<1}$

$0<f+G<2$

$\boxed{f+G=1}$ ——— (3)

i, To prove that I is an odd integer

$I+f+G=(7+4\sqrt{3})^n+(7-4\sqrt{3})^n$.

$=\left[{}^nC_0\, 7^n+{}^nC_1\, 7^{n-1}\cancel{(4\sqrt{3})^1}+{}^nC_2\, 7^{n-2}(4\sqrt{3})^2\right]+$

$\left[{}^nC_0\, 7^n-{}^nC_1\, 7^{n-1}\cancel{(4\sqrt{3})}+{}^nC_2\, 7^{n-1}(4\sqrt{3})^2\right]$

$=2\left[{}^nC_0\, 7^n+{}^nC_2\, 7^{n-2}(4\sqrt{3})^2+\cdots\right]$

$=2$[some integer] $=2$[even integer].

$I+f+G=$ even integer.

$I+1=$ even integer [from (3)]

$I=$ even integer $-1$

$\boxed{I = \text{odd integer}}$.

ii, $(I+f)(I-f)=1$

WKT from eq (3) $\quad f+G=1$

$\boxed{G=f-1}$

$=(I+f)(1-f)$

$=(I+f)(G)$

$=(7+4\sqrt{3})^n(7-4\sqrt{3})^n$.

$=(49-48)^n$

$=(1)^n$

$=1$

$=$ RHS.

# MEASURES OF DISPERSION

1. Calculate the variance and standard deviation of the following continous frequency distribution.

| class Interval | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 | 90-100 |
|---|---|---|---|---|---|---|---|
| frequency | 3 | 7 | 12 | 15 | 8 | 3 | 2 |

Sol:- Now construct the table with given data.

| class interval (C.I) | frequency $f_i$ | Mid point ($x_i$) | $y_i = \frac{x_i - 65}{10}$ | $y_i^2$ | $f_i y_i$ | $f_i y_i^2$ |
|---|---|---|---|---|---|---|
| 30-40 | 3 | 35 | -3 | 9 | -9 | 27 |
| 40-50 | 7 | 45 | -2 | 4 | -14 | 28 |
| 50-60 | 12 | 55 | -1 | 1 | -12 | 12 |
| 60-70 | 15 | 65 | 0 | 0 | 0 | 0 |
| 70-80 | 8 | 75 | 1 | 1 | 8 | 8 |
| 80-90 | 3 | 85 | 2 | 4 | 6 | 12 |
| 90-100 | 2 | 95 | 3 | 9 | 6 | 18. |
| | $N=50$ | | | | $\Sigma f_i y_i = -15$ | $\Sigma f_i y_i^2 = 5$ |

Here width of class interval $(n) = 10$ and

Assumed mean $(A) = 65$

$$\boxed{\text{Mean } (\bar{x}) = A + \left(\frac{\Sigma f_i y_i}{N}\right) \times n}$$

$= 65 + \left(\frac{-15}{50} \times 10\right) = 62$

$$\boxed{\text{variance } (\sigma_x^2) = \frac{h^2}{N^2}\left[N \sum f_i y_i^2 - \left(\sum f_i y_i\right)^2\right]}$$

$$= \frac{100}{2500}\left[50(105) - (-15)^2\right]$$

$$= \frac{1}{25}\left[5250 - 225\right] = 201$$

Standard deviation $(\sigma_x) = \sqrt{\text{variance}} = \sqrt{201} = 14.18$ (Approxi)

2. find the mean deviation from the median for the following data.

| $x_i$ | 6 | 9 | 3 | 12 | 15 | 13 | 21 | 22 |
|---|---|---|---|---|---|---|---|---|
| $f_i$ | 4 | 5 | 3 | 2 | 5 | 4 | 4 | 3 |

Sol Keeping the observations in ascending order.

| $x_i$ | $f_i$ | c.f | $\lvert x_i - \text{median}\rvert$ | $f_i \lvert x_i - \text{median}\rvert$ |
|---|---|---|---|---|
| 3 | 3 | 3 | 10 | 30 |
| 6 | 4 | 7 | 7 | 28 |
| 9 | 5 | 12 | 4 | 20 |
| 12 | 2 | 14 | 1 | 2 |
| 13 | 4 | 18 | 0 | 0 |
| 15 | 5 | 23 | 2 | 10 |
| 21 | 4 | 27 | 8 | 32 |
| 22 | 3 | 30 | 9 | 27 |

$\frac{\Sigma f_i}{2} = \frac{N}{2} = \frac{30}{2} = 15$

Median is the value of $x_i$ whose cummulative frequency is just greater than or equal to 15

median $= 13$.

$\therefore$ median of mean deviation $= \frac{\Sigma f_i |x_i - \text{median}|}{\Sigma f_i}$

$= \frac{149}{30}$

$= 4.97$

3. The following table gives the daily wages of workers in a factory compute the standard deviation and coefficient of variation of wages of workers.

| wages (RS) | 125-175 | 175-225 | 225-275 | 275-325 | 325-375 | 375-425 | 475-525 | 525-575 |
|---|---|---|---|---|---|---|---|---|
| No. of workers | 2 | 22 | 19 | 14 | 3 | 6 | 1 | 1 |

Sol

| wages | frequency $f_i$ | Mid point | $y_i = \frac{x_i - A}{h}$ | $f_i y_i$ | $f_i y_i^2$ |
|---|---|---|---|---|---|
| 125-175 | 2 | 150 | -3 | -6 | 18 |
| 175-225 | 22 | 200 | -2 | -44 | 88 |
| 225-275 | 19 | 250 | -1 | -19 | 19 |
| 275-325 | 14 | 300 | 0 | 0 | 0 |
| 325-375 | 3 | 350 | 1 | 3 | 3 |
| 375-425 | 4 | 400 | 2 | 8 | 16 |
| 425-475 | 6 | 450 | 3 | 18 | 54 |
| 475-525 | 1 | 500 | 4 | 4 | 16 |
| 525-575 | 1 | 550 | 5 | 5 | 25 |
| | $N = 72$ | | $\Sigma f_i y_i = -31$ | $\Sigma f_i y_i = -31$ | $\Sigma f_i y_i^2 = 239$ |

Here width of class interval $(h) = 50$

and Assumed mean $(A) = 300$

$\boxed{\text{mean } (\bar{x}) = A + \left(\frac{\Sigma f_i y_i}{N}\right) \times h} = 300 + \left(\frac{-31}{72}\right) 50$

$= 300 - \frac{1550}{72} = 278.47$

$\boxed{\text{Variance } (\sigma_x^2) = \frac{h^2}{N^2} = \left(N \Sigma f_i y_i^2 - (\Sigma f_i y_i)^2\right]}$

$= \frac{2500}{72 \times 72} \left[72(239) - 131 \times 31)\right]$

Standard deviation $= \sqrt{2500\left(\frac{239}{72} - \frac{961}{72 \times 72}\right)} = 88.52$

co-efficient of variation $= \frac{\sigma_x}{\bar{x}} \times 100 = \frac{88.52}{278.47} \times 100$

$= 31.79.$

4. find the mean deviation about the median from the following continous distribution.

| class interval | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
|---|---|---|---|---|---|---|---|---|
| frequency | 5 | 8 | 7 | 12 | 28 | 20 | 10 | 10 |

sol:-

| class interval | frequency $f_i$ | cummulative frequency $(x_i)$ | Mid point $(x_i)$ | $\lvert x_i - \text{median}\rvert$ $\lvert x_i - 46.43\rvert$ | $f_i \lvert x_i - \text{med}\rvert$ |
|---|---|---|---|---|---|
| 0-10 | 5 | 5 | 5 | 41.43 | 207.15 |
| 10-20 | 8 | 13 | 15 | 31.43 | 251.44 |
| 20-30 | 7 | 20 | 25 | 21.43 | 150.01 |
| 30-40 | 12 | 32 | 35 | 11.43 | 137.16 |
| 40-50 | 28 | 60 | 45 | 1.43 | 40.04 |

| | | | | | |
|---|---|---|---|---|---|
| 50-60 | 20 | 80 | 55 | 8.57 | 171.4 |
| 60-70 | 10 | 90 | 65 | 18.57 | 185.7 |
| 70-80 | 10 | 100 | 75 | 28.57 | 285.7 |

$\Sigma f_i|x_i - \text{median}| = 1428.6$

Here ; $\frac{N}{2} = \frac{100}{2} = 50$

lies in the class interval 40-50. This is medial class

$$\text{median} = L + \frac{\frac{N}{2} - P.C.f}{f} \times h$$

Here ; L = is the lower limit of median class = 40

P.C.f = preceeding cummulative frequency = 32

f = frequency cummulative = 28

h = width of median class = 10

$$\text{median} = 40 + \frac{50-32}{28} \times 10 = 40 + \frac{18}{28} \times 10 = 40 + 6.43$$

$$= 46.43$$

mean deviation about median $= \frac{1}{N} \sum_{i=1}^{8} f_i|x_i - \text{med}|$

$$= \frac{1}{100}(1428.6) = 14.286$$

$$= 14.29 \text{ (App)}$$

5. Find the mean deviation about the mean for the following continous distribution.

| Height (in cms) | 95-105 | 105-115 | 115-125 | 125-135 | 135-145 | 145-155 |
|---|---|---|---|---|---|---|
| No. of Boys | 9 | 13 | 26 | 30 | 12 | 10 |

| Sol: Height (C.I) | No. of Boys ($f_i$) | Mid point ($x_i$) | $f_i x_i$ | $\lvert x_i - \bar{x} \rvert$ | $f_i \lvert x_i - \bar{x} \rvert$ |
|---|---|---|---|---|---|
| 95-105 | 9 | 100 | 900 | 25.3 | 227.7 |
| 105-115 | 13 | 110 | 1430 | 15.3 | 198.9 |
| 115-125 | 26 | 120 | 3120 | 5.3 | 137.8 |
| 125-135 | 30 | 130 | 3900 | 4.7 | 141 |
| 135-145 | 12 | 140 | 1680 | 14.7 | 176.4 |
| 145-155 | 10 | 150 | 1500 | 24.7 | 247 |
| | N=100 | | $\Sigma f_i x_i = 12530$ | | $\Sigma f_i \lvert x_i - \bar{x} \rvert = 1128.8$ |

$$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{12530}{100} = 125.3.$$

Mean deviation from the mean $= \frac{1}{N} \Sigma f_i \lvert x_i - \bar{x} \rvert$

$$= \frac{1}{100} (1128.8)$$

$$= 11.288$$

$$= 11.29$$

6. find the mean and variance using the step deviation method.

| Age in Years | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 |
|---|---|---|---|---|---|---|---|
| No. of members | 3 | 61 | 132 | 153 | 140 | 51 | 2 |

25

Sol:

| class interval (C.I) | Mid point ($x_i$) | No. of members ($f_i$) | $y_i = \frac{x_i - 55}{h}$ | $f_i y_i$ | $f_i y_i^2$ |
|---|---|---|---|---|---|
| 20-30 | 25 | 3 | -3 | -9 | 27 |
| 30-40 | 35 | 61 | -2 | -12.2 | 244 |
| 40-50 | 45 | 132 | -1 | -132 | 132 |
| 50-60 | 55 | 153 | 0 | 0 | 0 |
| 60-70 | 65 | 140 | 1 | 140 | 140 |
| 70-80 | 75 | 51 | 2 | 102 | 204 |
| 80-90 | 85 | 2 | 3 | 6 | 18 |

$N = 542$ $\quad \Sigma f_i y_i = -15 \quad \Sigma f_i y_i^2 = 765$

A. Assumed value = 55, h = width of CI = 10

$$\text{Mean } (\bar{x}) = A + \frac{\Sigma f_i y_i}{N} \times h$$

$$55 + \frac{(-15)}{542} \times 10 = 55 - 0.28 = 54.72$$

$$\text{variance } (\sigma^2) = \frac{h^2}{N^2}[N \Sigma f_i y_i^2 - (\Sigma f_i y_i)^2]$$

$$\frac{100}{(542)^2}[542(765) - 225]$$

$$= \frac{100}{293764}[414630 - 225]$$

$$= \frac{100}{293764} \times 414405$$

$$141.067 = 141.07$$

7. find the mean deviation from the mean of the following data, using the step deviation.

| Marks | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 |
|---|---|---|---|---|---|---|---|
| No. of students | 6 | 5 | 8 | 15 | 7 | 6 | 5 |

Sol to find the required statics, we shall construct the following table

| Class interval | Midpoint ($x_i$) | No. of students ($f_i$) | $y_i = \frac{x_i - A}{h}$, $y_i = \frac{x_i - 35}{10}$ | $f_i y_i$ | $\lvert x_i - \bar{x}\rvert$ | $f_i \lvert x_i - \bar{x}\rvert$ |
|---|---|---|---|---|---|---|
| 0-10 | 5 | 6 | -3 | -18 | 28.4 | 170.4 |
| 10-20 | 15 | 5 | -2 | -10 | 18.4 | 92 |
| 20-30 | 25 | 8 | -1 | -8 | 8.4 | 67.2 |
| 30-40 | 35 | 15 | 0 | 0 | 1.6 | 24.0 |
| 40-50 | 45 | 7 | 1 | 7 | 11.6 | 81.2 |
| 50-60 | ~~65~~ 55 | 6 | 2 | 12 | 21.6 | 129.6 |
| 60-70 | 65 | 3 | 3 | 9 | 31.6 | 94.8 |
| | | $N = 50$ | | $\Sigma = -8$ | | 659.2 |

Here width of class interval $(h) = 10$ and assumed mean $(A) = 35$, $N = 50$

Here $N = 50$ mean $= A + \frac{(\Sigma f_i y_i)}{N} \times h = \frac{35}{1} + \frac{(-8)10}{50}$

$= 33.4$

mean deviation from mean $\frac{1}{N} \Sigma f_i \lvert x_i - \bar{x}\rvert$

$= \frac{1}{50}(659.2)$

$13.18$

# PROBABILITY

L·A·Q'S

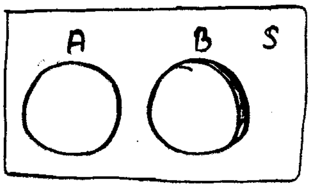
$L_{23}$ (1) State and prove addition theorem on probability

**Sol:-** Statement :- If A, B are only two events of a random experiment and P is a probability function then.

$$P(A\cup B) = P(A)+P(B)-P(A\cap B)$$

proof: if A, B are only two events of a random experiment and 'P' is probability function.

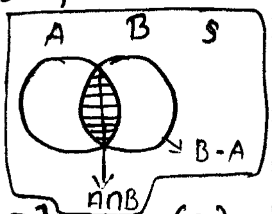
Case (i) $A\cap B=\phi$




$$P(A\cap B)=0$$
$$P(A\cup B) = P(A)+P(B)$$
$$= P(A)+P(B)-0$$
$$P(A\cup B) = P(A)+P(B)-P(A\cap B) \longrightarrow (1)$$

Case (ii) $A\cap B\neq\phi$

$$A\cup B = A\cup(B-A) \text{ and } A\cap B = A\cap(B-A)=\phi$$




$$P(A\cup B] = P[A\cup(B-A)]$$
$$= P(A)+P(B-A)$$
$$= P(A)+P(B-A\cap B)$$
$$P(A\cup B) = P(A)+P(B)-P(A\cap B] \longrightarrow (2)$$

From (1) & (2)

$$P(A\cup B) = P(A)+P(B)-P(A\cap B)$$

$L_{23}$ (2) Define conditional probability state and prove the multiplication theorem on probability.

**Sol:** Statement : If A and B are two events of a random experiment with $P(A)>0$ and

$P(B)>0$

By the definition and conditional probability

$$P\left(\frac{B}{A}\right) = \frac{P(A\cap B)}{P(A)}$$

$$P(B/A)P(A) = P(A\cap B) \longrightarrow (1)$$

Again $P(A/B) = \dfrac{P(A\cap B)}{P(B)}$

$$P(B)\,P(A/B) = P(A\cap B) \longrightarrow (2)$$

from (1) & (2)

$$P(A\cap B) = P(A)P(B/A) = P(B)\,P(A/B)$$

$L_{23}$ (3) State and prove Bay's theorem

Sol: Statement :- Suppose $E_1, E_2 \text{------} E_n$ are $n$ mutually exclusive and exhaustive events of a random experiment with $P(E_i) \neq 0$ for $i = 1,2 \text{----} n$ then for any event $A$ of the random experiment with $P(A) \neq 0$ $P\left(\frac{E_k}{A}\right) = \dfrac{P[E_k]\,P(A/E_k)}{\sum_{i=1}^{n} P(E_i)\,P(A/E_i)}$

proof: Suppose $E_1, E_2 \text{-----} E_n$ are $n$ mutually exclusive and exhaustive events of a random experiment

$$S = \bigcup_{i=1}^{n} E_i \text{ [Mutually exhaustive events]}$$

for any event $A$

$$A = A\cap S$$

$$= A\cap \left[\bigcup_{i=1}^{n} E_i\right]$$

$$A = \bigcup_{i=1}^{n} A\cap E_i$$

applying probability on both sides

$$P(A) = P\left[\bigcup_{i=1}^{n} A\cap E_i\right]$$

$$= \sum_{i=1}^{n} P(A\cap E_i)$$

$$P(A) = \sum_{i=1}^{n} P(E_i)\, P(A/E_i) \longrightarrow (1)$$

Now $P(E_k/A) = \dfrac{P(A \cap E_k)}{P(A)}$

$$= \frac{P(E_k)P(A/E_k)}{P(A)}$$

$$P\left(\frac{E_k}{A}\right) = \frac{P(E_k)P(A/E_k)}{\sum_{i=1}^{n} P(E_i)P(A/E_i)} \quad \text{from (i)}$$

from multiplication theorem $\boxed{P(A \cap B) = P(B)P(A/B)}$

from conditional Probability $P(B/A) = \dfrac{P(A \cap B)}{P(A)}$

4) Three boxes Numberd I, II, III contains the balls as follows

| Box | White | Black | Red |
|---|---|---|---|
| I | 1 | 2 | 3 |
| II | 2 | 1 | 1 |
| III | 4 | 5 | 3 |

One box is randomly selected and a ball is drawn from it. If the ball is red then find the probability that is from box II

Sol: Given

| Box | White | Black | Red | Total |
|---|---|---|---|---|
| I | 1 | 2 | 3 | 6 |
| II | 2 | 1 | 1 | 4 |
| III | 4 | 5 | 3 | 12 |

Let $A_1, A_2, A_3$ be the events selecting from boxes I, II, III respectively.

$$P(A_1) = 1/3,\ P(A_2) = 1/3,\ P(A_3) = 1/3$$

Let 'R' be the event to draw a Red ball from a box

$$P(R/A_1) = 3/6 = 1/2,\ P(R/A_2) = 1/4.\ P(R/A_3) = \frac{3}{12} = 1/4$$

the probability that to draw a red ball from box II.

from bayes theorem:

$$P(A_2/R) = \frac{P(A_2)\, P(R/A_2)}{P(A_1)P(R/A_1) + P(A_2)P(R/A_2) + P(A_3)P(R/A_3)}$$

$$= \frac{\left(\frac{1}{3}\right)\left(\frac{1}{4}\right)}{\frac{1}{3}\left(\frac{1}{2}\right)+\frac{1}{3}\left(\frac{1}{4}\right)+\frac{1}{3}\left(\frac{1}{4}\right)}$$

$$= \frac{\cancel{\frac{1}{3}}\left(\frac{1}{4}\right)}{\cancel{\frac{1}{3}}\left[\frac{1}{2}+\frac{1}{4}+\frac{1}{4}\right]}$$

$$= \frac{\frac{1}{4}}{\frac{2+1+1}{4}}$$

$$= \frac{1}{4}$$

5) Three boxes $B_1$, $B_2$ and $B_3$ contain balls with different colours as shown below.

| Box | White | black | Red |
|---|---|---|---|
| $B_1$ | 2 | 1 | 2 |
| $B_2$ | 3 | 2 | 4 |
| $B_3$ | 4 | 3 | 2 |

A die is thrown $B_1$ is choosen if either 1 or 2 turn ups. $B_2$ is choosen if 3 or 4 turn ups and $B_3$ is choosen if 5 or 6 turn up. Having choosen a box in this way a ball is choosen at random from this box. If the ball drawn is found to be red, find the probability that it is drawn from box $B_2$.

Sol: Given

| Box | White | Black | Red | total |
|---|---|---|---|---|
| $B_1$ | 2 | 1 | 2 | 5 |
| $B_2$ | 3 | 2 | 4 | 9 |
| $B_3$ | 4 | 3 | 2 | 9 |

Let $A_1, A_2, A_3$ be the events of selecting from boxes $B_1, B_2, B_3$ respectively. A die is thrown $B_1$ is selecting either 1 (or) 2 turns ups.

$B_2$ is selecting if 3 (or) 4 turns ups. $B_3$ is selecting if 5 (or) 6 turns ups.

$$P(A_1) = 2/6 = 1/3, \quad P(A_2) = \frac{2}{6} = \frac{1}{3}, \quad P(A_3) = \frac{2}{6} = \frac{1}{3}$$

(28)

Let 'R' be the events drawn a red ball from a box

$P(R/A_1) = 2/5$ , $P(R/A_2) = 4/9$ , $P(R/A_3) = 2/9$

the probability that to draw a ball from box.

By using bayes theorem:

$$P\left(\frac{A_2}{R}\right) = \frac{P(A_2)\,P(R/A_2)}{P(A_1)P(R/A_1) + P(A_2)P(R/A_2) + P(A_3)P(R/A_3)}$$

$$= \frac{\frac{1}{3}\left(\frac{4}{9}\right)}{\frac{1}{3}\cdot\frac{2}{5} + \frac{1}{3}\cdot\frac{4}{9} + \frac{1}{2}\cdot\frac{2}{9}} = \frac{\cancel{\frac{1}{3}}\left(\frac{4}{9}\right)}{\cancel{\frac{1}{3}}\left[\frac{2}{5} + \frac{4}{9} + \frac{2}{9}\right]}$$

$$= \frac{\frac{4}{\cancel{9}}}{\frac{18+20+10}{\cancel{45}\,5}} = \frac{20}{48} = \frac{5}{12}$$

6) Three urns have the following composition of balls urn I : 1 white, 2 black, urn II : 2 white, 1 black, urn III : 2 white, 2 black. one of the urn is selected at random and a ball is drawn, It turns out to be white.

Sol: Given:

| urn | white | Black | Total |
|---|---|---|---|
| I | 1 | 2 | 3 |
| II | 2 | 1 | 3 |
| III | 2 | 2 | 4 |

Let $A_1, A_2, A_3$ be the events of selecting from urn I, II, III respectively $P(A_1) = \frac{1}{3}$, $P(A_2) = \frac{1}{3}$, $P(A_3) = \frac{1}{3}$

let 'w' be the event to draw white ball from a urn $P(W/A_1) = \frac{1}{3}$, $P(W/A_2) = \frac{2}{3}$, $P(W/A_3) = 2/4 = 1/2$

the probability that to draw a white ball from the urn III.

from Bayes theorem:

$$P\left(\frac{A_3}{W}\right) = \frac{P(A_3)\,P(W/A_3)}{P(A_1)P(W/A_1) + P(A_2)P(W/A_2) + P(A_3)P(W/A_3)}$$

$$= \frac{\cancel{\frac{1}{3}}\cdot\frac{1}{2}}{1/3 \cdot 1/3 + 1/3 \cdot 2/3 + \cancel{1/3}\cdot 1/2}$$

$$= \frac{\cancel{\frac{1}{3}}\left(\frac{1}{2}\right)}{\cancel{\frac{1}{3}}\left[\frac{1}{3}+\frac{2}{3}+\frac{1}{2}\right]} = \frac{1/\cancel{2}}{\frac{2+4+3}{\cancel{6}3}} = \frac{3}{9} = \frac{1}{3}$$

7) If one ticket is randomly selected from the ticket numbered 1 to 30, then find the probability that the number on the ticket is :

i) a multiple of 5 or 7  ii) a multiple of 3 or 5.

Sol: The no.of ways of selecting from ticket numbers 1 to 30 is $n(S) = 30$

i) A multiple of 5(or)7:

Let event 'A' and 'B' are the multiple of 5 and 7 respectively.

$$A = \{5,10,15,20,25,30\}$$

$$n(A) = 6$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{30} = \frac{1}{5}$$

$$B = \{7,14,21,28\}$$

$$n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{30}$$

$A \cap B = \phi$

$P(A \cap B) = 0$

the probability that the number on the tickets is a multiple of 5 (or) 7.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{6}{30} + \frac{4}{30} - 0$$

$$= \frac{10}{30} = \frac{1}{3}$$

ii) A multiple of 3(or)5 :

Let events A and B are the multiple of 3(or)5.respectively

$$A = \{3,6,9,12,15,18,21,24,27,30\}$$

$$n(A) = 30$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{10}{30}$$

$$B = \{5,10,15,20,25,30\}$$

$$n(B) = 6$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{6}{30}$$

$A \cap B$ $\{15,30\}$. $n(A \cap B) = 2$

$$P(A\cap B) = \frac{n(A\cap B)}{n(S)} = \frac{2}{30}$$

the probability that the number on the ticket is the multiple of 3(or) 5.

$$P(A\cup B) = P(A)+P(B)-P(A\cap B)$$
$$= \frac{10}{30}+\frac{6}{30}-\frac{2}{30}$$
$$= \frac{16-2}{30}$$
$$= \frac{14}{30}$$

8) Define conditional probability bag $B_1$ contains 4 white and 2 black balls. Bag $B_2$ contains 3 white and 4 black balls. A bag is drawn at random and a ball is choosen at random from it. What is the probability that the ball drawn is white.

Sol: Conditional probability: If A and B are two events of a sample space then the probability of B after the event A has occured is called conditional probability of B given A, and is denoted by $P(B/A)$.

We define $P(B/A) = \frac{P(B\cap A)}{P(A)}$, $P(A)\neq 0$

| Bag | White | Black | total |
|---|---|---|---|
| $B_1$ | 4 | 2 | 6 |
| $B_2$ | 3 | 4 | 7 |

let $A_1, A_2$ are the events of selecting from bags $B_1, B_2$ respectively $P(A_1) = \frac{1}{2}$, $P(A_2) = \frac{1}{2}$

let 'w' be the events drawn a white ball from the box

$P(W/A_1) = \frac{4}{6}$, $P(W/A_2) = 3/7$

the probability to draw a white ball

$$W = (W\cap A_1)\cup(W\cap A_2)$$
$$P(W) = P[(W\cap A_1)\cup(W\cap A_2)]$$
$$= P(W\cap A_1) + P(W\cap A_2)$$
$$= P(A_1)P(W/A_1) + P(A_2)P(W/A_2)$$
$$= \frac{1}{2}\cdot\frac{4}{6}+\frac{1}{2}\cdot\frac{3}{7}$$

$$= \frac{1}{2}\left[\frac{4}{6}+\frac{3}{7}\right]$$

$$= \frac{1}{2}\left[\frac{28+18}{42}\right]$$

$$= \frac{1}{\cancel{2}}\left[\frac{\cancel{46}^{23}}{42}\right]$$

$$= \frac{23}{42}$$

9) Three are 3 blacks and 4 white balls in one bag, 4 black and 3 white balls in second bag. A die is rolled and the first bag is selected if the die shows up 1(or) 3, and the second bag for the rest. Find the probability of drawing a black ball, from the bag thus selected.

Ans:

| Bag | Blacks | White | Total |
|---|---|---|---|
| $B_1$ | 3 | 4 | 7 |
| $B_2$ | 4 | 3 | 7 |

Let $A_1, A_2$ be the events selecting from bag $B_1, B_2$ respectively 1st bag is selected the die show up 1(or) 3 and 2nd Bag for the rest. $P(A_1) = \frac{2}{6} = \frac{1}{3}$ $P(A_2) = \frac{4}{6} = \frac{2}{3}$

Let B be the event drawn a black ball from a bag. $P(B/A_1) = \frac{3}{7}$ $P(B/A_2) = \frac{4}{7}$

the probability of drawing a black ball from the bag.

$$B = (B \cap A_1) \cup (B \cap A_2)$$

$$P(B) = P[(B \cap A_1) \cup (B \cap A_2)]$$

$$P(B) = P(B \cap A_1) + P(B \cap A_2)$$

$$= P(A_1) P(B/A_1) + P(A_2) P(B/A_2)$$

$$= \frac{1}{3} \cdot \frac{3}{7} + \frac{2}{3} \cdot \frac{4}{7}$$

$$= \frac{3}{21} + \frac{8}{21}$$

$$= \frac{3+8}{21}$$

$$P(B) = \frac{11}{21}$$

$$\boxed{(B \cap A_1) \cap (B \cap A_2) = \phi \quad P[(B \cap A_1) \cap (B \cap A_2)] = 0}$$

$$\boxed{P(A \cap B) = P(B)\, P(A/B)}$$

10) A,B,C are 3 newspapers from a city 20% of the population read A, 16% read B, 14% read C, 8% read both A and B, 5% read both A and C 4% read both B and C and 2% read all the three. Find the percentage of the population who read atleast one newspaper.

Ans: Let A,B,C are the events reading newspaper A, B, C respectively.

$P(A) = 20\% = \frac{20}{100}$, $P(B) = 16\% = \frac{16}{100}$, $P(C) = 14\% = \frac{14}{100}$

$P(A\cap B) = 8\% = \frac{8}{100}$, $P(A\cap C) = 5\% = \frac{5}{100}$, $P(B\cap C) = 4\% = \frac{4}{100}$

$P(A\cap B\cap C) = 2\% = \frac{2}{100}$

$$\boxed{P(A\cup B\cup C) = P(A)+P(B)+P(C) - P(A\cap B) - P(B\cap C) - P(C\cap A) + P(A\cap B\cap C)}$$

$$= \frac{20}{100} + \frac{16}{100} + \frac{14}{100} - \frac{8}{100} - \frac{5}{100} - \frac{4}{100} + \frac{2}{100}$$

$$= \frac{52-17}{100} = \frac{35}{100} = 35\%.$$

∴ The percentange of the populations read atleast 1 newspapers is 35%.

11) In a box containing 15 bulbs, 5 are defective. If 5 bulbs are selected at random from the box. Find the probability of the event that i) None of them is defective ii) only one of them is defective iii) atleast one of them is defective.

Ans: No. of bulbs = 15

Defective bulbs = 5 ; Non defective bulbs = 10

Let s be the sample space in the experiment of 5 bulbs are selected from the 15 bulbs.

$$n(S) = 15C_5 = \frac{\cancel{15}^{3} \times \cancel{14}^{7} \times 13 \times \cancel{12} \times 11}{1\times \cancel{2} \times \cancel{3} \times \cancel{4} \times \cancel{5}} = 3003$$

i) None of them is defective:

Let A be the event of getting of them defective bulb is $n(A) = 10C_5 = \frac{\cancel{10} \times \cancel{9}^{3} \times \cancel{8}^{2} \times 7 \times 6}{1\times \cancel{2} \times \cancel{3} \times \cancel{4} \times \cancel{5}} = 252$

$$P(A) = \frac{n(A)}{n(S)} = \frac{252}{3003} = \frac{12}{143}$$

∴ The probability that none of them defective bulb is $\frac{12}{143}$

ii) Only one of them is defective :

Let B be the event of getting one of them is defective and 4 are non defective bulbs

$$n(B) = 5c_1 \times 10c_4 = 5 \times \frac{10 \times \cancel{9}^{3} \times \cancel{8} \times 7}{1 \times \cancel{2} \times \cancel{3} \times \cancel{4}} = 1050$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{1050}{3003} = \frac{50}{143}$$

∴ the probability that the only one of them is defective bulb is $\frac{50}{143}$

iii) atleast one of them is defective :

Let C be the event of getting atleast of them is defective bulb.

$P(C) = 1 - P(\text{none of them is defective bulbs})$

$$P(C) = 1 - \frac{12}{143}$$

$$P(C) = \frac{143-12}{143} = \frac{131}{143}$$

12) Two persons A and B are rolling a die on the condition that the persons who gets 3 will win the game. If A starts the games then the find the probabilities of A and B respectively to win the game

Ans: Let P be the probability of Success and Q be the probability of failure, we will get 3 and not getting 3 respectively. $\boxed{P = \frac{1}{6}}$

$$P + q = 1$$

$$q = 1 - P \Rightarrow q = 1 - \frac{1}{6}$$

$$\boxed{q = 5/6}$$

Two persons A and B are rolling a die. If A start the game.

| A | B |
|---|---|
| $p$ | $qp$ |
| $qqp$ | $qqqp$ |
| $qqqqp$ | $\vdots$ |
| $\vdots$ | |

$$P(A \text{ to win the game}) = p + qqp + qqqqp + \cdots$$
$$= p + (q^2)^1 p + (q^2)^2 p + \cdots$$

$\boxed{a + ar^1 + ar^2 + \cdots \text{ are in G.P then } S_\infty = \frac{a}{1-r}}$

$$= \frac{p}{1-q^2} = \frac{\frac{1}{6}}{1-\frac{25}{36}} = \frac{\frac{1}{\cancel{6}}}{\frac{36-25}{\cancel{36}\,6}}$$

$$\boxed{P(A) = \frac{6}{11}}$$

$$P(B \text{ to win the game}) = 1 - P(A)$$
$$= 1 - \frac{6}{11}$$
$$= \frac{11-6}{11}$$
$$\boxed{P(B) = \frac{5}{11}}$$

13) In an experiment of drawing a card at random from a pack, the event of getting a spade is denoted by A and getting a pictured card (king, queen (or) jack) is denoted by B. Find the probabilities of A, B, $A \cap B$, $A \cup B$.

Ans: the no. of ways of drawing a plain cards from a pack $n(s) = 52$

The event of getting a spade is denoted by A.

$$n(A) = 13$$
$$P(A) = \frac{n(A)}{n(s)} = \frac{13}{52}$$

The event of getting a picture card $\{K, Q, J\}$ is denoted by B.

$n(B) = 12$

$P(B) = \frac{n(B)}{n(S)} = \frac{12}{52}$

$A \cap B = \{J, k, Q\} = 3$

$n(A \cap B) = 3$

$\therefore P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{3}{52}$

$\therefore$ the probability of getting spade (or) picture card is $\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$

$= \frac{13}{52} + \frac{12}{52} - \frac{3}{52}$

$= \frac{25-3}{52} = \frac{22}{52}$

$P(A \cup B) = \frac{11}{26}$

14) In a shooting test the probability of A, B, C hitting the target 6 are $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$ respectively. If all of them fire at the same target. Find the probability that.

i) only one of them hit the tangent ii) atleast one of them hits the target.

Ans: Let A, B, C be the events of shooting test hit the targets A, B, C respectively.

$P(A) = \frac{1}{2}$ $P(B) = \frac{2}{3}$ $P(C) = \frac{3}{4}$

$P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{2}$ $\quad P(\bar{B}) = 1 - P(B)$ $\quad P(\bar{C}) = 1 - P(C)$

$P(\bar{A}) = \frac{1}{2}$ $\quad P(\bar{B}) = 1 - \frac{2}{3}$ $\quad = 1 - \frac{3}{4}$

$P(\bar{B}) = \frac{1}{3}$ $\quad P(\bar{C}) = \frac{1}{4}$

i) only one of them hit the target:

the probability that only one of them hit the target $= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap \bar{C})$

$= P(A)P(\bar{B})P(\bar{C}) + P(\bar{A})P(B)P(\bar{C}) + P(\bar{A})P(\bar{B})P(C)$

$= \frac{1}{2}\cdot\frac{1}{3}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{1}{3}\cdot\frac{3}{4}$

$= \frac{1+2+3}{24} = \frac{6}{24} = \frac{1}{4}$

ii) Atleast one of them hit the target:

the probability that atleast one of them hit the target.

$= 1 - P(\text{none of them hit the target})$

$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$

$= 1 - P(\bar{A})P(\bar{B})P(\bar{C})$

$= 1 - \left(\frac{1}{2}\right)\cdot\left(\frac{1}{3}\right)\cdot\left(\frac{1}{4}\right)$

$= 1 - \frac{1}{24}$

$= \frac{23}{24}$

# RANDOM VARIABLES

① A random variable $x$ has the following probability distribution

| $X=x_i$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X=x_i)$ | k | 2k | 3k | 4k | 5k |

Find (i) k (ii) Mean (iii) Variance of K

Sol: (i) Sum of probability = 1

$$K+2k+3k+4k+5k=1$$
$$15k=1$$
$$k=\frac{1}{15}$$

$$\boxed{\sum_{i=1}^{5} P(X=x_i)=1}$$

(ii) Mean $(\mu) = \sum_{i=1}^{5} x_i \, P(X=x_i)$

$$\boxed{\mu = \sum_{i=1}^{n} x_i \, . \, P(X=x_i)}$$

$$1(k)+2(2k)+3(3k)+4(4k)+5(5k)$$
$$K+4k+9k+16k+25k$$
$$55k$$
$$55\left(\frac{1}{15}\right)$$
$$=\frac{11}{33}$$

(iii) Variance $(\sigma^2) = \sum_{i=1}^{5} x_i^2 \, P(x=x_i) - \mu^2$

$$1(k)+4(2k)+9(3k)+16(4k)+25(5k)-\left(\frac{11}{3}\right)^2$$
$$K+8k+27k+64k+125k-\frac{121}{9}$$
$$225k-\frac{121}{9}$$
$$=\cancel{225}^{15}\left(\frac{1}{\cancel{15}}\right)-\frac{121}{9}$$
$$=15-\frac{121}{9}$$
$$=\frac{135-121}{9}$$
$$=\frac{14}{9}$$

② A random variable $x$ has the following probability distribution

| $X=x$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $P(X=x)$ | 0 | k | 2k | 2k | 3k | $k^2$ | $2k^2$ | $7k^2+k$ |

Find (i) K (ii) The mean of X (iii) $P(0<x<5)$

Sol: (i) Sum of probability = 1

$$\sum_{i=0}^{7} P(x=x_i)=1$$

$$0+k+2k+2k+3k+k^2+2k^2+7k^2+k=1$$
$$10k^2+9k-1=0$$
$$10k^2+10k-1k-1=0$$

$10k(k+1) - 1(k+1) = 0$

$(10k-1)(k+1) = 0$

$10k - 1 = 0 \qquad k+1 = 0$

$10k = 1 \qquad k = -1$

$k = \frac{1}{10}$

(ii) The mean of x

$= 0 + 1(k) + 2(2k) + 3(2k) + 4(3k) + 5(k^2) + 6(2k^2) + 7(7k^2 + k)$

$= k + 4k + 6k + 12k + 5k^2 + 12k^2 + 49k^2 + 7k$

$\mu = 30k + 66k^2$

$\mu = 30\left(\frac{1}{10}\right) + 66\left(\frac{1}{10^2}\right)$

$\mu = \frac{30}{10} + \frac{66}{100}$

$= 3 + 0.66$

$= 3.66$

Mean $(\mu) = 3.66$

(iii) $P(0 < x < 5) = P(x=1) + P(x=2) + P(x=3) + P(x=4)$

$= k + 2k + 2k + 3k$

$= 8k$

$= 8\left(\frac{1}{10}\right)$

$= \frac{8}{10} = 0.8$

---

③ A random variable $x$ has the following probability distribution

| $X = x_i$ | -2 | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|
| $P(X = x_i)$ | 0.1 | k | 0.2 | 2k | 0.3 | k |

Find 'k' mean and variance of x

Sol: Sum of Probabilities = 1

$$\sum_{i=-2}^{3} P(x = x_i) = 1$$

$0.1 + k + 0.2 + 2k + 0.3 + k = 1$

$4k + 0.6 = 1$

$4k = 1 - 0.6$

$4k = 0.4$

$k = 0.1$

Mean $(\mu)$ $\sum_{i=-2}^{3} x_i P(x = x_i)$

$= -2(0.1) + 0(0.2) + 1(2k) + 2(0.3) + 3(k) - 1(k)$

$= -0.2 + 2k + 0.6 + 3k - k$

$= 4k + 0.4$

$= 4(0.1) + 0.4 = 0.4 + 0.4 = 0.8$

Variance ($\sigma^2$): $\sum_{i=-2}^{3} x_i^2 \, P(x=x_i) - \mu^2$

$= 4(0.1)+1(k)+0(0.2)+1(2k)+4(0.3)+9(k)-(0.8)^2$

$= 0.4+k+2k+1.2+9k-(0.8)^2$

$= 12k+1.6-0.64$

$= 12(0.1)+1.6-0.64$

$= 1.2+1.6-0.64$

$= 2.8-0.64$

$= 2.16$

④ let X be a random variable such that $P(X=-2)=P(X=-1)=P(X=2)=P(X=1)=1/6$ and $P(X=0)=1/3$. Find mean and variance of X.

Sol: Given $P(X=-2)=P(X=-1)=P(X=2)=P(X=1)=1/6$ and $P(X=0)=1/3$

| $X=x_i$ | -2 | -1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $P(X=x_i)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{3}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

Mean ($\mu$) = $\sum_{i=-2}^{2} x_i \, P(x=x_i)$

$= -2(1/6)-1(1/6)+0(1/3)+1(1/6)+2(1/6)$

$= -\cancel{\frac{2}{6}}-\cancel{\frac{1}{6}}+0+\cancel{\frac{1}{6}}+\cancel{\frac{2}{6}}$

$= 0$

Variance ($\sigma^2$) = $\sum_{i=-2}^{2} x_i \, P(x=x_i) - \mu^2$

$= 4\left(\frac{1}{6}\right)+1\left(\frac{1}{6}\right)+1\left(\frac{1}{6}\right)+4\left(\frac{1}{6}\right)-0$

$= \frac{4}{6}+\frac{1}{6}+\frac{1}{6}+\frac{4}{6}$

$= \frac{10}{6} = \frac{5}{3} = 1.6$

⑤

| $X=x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| No. of workers | $\frac{1}{9}$ | $\frac{1}{9}$ | $\frac{1}{9}$ | $\frac{1}{3}$ | $\frac{1}{9}$ | $\frac{1}{9}$ | $\frac{1}{9}$ |

is a probability distribution of a random variable $x$

Sol: Mean ($\mu$) = $\sum_{i=-3}^{3} x_i \, P(x=x_i)$

$= -3\left(\cancel{\frac{1}{9}}\right)-2\left(\cancel{\frac{1}{9}}\right)-1\left(\cancel{\frac{1}{9}}\right)+0+1\left(\cancel{\frac{1}{9}}\right)+2\left(\cancel{\frac{1}{9}}\right)+3\left(\cancel{\frac{1}{9}}\right)$

$\mu = 0$

Variance ($\sigma^2$) = $\sum_{i=-3}^{3} x_i^2 \, P(x=x_i) - \mu^2$

$= 9\left(\frac{1}{9}\right)+4\left(\frac{1}{9}\right)+1\left(\frac{1}{9}\right)+0\left(\frac{1}{3}\right)+1\left(\frac{1}{9}\right)+4\left(\frac{1}{9}\right)+9\left(\frac{1}{9}\right)-0$

$= \frac{9}{9}+\frac{4}{9}+\frac{1}{9}+0+\frac{1}{9}+\frac{4}{9}+\frac{9}{9}$

$= \frac{9+4+1+1+4+9}{9} = 28/9$

⑥ A cubical die is thrown. Find the mean and variance of x, giving the number on the face that shows up.

Sol:

| $X=x_i$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $P(X=x_i)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

let 'S' be sample space and 'x' be the given random variable associated with $x$ where $P(x)$ is given by the following table.

$$\text{Mean } (\mu) = \sum_{i=1}^{6} (x_i) P(X=x_i)$$

$$= 1\left(\tfrac{1}{6}\right)+2\left(\tfrac{1}{6}\right)+3\left(\tfrac{1}{6}\right)+4\left(\tfrac{1}{6}\right)+5\left(\tfrac{1}{6}\right)+6\left(\tfrac{1}{6}\right)$$

$$= \frac{1+2+3+4+5+6}{6}$$

$$= \frac{21}{6} = \frac{7}{2}$$

$$\text{Variance } (\sigma^2) = \sum_{i=1}^{6} (x_i^2)\ P(x=x_i) - \mu^2$$

$$= 1\left(\tfrac{1}{6}\right)+4\left(\tfrac{1}{6}\right)+9\left(\tfrac{1}{6}\right)+16\left(\tfrac{1}{6}\right)+25\left(\tfrac{1}{6}\right)+36\left(\tfrac{1}{6}\right)-\left(\tfrac{7}{2}\right)^2$$

$$= \frac{1+4+9+16+25+36}{6} - \frac{49}{4}$$

$$= \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

⑦ The range of a random variable x is $\{0,1,2\}$ and given that $P(X=0) = 3c^3$, $P(X=1) = 4c-10c^2$, $P(X=2) = 5c-1$. Find (i) The value of c (ii) $P(x<1)$, $P(1<x\leq 2)$ and $P(0<x<3)$

Sol: Given $P(X=0) = 3c^3$, $P(X=1) = 4c-10c^2$, $P(X=2) = 5c-1$

(i) Sum of probabilities $= 1$

$$\sum_{i=0}^{2} P(X=x_i) = 1$$

$$P(X=0) + P(X=1) + P(X=2) = 1$$

$$3c^3 + 4c - 10c^2 + 5c - 1 - 1 = 0$$

$$3c^3 - 10c^2 + 9c - 2 = 0$$

$c=1$ is one of the root of above eq.

By using synthetic division.

$$\begin{array}{r|rrrr} 1 & 3 & -10 & 9 & -2 \\ & 0 & 3 & -7 & 2 \\ \hline & 3 & -7 & 2 & \boxed{0} \end{array}$$

$$3c^2 - 7c + 2 = 0$$

$$3c^2 - c - 6c + 2 = 0$$

$$c(3c-1) - 2(3c-1) = 0$$

$(3c-1)(c-2)=0$

$3c=1 \qquad c-2=0$

$c=\frac{1}{3} \qquad c=2$

(ii) $P(X<1) = P(X=0)$

$3c^3 = 3\left(\frac{1}{3}\right)^3 = \cancel{3}\left(\frac{1}{\cancel{27}_9}\right) = \frac{1}{9}$

$P(1<X\leq 2) = P(X=2)$

$= 5c-1 = 5\left(\frac{1}{3}\right)-1$

$= 5/3-1$

$= \frac{2}{3}$

$P(0<x\leq 3) = P(x=1)+P(x=2)$

$= 4c-10c^2+5c-1$

$= 9c-10c^2-1$

$= \cancel{9}^3\left(\frac{1}{\cancel{3}}\right)-10\left(\frac{1}{9}\right)-1$

$= 3-\frac{10}{9}-1$

$= \frac{8}{9}$

(8) One in 9 ships is likely to be wrecked, when they are set on the sail when 6 ships are on sail find the probability of

(i) Atleast one will arrive safely

(ii) Exactly three will arrive safely

Sol: let P and Q are the events that probabilities of ship will arrive safely and wrecked respectively $n=6$

$p+q=1$

$p+\frac{1}{9}=1$

$p = 1-\frac{1}{9}$

$p = \frac{8}{9}$

(i) The probability that atleast one will arrive safe

$= 1-$ (The probability that no one will arrive safe)

$= 1-P(X=0)$

$= 1-{}^6C_0\left(\frac{8}{9}\right)^0\left(\frac{1}{9}\right)^6$

$= 1-\left(\frac{1}{9}\right)^6$

$\therefore$ At least one will arrive safely $= 1-\frac{1}{9^6}$

(ii) The probability that exactly three will arrive safely

$$P(X=3) = {}^{6}C_3\left(\frac{8}{9}\right)^3\left(\frac{1}{9}\right)^{6-3}$$

$$= \frac{6\times5\times4}{1\times2\times3}\left(\frac{8}{9}\right)^3\left(\frac{1}{9}\right)^3$$

$$= 20\left(\frac{8}{9}\right)^3\left(\frac{1}{9}\right)^3$$

(9) In the experiment of tossing a coin 'n' times, if the variable X denotes the number of head, and $P(X=4)$, $P(X=5)$ and $P(X=6)$ are in A.P then find 'n'.

Sol: $p=\frac{1}{2}$, $q=\frac{1}{2}$

$P(X=4)$, $P(X=5)$, $P(X=6)$ are in A.P

If $a, b, c$ are in A.P then $2b = a+c$

$$2P(X=5) = P(X=4)+P(X=6)$$

$$2\,{}^{n}C_5\left(\tfrac{1}{2}\right)^5\left(\tfrac{1}{2}\right)^{n-5} = {}^{n}C_4\left(\tfrac{1}{2}\right)^4\left(\tfrac{1}{2}\right)^{n-4} + {}^{n}C_6\left(\tfrac{1}{2}\right)^6\left(\tfrac{1}{2}\right)^{n-6}$$

$$2\,{}^{n}C_5\left(\tfrac{1}{2}\right)^n = {}^{n}C_4\left(\tfrac{1}{2}\right)^n + {}^{n}C_6\left(\tfrac{1}{2}\right)^n$$

$$2\,{}^{n}C_5\left(\tfrac{1}{2}\right)^n = \left(\tfrac{1}{2}\right)^n\left({}^{n}C_4 + {}^{n}C_6\right)$$

$$2 = \frac{{}^{n}C_4}{{}^{n}C_5} + \frac{{}^{n}C_6}{{}^{n}C_5}$$

$$2 = \frac{5}{n-5+1} + \frac{n-6+1}{6}$$

$$2 = \frac{5}{n-4} + \frac{n-5}{6}$$

$$2 = \frac{30+(n-5)(n-4)}{6(n-4)}$$

$$12(n-4) = 30+n^2-5n-4n+20$$

$$12n-48 = n^2-9n-50$$

$$n^2-21n+98=0$$

$$n^2-7n-14n+98=0$$

$$n(n-7)(n-14)=0$$

$$n=7 \qquad n=14$$

(10) Two dice are rolled at random. Find the probability distribution of sum of the numbers on them. Find the mean of the random variable.

Sol:

| $X=x_i$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $P(X=x_i)$ | $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{3}{36}$ | $\frac{4}{36}$ | $\frac{5}{36}$ | $\frac{6}{36}$ | $\frac{5}{36}$ | $\frac{4}{36}$ | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ |

When two dice are rolled, the sample spaces consists of 36.

$S = (1,1)(1,2) --- (6,6)$

let $x$ be denote the sum of the numbers on the two dice, then the range of $x = 2, 3, 4 --- 12$

$$\text{Mean}(\mu) = \sum_{i=2}^{12} x_i P(X = x_i)$$

$$= 2\left(\frac{1}{36}\right) + 3\left(\frac{2}{36}\right) + 4\left(\frac{3}{36}\right) + 5\left(\frac{4}{36}\right) + 6\left(\frac{5}{36}\right) + 7\left(\frac{6}{36}\right) + 8\left(\frac{5}{36}\right) + 9\left(\frac{4}{36}\right) + 10\left(\frac{3}{36}\right) + 11\left(\frac{2}{36}\right) + 12\left(\frac{1}{36}\right)$$

$$= \frac{2+6+12+20+30+42+40+36+30+22+12}{36}$$

$$= \frac{112+140}{36}$$

$$= \frac{252}{36}$$

$$= 7$$

$\text{Mean}(\mu) = 7$

# COMPLEX NUMBERS (SA·Q'S)

1. If $\dfrac{z_3-z_1}{z_2-z_1}$ is a real number, show that the points represented by the complex numbers $z_1, z_2, z_3$ are collinear.

Sol:- let $\dfrac{z_3-z_1}{z_2-z_1} = K$

$$z_3 - z_1 = Kz_2 - Kz_1$$

$$z_3 - Kz_2 = z_1 - Kz_1$$

$$z_3 - Kz_2 = z_1(1-K)$$

$$\frac{z_3-Kz_2}{1-K} = z_1$$

$\therefore$ $z_1$ divides $z_2$ & $z_3$ in the ratio $1:K$ externally

$\therefore$ $z_1, z_2, z_3$ are collinear.

---

2. If $z = x+iy$ & if the point P in the argand plane represents 'z'. Find the locus of z satisfying the equation $|z-3+i| = 4$

Sol:- let $z = x+iy$

$$|x+iy-3+i| = 4$$

$$|(x-3)+i(y+1)| = 4$$

$$\sqrt{(x-3)^2+(y+1)^2} = 4$$

S.O.B.S

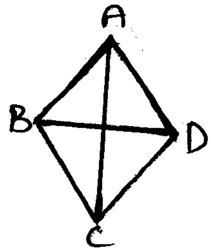
$$(x-3)^2+(y+1)^2 = 16$$

$$x^2+9-6x+y^2+1+2y-16=0$$

$$x^2+y^2-6x+2y-6=0$$

3) i, Show that points in the argand plane represented the complex numbers $-2+7i$, $\frac{-3}{2}+\frac{1}{2}i$, $4-3i$, $\frac{7}{2}(1+i)$ are vertices of a rhombus?

Sol: $A=(-2,7)$ $B=\left(\frac{-3}{2},\frac{1}{2}\right)$ $C=(4,-3)$ $D=\left(\frac{7}{2},\frac{7}{2}\right)$




$AB=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$=\sqrt{\left(\frac{-3}{2}+2\right)^2+\left(\frac{1}{2}-7\right)^2}=\sqrt{\left(\frac{-3+4}{2}\right)^2+\left(\frac{1-14}{2}\right)^2}$

$=\sqrt{\frac{1}{4}+\frac{169}{4}}=\sqrt{\frac{170}{4}}$

$BC=\sqrt{\left(4+\frac{3}{2}\right)^2+\left(-3-\frac{1}{2}\right)^2}=\sqrt{\left(\frac{8+3}{2}\right)^2+\left(\frac{-6-1}{2}\right)^2}=\sqrt{\frac{121}{4}+\frac{49}{4}}=\sqrt{\frac{170}{4}}$

$CD=\sqrt{\left(\frac{7}{2}-4\right)^2+\left(\frac{7}{2}+3\right)^2}=\sqrt{\left(\frac{7-8}{2}\right)^2+\left(\frac{7+6}{2}\right)^2}=\sqrt{\frac{1}{4}+\frac{169}{4}}=\sqrt{\frac{170}{4}}$

$DA=\sqrt{\left(\frac{7}{2}+2\right)^2+\left(\frac{7}{2}-7\right)^2}=\sqrt{\left(\frac{7+4}{2}\right)^2+\left(\frac{7-14}{2}\right)^2}=\sqrt{\frac{121}{4}+\frac{49}{4}}=\sqrt{\frac{170}{4}}$

$AC=\sqrt{(4+2)^2+(-3-7)^2}=\sqrt{36+100}=\sqrt{136}$

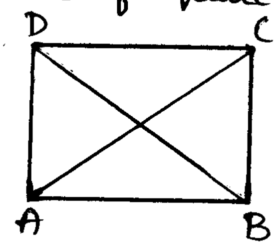
$BD=\sqrt{\left(\frac{7}{2}+\frac{3}{2}\right)^2+\left(\frac{7}{2}-\frac{1}{2}\right)^2}=\sqrt{\left(\frac{10}{2}\right)^2+\left(\frac{6}{2}\right)^2}=\sqrt{\frac{100}{4}+\frac{36}{4}}=\sqrt{\frac{136}{4}}$

$\therefore AB=BC=CD=DA$ & $AC\neq BD$

$\therefore$ They forms Rhombus

ii, Show that the four points in the argand plane represented by the complex numbers. $2+i$, $4+3i$, $2+5i$, $3i$ are the vertices of square.

Sol: $A=(2,1)$ $B=(4,3)$ $C=(2,5)$ $D=(0,3)$




Distance b/w two points $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$AB=\sqrt{(4-2)^2+(3-1)^2}=\sqrt{4+4}=\sqrt{8}$

$BC=\sqrt{(2-4)^2+(5-3)^2}=\sqrt{4+4}=\sqrt{8}$

$CD=\sqrt{(0-2)^2+(3-5)^2}=\sqrt{4+4}=\sqrt{8}$

$DA=\sqrt{(0-2)^2+(3-1)^2}=\sqrt{4+4}=\sqrt{8}$

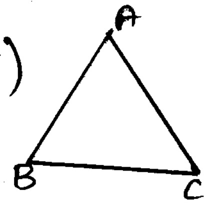
$AC = \sqrt{(2-2)^2+(5-1)^2} = \sqrt{0+16} = \sqrt{16} = 4$

$BD = \sqrt{(0-4)^2+(3-3)^2} = \sqrt{16+0} = 4$

$\therefore AB = BC = CD = DA$ & $AC = BD$

$\therefore$ ABCD forms a Square.

iii, show that the points in the argand diagram represented by the complex numbers $2+2i$, $-2-2i$, $-2\sqrt{3}+2\sqrt{3}i$ are the vertices of an equilateral $\Delta^{le}$.




Sol: $A = (2,2)$ $B = (-2,-2)$ $C = (-2\sqrt{3}, 2\sqrt{3})$

$\boxed{\text{Distance b/w two points} = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}}$

$AB = \sqrt{(-2-2)^2+(-2-2)^2} = \sqrt{16+16} = \sqrt{32}$

$BC = \sqrt{(-2\sqrt{3}+2)^2+(2\sqrt{3}+2)^2} = \sqrt{(12+4-\cancel{8\sqrt{3}}+12+4+\cancel{8\sqrt{3}})} = \sqrt{32}$

$CA = \sqrt{(-2\sqrt{3}-2)^2+(2\sqrt{3}-2)^2} = \sqrt{12+4-\cancel{8\sqrt{3}}+12+4+\cancel{8\sqrt{3}}} = \sqrt{32}$

$AB = CA = BC$

Given vertices ABC forms an equilateral $\Delta^{le}$.

4. The points P, Q denote the complex number $z_1$ $z_2$ in the argand diagram. O is the origin. If $z_1 \cdot \bar{z}_2 + \bar{z}_1 \cdot z_2 = 0$ then S.T $\angle POQ = 90°$

Sol: Given points P, Q denotes the complex numbers be $z_1, z_2$

let $z_1 = x_1+iy_1$, $z_2 = x_2+iy_2$ ; $\bar{z}_1 = x_1 - iy_1$ $\bar{z}_2 = x_2 - iy_2$

$P(x_1, y_1)$ $Q(x_2, y_2)$ $O(0,0) \Rightarrow$ Let 'O' be the origin

slope of $OP = \frac{y_1}{x_1}$ slope of $OQ = \frac{y_2}{x_2}$

$z_1 \cdot \bar{z}_2 + \bar{z}_1 \cdot z_2 = 0$

$(x_1+iy_1)(x_2-iy_2)+(x_1-iy_1)(x_2+iy_2) = 0$

$x_1x_2 - ix_1y_2 + ix_2y_1 + y_1y_2 + x_1x_2 + ix_1y_2 - iy_1x_2 + y_1y_2 = 0$

$2x_1x_2 + 2y_1y_2 = 0 \Rightarrow x_1x_2 + y_1y_2 = 0$

$y_1y_2 = -x_1x_2$

$\frac{y_1}{x_1} \times \frac{y_2}{x_2} = 1$

Slope of OP + Slope of OQ = -1

$\angle POQ = 90^\circ$

5. Determine the locus of $z$; $z \neq 2i$ such that $Re\left(\frac{z-4}{z-2i}\right) = 0$

Sol:- let $z = x+iy$

$$\frac{z-4}{z-2i} = \frac{x+iy-4}{x+iy-2i}$$

$$= \frac{(x-4)+iy}{x+i(y-2)} + \frac{x-i(y-2)}{x-i(y-2)}$$

$$= \frac{x(x-4) - i(x-4)(y-2) + ixy + y(y-2)}{x^2 + (y-2)^2}$$

$$= \frac{x(x-4)+y(y-2)}{x^2+(y-2)^2} + i\,\frac{xy-(x-4)(y-2)}{x^2+(y-2)^2}$$

Given real part is zero.

$$\frac{x(x-4)+y(y-2)}{x^2+(y-2)^2} = 0$$

$$x^2-4x+y^2-2y=0$$

$\therefore$ The required locus of $z$ is $x^2+y^2-4x-2y=0$

6) Show that the $z_1 = \frac{2+11i}{25}$ $z_2 = \frac{-2+i}{(1-2i)^2}$ are conjugate to each other.

Sol:- $z_1 = \frac{2+11i}{25}$ $z_2 = \frac{-2+i}{(1-2i)^2}$

$$z_2 = \frac{-2+i}{1+4i^2-4i} = \frac{-2+i}{1-4-4i} = \frac{-2+i}{-3-4i}$$

$$z_2 = \frac{-(2-i)}{-(3+4i)} = \frac{2-i}{3+4i} = \frac{3-4i}{3-4i}$$

$$z_2 = \frac{6-8i-3i+4i^2}{3^2+16} = \frac{6+11i-4}{25} = \frac{2-11i}{25}$$

The conjugateve of $x+iy$ is $x-iy$

$\therefore z_1 \& z_2$ are conjugative to each other.

7) i. If $x+iy = \dfrac{1}{1+\cos\theta+i\sin\theta}$ then show that $(4x^2-1)=0$

Sol:- $x+iy = \dfrac{1}{1+\cos\theta+i\sin\theta}$

$$x+iy = \frac{1}{1+\cos\theta+i\sin\theta} \times \frac{1+\cos\theta-i\sin\theta}{1+\cos\theta-i\sin\theta} \qquad \boxed{i^2=-1}$$

$$x+iy = \frac{1+\cos\theta-i\sin\theta}{(1+\cos\theta)^2+\sin^2\theta} = \frac{1+\cos\theta-i\sin\theta}{1+2\cos\theta+\cos^2\theta+\sin^2\theta}$$

$$x+iy = \frac{1+\cos\theta-i\sin\theta}{2+2\cos\theta} = \frac{1+\cos\theta}{2(1+\cos\theta)} + i\,\frac{-\sin\theta}{2(1+\cos\theta)}$$

Comparing real part on Both sides

$$x = \frac{1+\cancel{\cos\theta}}{2(\cancel{1+\cos\theta})}$$

$$x = \frac{1}{2}$$

$$2x = 1$$

S.O.B.S

$$4x^2 = 1$$

$$4x^2-1=0$$

ii. If $u+iv = \dfrac{2+i}{z+3}$ and $z = x+iy$ then find $u, v$.

Sol:- Given $u+iv = \dfrac{2+i}{z+3}$

given $z = x+iy$

$$u+iv = \frac{2+i}{x+iy+3}$$

$$= \frac{2+i}{(x+3)+iy} \times \frac{(x+3)-iy}{(x+3)-iy} \qquad \boxed{i^2=-1}$$

$$= \frac{2(x+3)-i2y+i(x+3)+y}{(x+3)^2+y^2}$$

$$= \frac{2(x+3)+y}{(x+3)^2+y^2} + i\,\frac{x+3-2y}{(x+3)^2+y^2}$$

Comparing real & imaginary part on b.s

$$u = \frac{2(x+3)+y}{(x+3)^2+y^2}, \quad v = \frac{x+3-2y}{(x+3)^2+y^2}$$

$$u = \frac{2x+y+6}{(x+3)^2+y^2} \qquad v = \frac{x-2y+3}{(x+3)^2+y^2}$$

8. If $x+iy = \frac{3}{2+\cos\theta+i\sin\theta}$ then Show that $x^2+y^2 = 4x-3$

Sol:- given $x+iy = \frac{3}{2+\cos\theta+i\sin\theta}$

$$= \frac{3}{(2+\cos\theta)+i\sin\theta} \times \frac{2+\cos\theta-i\sin\theta}{2+\cos\theta-i\sin\theta} \qquad \boxed{i^2=-1}$$

$$= \frac{6+3\cos\theta-i3\sin\theta}{(2+\cos\theta)^2+\sin^2\theta} \qquad \boxed{\sin^2\theta+\cos^2\theta=1}$$

$$= \frac{6+3\cos\theta-i3\sin\theta}{4+\cos^2\theta+4\cos\theta+\sin^2\theta}$$

$$= \frac{6+3\cos\theta-i3\sin\theta}{5+4\cos\theta}$$

$$x+iy = \frac{6+3\cos\theta}{5+4\cos\theta} + i\,\frac{-3\sin\theta}{5+4\cos\theta}$$

$$x = \frac{6+3\cos\theta}{5+4\cos\theta}, \quad y = \frac{-3\sin\theta}{5+4\cos\theta}$$

$$L.H.S = x^2+y^2 = \left(\frac{6+3\cos\theta}{5+4\cos\theta}\right)^2 + \left(\frac{-3\sin\theta}{5+4\cos\theta}\right)^2$$

$$x^2+y^2 = \frac{36+9\cos^2\theta+36\cos\theta+9\sin^2\theta}{(5+4\cos\theta)^2}$$

$$= \frac{36+9(\sin^2\theta+\cos^2\theta)+36\cos\theta}{(5+4\cos\theta)^2}$$

$$= \frac{45+36\cos\theta}{(5+4\cos\theta)^2} = \frac{9(5+\cancel{4\cos\theta})}{(5+4\cos\theta)^{\cancel{2}}}$$

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$$x^2+y^2 = \frac{9}{5+4\cos\theta} \longrightarrow ①$$

$$R.H.S = 4x-3 = 4\left(\frac{6+3\cos\theta}{5+4\cos\theta}\right)-3$$

$$= \frac{24+12\cancel{\cos\theta}-15-12\cancel{\cos\theta}}{5+4\cos\theta}$$

$$4x-3 = \frac{9}{5+4\cos\theta} \longrightarrow ②$$

from ① and ②

$$x^2+y^2 = 4x-3$$

# QUADRATIC EXPRESSIONS

1. If the Expansion $\frac{x-P}{x^2-3x+2}$ takes all real values for $x \in R$ then find the limits for 'P'.

Sol:- let $y = \frac{x-P}{x^2-3x+2}$

$$yx^2 - 3yx + 2y = x - P$$

$$yx^2 - 3yx + 2y - x + P = 0$$

$$yx^2 - (3y+1)x + 2y + P = 0$$

It is in the form of $ax^2+bx+c=0$

$$a = y,\ b = -(3y+1),\ c = 2y+P$$

Since 'x' is real $\Rightarrow \Delta \geq 0$

$$b^2 - 4ac \geq 0$$

$$(3y+1)^2 - 4y(2y+P) \geq 0$$

$$9y^2 + 1 + 6y - 8y^2 - 4Py \geq 0$$

$$y^2 + (6-4P)y + 1 \geq 0$$

$$\boxed{\text{since expression is } \geq 0 \text{ and coefficient of } x^2 > 0 \Rightarrow \Delta \geq 0}$$

$$b^2 - 4ac < 0$$

$$(6-4P)^2 - 4(1)(1) < 0$$

$$36 + 16P^2 - 48P - 4 < 0$$

$$16P^2 - 48P + 32 < 0$$

$$16(P^2 - 3P + 2) < 0$$

$$P^2 - P - 2P + 2 < 0$$

$$P(P-1) - 2(P-1) < 0$$

$$(P-1)(P-2) < 0$$

$$\boxed{\text{If } (x-\alpha)(x-\beta) < 0 \text{ then } x \in (\alpha, \beta)}$$

$\therefore P \in (1, 2)$ or $1 < P < 2$

2. If 'x' is real then prove that $\frac{x}{x^2-5x+9}$ lies between $\frac{-1}{11}$ and 1.

Sol:- let $y = \frac{x}{x^2-5x+9}$

$yx^2-5yx+9y-x=0$

$yx^2-(5y+1)x+9y=0$

It is in the form of $ax^2+bx+c=0$

Here $a=y$, $b=-(5y+1)$, $c=9y$

Since 'x' is real $\Rightarrow \Delta \geq 0$

$b^2-4ac \geq 0$

$(5y+1)^2-4(y)(9y) \geq 0$

$25y^2+1+10y-36y^2 \geq 0$

$-11y^2+10y+1 \geq 0$

$11y^2-10y-1 \leq 0$

$11y^2+y-11y-1 \leq 0$

$y(11y+1)-1(11y+1) \leq 0$

$(11y+1)(y-1) \leq 0$

$\left(y-\left(\frac{-1}{11}\right)\right)(y-1) \leq 0$

If $(x-\alpha)(x-\beta) \leq 0$ then $x \in [\alpha, \beta]$

$\therefore y \in \left[\frac{-1}{11}, 1\right]$

$\therefore$ y lies b/w $-1/11$ and 1

3. If '$x$' is real show that the value of Expression $\frac{x^2+34x-71}{x^2+2x-7}$ does not lie between 5 and 9.

Sol:- let $y = \frac{x^2+34x-71}{x^2+2x-7}$

$$yx^2+2yx-7y = x^2+34x-71$$

$$yx^2+2yx-7y-x^2-34x+71=0$$

$$(y-1)x^2+(2y-34)x+71-7y=0$$

It is in the form of $ax^2+bx+c=0$

Here, $a=y-1$, $b=2(y-17)$, $c=71-7y$

Since '$x$' is real $\Rightarrow \Delta \geq 0$

$$b^2-4ac \geq 0$$

$$4(y-17)^2-4(y-1)(71-7y) \geq 0$$

$$4\left[(y^2+289-34y)-(71y-7y^2-71+7y)\right] \geq 0$$

$$y^2+289-34y-71y+7y^2+71-7y \geq 0$$

$$8y^2-112y+360 \geq 0$$

$$8(y^2-14y+45) \geq 0$$

$$y^2-5y-9y+45 \geq 0$$

$$y(y-5)-9(y-5) \geq 0$$

$$(y-5)(y-9) \geq 0$$

$$\boxed{\text{If } (x-\alpha)(x-\beta) \geq 0 \text{ then } x \in [-\infty,\alpha] \cup [\beta,\infty]}$$

$$\therefore y \in [-\infty, 5] \cup [9, \infty]$$

$\therefore y$ does not lies between 5 and 9.

4. Prove that $\frac{1}{3x+1} + \frac{1}{x+1} - \frac{1}{(3x+1)(x+1)}$ does not lies between 1 and 4, If x is real.

Sol:- let $y = \frac{1}{3x+1} + \frac{1}{x+1} - \frac{1}{(3x+1)(x+1)}$

$$= \frac{x+1+3x+\cancel{1}-\cancel{1}}{(3x+1)(x+1)}$$

$$= \frac{4x+1}{3x^2+3x+x+1}$$

$$y = \frac{4x+1}{3x^2+4x+1}$$

$$3yx^2+4yx+y-4x-1=0$$

$$3yx^2+4(y-1)x+(y-1)=0$$

It is in the form of $ax^2+bx+c=0$

Here $a=3y$, $b=4(y-1)$, $c=(y-1)$

Since '$x$' is real $\Rightarrow \Delta \geq 0$

$$b^2-4ac \geq 0$$

$$16(y-1)^2-4(3y)(y-1) \geq 0$$

$$4(y-1)\left[4(y-1)-3y\right] \geq 0$$

$$(y-1)(4y-4-3y) \geq 0$$

$$(y-1)(y-4) \geq 0$$

$$\boxed{\text{If } (x-\alpha)(x-\beta) \geq 0 \text{ then } x \in (-\infty, \alpha) \cup (\beta, \infty)}$$

$$\therefore y \in [-\infty, 1] \cup [4, \infty]$$

$\therefore$ y doesnot lies between 1 and 4

5. If '$x$' is real, find the maximum and minimum values of the Expression $\dfrac{x^2+14x+9}{x^2+2x+3}$

sol:- let $y = \dfrac{x^2+14x+9}{x^2+2x+3}$

$yx^2+2yx+3y = x^2+14x+9$

$yx^2+2yx+3y-x^2-14x-9=0$

$(y-1)x^2+(2y-14)x+3y-9=0$

It is in the form of $ax^2+bx+c=0$

Here $a=y-1$, $b=2(y-7)$, $c=3y-9$

Since '$x$' is real $\Rightarrow \Delta \geq 0$

$b^2-4ac \geq 0$

$4(y-7)^2-4(y-1)(3y-9) \geq 0$

$4\left[y^2+49-14y-(3y^2-9y-3y+9)\right] \geq 0$

$y^2+49-14y-3y^2+9y+3y-9 \geq 0$

$-2y^2-2y+40 \geq 0$

$2y^2+2y-40 \leq 0$

$2(y^2+y-20) \leq 0$

$y^2-4y+5y-20 \leq 0$

$y(y-4)+5(y-4) \leq 0$

$(y-4)(y+5) \leq 0$

$(y-4)(y-(-5)) \leq 0$

$\boxed{\text{If } (x-\alpha)(x-\beta) \leq 0 \text{ then } x \in [\alpha, \beta]}$

$\therefore y \in [-5, 4]$

Min value $= -5$

Max value $= 4$

6. Determine the range of $\frac{x+2}{2x^2+3x+6}$ for $x \in R$

Sol:- let $y = \frac{x+2}{2x^2+3x+6}$

$2yx^2+3yx+6y = x+2$

$2yx^2+3yx+6y-x-2=0$

$2yx^2+(3y-1)x+6y-2=0$

It is in the form of $ax^2+bx+c=0$

Here $a=2y$, $b=3y-1$, $c=6y-2$

Since 'x' is real $\Rightarrow \Delta \geq 0$

$b^2-4ac \geq 0$

$(3y-1)^2-4(2y)(6y-2) \geq 0$

$9y^2+1-6y-8y(6y-2) \geq 0$

$9y^2+1-6y-48y^2+16y \geq 0$

$-39y^2+10y+1 \geq 0$

$39y^2-10y-1 \leq 0$

$39y^2+3y-13y-1 \leq 0$

$3y(13y+1)-1(13y+1) \leq 0$

$(13y+1)(3y-1) \leq 0$

$\left(y-\left(\frac{-1}{13}\right)\right)\left(y-\frac{1}{3}\right) \leq 0$

$\boxed{\text{If } (x-\alpha)(x-\beta) \leq 0 \text{ then } x \in [\alpha, \beta]}$

$\therefore y \in \left(\frac{-1}{13}, \frac{1}{3}\right)$

7. Determine the range of the expression $\frac{x^2+x+1}{x^2-x+1}$ for $x\in R$

Sol: Let $y = \frac{x^2+x+1}{x^2-x+1}$

$yx^2 - yx + y = x^2 + x + 1$

$yx^2 - yx + y - x^2 - x - 1 = 0$

$(y-1)x^2 - (y+1)x + y - 1 = 0$

It is in the form of $ax^2+bx+c=0$

Here $a = y-1$, $b = -(y+1)$, $c = y-1$

Since '$x$' is real $\Rightarrow \Delta \geq 0$

$b^2 - 4ac \geq 0$

$(y+1)^2 - 4(y-1)(y-1) \geq 0$

$(y+1)^2 - 4(y-1)^2 \geq 0$

$y^2+1+2y-4(y^2+1-2y) \geq 0$

$y^2+1+2y-4y^2-4+8y \geq 0$

$-3y^2+10y-3 \geq 0$

$3y^2-10y+3 \leq 0$

$3y^2-y-9y+3 \leq 0$

$y(3y-1)-3(3y-1) \leq 0$

$(3y-1)(y-3) \leq 0$

$(y-\frac{1}{3})(y-3) \leq 0$

$$\boxed{\text{If } (x-\alpha)(x-\beta) \leq 0 \text{ then } x \in [\alpha, \beta]}$$

$\therefore y \in [\frac{1}{3}, 3]$.

8. let $a, b, c \in R$ and $a \neq 0$ such that the equation $ax^2 + bx + c = 0$ has real roots $\alpha$ and $\beta$ with $\alpha < \beta$ then

i, for $\alpha < x < \beta$, $ax^2+bx+c$ and 'a' have opposite Signs

ii, for $x < \alpha$ (or) $x > \beta$ $ax^2+bx+c$ and 'a' have same sign.

Proof: Let $\alpha, \beta$ are roots of $ax^2+bx+c=0$ then

$$\alpha+\beta = -b/a, \quad \alpha\beta = c/a$$

Now
$$ax^2+bx+c = a\left[x^2 + \left(\frac{b}{a}\right)x + \frac{c}{a}\right]$$
$$= a\left[x^2 - \left(\frac{-b}{a}\right)x + \frac{c}{a}\right]$$
$$= a\left[x^2 - (\alpha+\beta)x + \alpha\beta\right]$$
$$\Rightarrow \frac{ax^2+bx+c}{a} = (x-\alpha)(x-\beta) \rightarrow (1)$$

$\boxed{\because (x-\alpha)(x-\beta) = x^2 - (\alpha+\beta)x + \alpha\beta}$

i, Given, $\alpha < x < \beta$

$\Rightarrow \alpha < x, \; x < \beta$

i.e $x > \alpha, \; x < \beta$

$(x-\alpha) > 0, \; (x-\beta) < 0$

$(x-\alpha)(x-\beta) < 0$

from (1) $\dfrac{ax^2+bx+c}{a} < 0$

$\Rightarrow ax^2+bx+c$ and 'a' have opposite sign.

ii, case(i) :- $x < \alpha$ and we have $\alpha < \beta$

$\Rightarrow x < \alpha \,,\ \alpha < \beta$

$x < \beta$

$(x-\alpha) < 0 \quad (x-\beta) < 0$

$(x-\alpha)(x-\beta) > 0$

Case(ii) :- $x > \beta$ and $\beta > \alpha$

$\Rightarrow x > \beta$ and $\beta > \alpha \Rightarrow x > \alpha$

$\Rightarrow x > \alpha, \ x > \beta$

$(x-\alpha) > 0 \,; \ (x-\beta) > 0$

$(x-\alpha)(x-\beta) > 0$

Hence from ① we have $\dfrac{ax^2+bx+c}{a} > 0$

$\therefore$ $ax^2+bx+c$ and 'a' have the same sign.

9. Solve $2x^4 + x^3 - 11x^2 + x + 2 = 0$.

Sol:- Given $2x^4 + x^3 - 11x^2 + x + 2 = 0$

Dividing on both sides by $x^2$

$$\Rightarrow \frac{2x^4}{x^2} + \frac{x^3}{x^2} - \frac{11x^2}{x^2} + \frac{x}{x^2} + \frac{2}{x^2} = 0$$

$$2x^2 + x - 11 + \frac{1}{x} + \frac{2}{x^2} = 0$$

$$2\left[x^2 + \frac{1}{x^2}\right] + \left[x + \frac{1}{x}\right] - 11 = 0$$

$$2\left[\left(x+\frac{1}{x}\right)^2 - 2\right] + \left[x + \frac{1}{x}\right] - 11 = 0$$

let $x+\frac{1}{x}=a$

$2[a^2-2]+a-11=0$

$2a^2-4+a-11=0$

$2a^2+6a-5a-15=0$

$2a(a+3)-5(a+3)=0$

$(2a-5)(a+3)=0$

$a+3=0$ (or) $2a-5=0$

$a=-3$ (or) $a=5/2$

Case i, $a=-3$

$x+\frac{1}{x}=-3$

$\frac{x^2+1}{x}=-3$

$x^2+1=-3x$

$x^2+3x+1=0$

$x=\frac{-3\pm\sqrt{9-4}}{2}$

$x=\frac{-3\pm\sqrt{5}}{2}$

Case ii, $a=5/2$

$x+\frac{1}{x}=\frac{5}{2}$

$\frac{x^2+1}{x}=5/2$

$2x^2+2=5x$

$2x^2-5x+2=0$

$2x^2-4x-x+2=0$

$2x(x-2)-1(x-2)=0$

$(x-2)(2x-1)=0$

$x=2$ (or) $x=1/2$

$\therefore$ The roots are $2, \frac{1}{2}, \frac{-3\pm\sqrt{5}}{2}$

# PERMUTATIONS & COMBINATIONS

## SAQ's

1) i) simplify $^{34}C_5 + \sum_{r=0}^{4} {}^{(38-r)}C_4$

ii) Prove that : $^{25}C_4 + \sum_{r=0}^{4} {}^{(29-r)}C_3 = {}^{30}C_4$.

i) Sol:- $\left[{}^{34}C_5 + {}^{34}C_4\right] + {}^{35}C_4 + {}^{36}C_4 + {}^{37}C_4 + {}^{38}C_4$.

$= \left[{}^{35}C_5 + {}^{35}C_4\right] + {}^{36}C_4 + {}^{37}C_4 + {}^{38}C_4$.

$= \left[{}^{36}C_5 + {}^{36}C_4\right] + {}^{37}C_4 + {}^{38}C_4$

$= \left[{}^{37}C_5 + {}^{37}C_4\right] + {}^{38}C_4$

$= {}^{38}C_5 + {}^{38}C_4$

$= {}^{39}C_5$.

ii) Sol:- $L.H.S = \left[{}^{25}C_4 + {}^{25}C_3\right] + {}^{26}C_3 + {}^{27}C_3 + {}^{28}C_3 + {}^{29}C_3$

$= \left[{}^{26}C_4 + {}^{26}C_3\right] + {}^{27}C_3 + {}^{28}C_3 + {}^{29}C_3$.

$= \left[{}^{27}C_4 + {}^{27}C_3\right] + {}^{28}C_3 + {}^{29}C_3$

$= \left[{}^{28}C_4 + {}^{28}C_3\right] + {}^{29}C_3$.

$= {}^{29}C_4 + {}^{29}C_3$

$= {}^{30}C_4$.

$= R.H.S$

2) Prove that $\dfrac{{}^{4n}C_{2n}}{{}^{2n}C_n} = \dfrac{1.3.5....(4n-1)}{\{1.3.5....(2n-1)\}^2}$.

Sol:- $L.H.S = \dfrac{{}^{4n}C_{2n}}{{}^{2n}C_n}$

$= \dfrac{\dfrac{4n!}{(4n-2n)!\,2n!}}{\dfrac{2n!}{(2n-n)!\,n!}}$

$\boxed{\because\ {}^{n}C_r = \dfrac{n!}{(n-r)!\,n!}}$

$$= \frac{\frac{4n!}{2n!\,2n!}}{\frac{2n!}{n!\,n!}}$$

$$= \frac{4n!}{(2n!)^2} \times \frac{(n!)^2}{2n!}$$

$$= \frac{(4n)(4n-1)(4n-2)\cdots 5\cdot4\cdot3\cdot2\cdot1}{[2n(2n-1)(2n-2)\cdots 5\cdot4\cdot3\cdot2\cdot1]^2} \times \frac{(n!)^2}{2n!}$$

$$= \frac{[(4n-1)(4n-3)\cdots 5\cdot3\cdot1][(4n)(4n-2)\cdots 4\cdot2]}{[(2n-1)(2n-3)\cdots 5\cdot3\cdot1]^2[(2n)\cdot(2n-2)\cdots 4\cdot2]^2}$$

$$= \frac{[(4n-1)(4n-3)\cdots 5\cdot3\cdot1][2(2n)\,2(2n-1)\cdots 2\times2\times2\times1]}{[(2n-1)(2n-3)\cdots 5\cdot3\cdot1]^2[2\cdot n\cdot2(n-1)\cdots 2\times2\times2\times1]^2} \times \frac{(n!)^2}{2n!}$$

$$= \frac{[(4n-1)(4n-3)\cdots 5\cdot3\cdot1]\,\cancel{2^{2n}}\cdot\cancel{2n!}}{[(2n-1)(2n-3)\cdots 5\cdot3\cdot1]^2\,\cancel{2^{2n}}\cancel{(n!)^2}} \times \frac{\cancel{(n!)^2}}{\cancel{2n!}}$$

$$= \frac{1\cdot3\cdot5\cdots(4n-3)(4n-1)}{[(1\cdot3\cdot5\cdots(2n-3)(2n-1)]^2}.$$

$= R.H.S.$

③ If the letters of the word MASTER are permuted in all possible ways and the words thus formed are arranged in the dictionary order. Then find the rank of the word · i) REMAST ii) MASTER.

i) REMAST ·

Sol:- The alphabetical order is A,E,M,R,S,T ·

The no. of words begin with $\underline{A}$ _ _ _ _ _ $= 5!$

The no. of words begin with $\underline{E}$ _ _ _ _ _ $= 5!$

The no. of words begin with $\underline{M}$ _ _ _ _ _ $= 5!$

The no. of words begin with $\underline{R}\ \underline{A}$ _ _ _ _ $= 4!$

The no. of words begin with $\underline{R}\ \underline{E}\ \underline{A}$ _ _ _ $= 3!$

The no. of words begin with $\underline{R}\ \underline{E}\ \underline{M}\ \underline{A}\ \underline{S}\ \underline{T}$ $= 0!$

$\therefore$ The rank of the word REMAST $= 5!+5!+5!+4!+3!+0!$

$= 120+120+120+24+6+1$

$= 391$.

ii) MASTER .

Sol:- The alphabetical order is A, E, M, R, S, T

The no. of words begin with A - - - - - $= 5!$

The no. of words begin with E - - - - - $= 5!$

The no. of words begin with M A E - - - $= 3!$

The no. of words begin with M A R - - - $= 3!$

The no. of words begin with M A S E - - $= 2!$

The no. of words begin with M A S R - - $= 2!$

The no. of words begin with M A S T E R $= 0!$

$\therefore$ the rank of the word MASTER is $= 5!+5!+3!+3!+2!+2!+0!$

$= 120+120+6+6+2+2+1$

$= 257$

④ If the letters of the word PRISON are permutated in all possible ways and words formed arranged in dictionary order then find rank of PRISON.

Sol:- The alphabetical order is I, N, O, P, R, S .

The no. of words begin with I - - - - - $= 5!$

The no. of words begin with N - - - - - $= 5!$

The no. of words begin with O - - - - - $= 5!$

The no. of words begin with P I - - - - $= 4!$

The no. of words begin with P N - - - - $= 4!$

The no. of words begin with P O - - - - $= 4!$

The no. of words begin with P R I N - - $= 2!$

The no. of words begin with P R I O - - $= 2!$

The no. of words begin with P R I S N - $= 1!$

The no. of words begin with P R I S O N $= 0!$

$\therefore$ The rank of the word PRISON is $= 5!+5!+5!+4!+4!+4!+2!+2!+1!+0!$

$= 120+120+120+24+24+24+2+2+1+1$

$= 438$ .

⑤ If the possible letters of word · BRING are permutated in all possible ways and the words thus formed are arranged in dictionary order, then find $59^{th}$ word.

Sol:- Given BRING $59^{th}$ word.

The alphabetical order is B, G, I, N, R.

The no. of words Begin with B _ _ _ _ $= 4! = 24$

The no. of words Begin with G _ _ _ _ $= 4! = 24$

The no. of words Begin with I B _ _ _ $= 3! = 6$

The no. of words Begin with I G B _ _ $= 2! = 2$

The no. of words Begin with I G N _ _ $= 2! = 2$

The no. of words Begin with I G R B N $= 0! = 1$.

∴ The $59^{th}$ word is IGRBN.

⑥ If the letters of the word EAMCET are permuted in all possible ways and if the words thus formed are arranged in dictionary order, find the rank of the word EAMCET.

Sol:- The alphabetical order is A, C, E, E, M, T.

The no. of words begin with A _ _ _ _ _ $= \frac{5!}{2!}$

The no. of words begin with C _ _ _ _ _ $= \frac{5!}{2!}$

The no. of words begin with E A C _ _ _ $= 3!$

The no. of words begin with E A E _ _ _ $= 3!$

The no. of words begin with E A M C E T $= 0!$

∴ The rank of the word EAMCET is $= \frac{5!}{2!} + \frac{5!}{2!} + 3! + 3! + 0!$

$= \frac{120}{2} + \frac{120}{2} + 6 + 6 + 1$

$= 60 + 60 + 6 + 6 + 1$

$= 133.$

⑦ If the letters of the word AJANTA are permuted in all possible ways and the words thus formed are arranged in the dictionary order, find the ranks of the words.

i) AJANTA ii) JANATA.

i) AJANTA.

Sol:- The alphabetical order is A, A, A, J, N, T.

The no. of words begin with A A _ _ _ _ = $4!$

The no. of words begin with A J A A _ _ = $2!$

The no. of words begin with A J A N A _ = $1!$

The no. of words begin with A J A N T A = $0!$

$\therefore$ The rank of the word is AJANTA $= 4!+2!+1!+0!$

$= 24+2+1+1$

$= 28.$

ii) JANATA.

The alphabetical order is A, A, A, J, N, T.

The no. of words begin with A _ _ _ _ _ = $\frac{5!}{2!}$

The no. of words begin with J A A _ _ _ = $3!$

The no. of words begin with J A N A A _ = $1!$

The no. of words begin with J A N A T A = $0!$

$\therefore$ The rank of the word is JANATA $= \frac{5!}{2!}+3!+1!+0!$

$= \frac{120}{2}+6+1+1$

$= 60+6+1+1$

$= 68.$

(8) Find the permutations of the letters word PICTURE so that i) all vowels come together ii) No two vowels come together.

Sol:- Given PICTURE is the word.

They are 3 vowels = I, U, E.

4 consonants = P, C, T, R

i) all vowels come together.

First treat

$n = 1$ unit (3 vowels) + 4 consonants = 5

$5!$ => (arrangements).

The vowels 3 are arranging among themselves is $3!$

$\therefore 5!\ 3! \Rightarrow 720$ ways.

ii) No two vowels come together:-

First arranging 4 consonants in $4!$ ways × C × C × C × C ×

Then we have to find 5 gaps between them.
The 5 gaps fill with the 3 vowels in $^5P_3$ ways.
The required no. of arrangements $= 4! \times {}^5P_3$
$= 1440$.

(9) Find the number of ways of seating 5 Indians, 4 Americans and 3 Russians at a round table so that i) all Indians are sit together. ii) persons of same nationality sit together iii) No two Russians sit together.

Sol:- Given 5 Indians + 4 Americans + 3 Russians = 12.
These can be arranged in a round table is $(n-1)! = (12-1)!$
$= 11!$

i) All Indians sit together:-
First 5 Indians treat as 1 unit
$n = 1\text{unit}(5I) + 4A + 3R = 8$.
These can be arranged in round table $(n-1)! = (8-1)! = 7!$
These 5 I are arranging among themselves is $5!$ ways.
∴ The required no. of arrangement $= 7! \times 5!$

ii) No two Russians sit together:-
First arranging $5I + 4A = 9$.
These can be arranged in a round table $(n-1)! = (9-1)! = 8!$.
Then we have to find 9 gaps between them.
The 9 gaps can be filled with 3R in $^9P_3$ ways.
∴ The required number of arrangement $= 8! \times {}^9P_3$.

iii) Persons of same nationality sit together:-
5I treat as 1unit, 4 A treat as 1unit, 3R treat as 1unit,
$n = 1\text{unit}(5I) + 1\text{unit}(4A) + 1\text{unit}(3R) = 3$.
These can be arranged in a round table $(n-1)! = (3-1)! = 2!$
5 Indians are arranging among themselves is $5!$
4 Americans are arranging among themselves is $4!$
3 Russians are arranging among themselves is $3!$
∴ The required no. of arrangements $= 5! \times 4! \times 3!$

(10) Find the number of ways of arranging 6 Red roses, and 3 yellow roses of different sizes into a garland In How many of them.

i) all the roses are together.

ii) no two yellow roses are together.

Sol:- Given 6 Red roses + 3 yellow roses = 9

These can be arranged in a garland is $\frac{(n-1)!}{2} = \frac{(9-1)!}{2} = \frac{8!}{2}$.

i) All the yellow roses are together:-

First 3 yellow roses treat as one unit.

$n = 1$ unit of $(3Y) + 6R = 7$.

This can be arranged in a garland is $\frac{(n-1)!}{2} = \frac{(7-1)!}{2} = \frac{6!}{2}$.

The 3 yellow roses are arranging among themselves is $3!$

∴ The required no. of arrangement $\frac{6!}{2} \times 3!$

ii) No two yellow roses are together:-

First arranging the 6 Red roses in a garland is $\frac{(n-1)!}{2} = \frac{(6-1)!}{2} = \frac{5!}{2}$.

Then we have to find 6 gaps between them.

The 6 gaps can be fill with 3 yellow roses is $^6P_3$ ways.

∴ The required no. of arrangement $\frac{5!}{2} \times {}^6P_3$.

(11) Find the number of ways of arranging 6 boys and 6 girls in a row. In how many of these arrangement.

i) all the girls are together ii) no two girls are together.

iii) Boys and girls come ~~togeth~~ alternately.

Sol:- 6 Boys + 6 girls = 12 persons

They can be arrange in a row in $12!$ ways.

i) all the girls are come together:

First treat as a 6 girls as a 1 unit and 6 boys as 6 units.

$n = 1$ unit $(6G) + 6B = 7$.

These can be arrange in a row $= 7!$ ways.

The 6 girls can arranging among themselves is $6!$ ways.

∴ the require no. of arrangements $7!6!$

ii) No two girls are together :-

The first arrange in the 6 boys in a row is $6!$ $\quad -B_1-B_2-B_3-B_4-B_5-B_6-$

Then we have to find 7 gaps between them.

The 7 gaps fill with the 6 girls is ${}^7P_6$.

∴ The required no. of arrangements $6! \times {}^7P_6$

$= 6!\,7!$

iii) Boys and Girls comes alternatively :-

Let us take 12 places.

The row may began with either a boy or girl that is 2 ways.

If its begin with a boy the odd places (1,3,5,7,9,11) will be occupied by boys and the even places (2,4,6,8,12) by girls.

The 6 boys can be arranged in the 6 odd places is $6!$

The 6 girls can be arranged in the 6 even places is $6!$ ways.

The required no. of arrangement $= 2 \times 6! \times 6!$

$= 1038800$.

(12) Find the number of ways of arranging 6 boys and 6 girls around a circular table. So that i) all the girls sit together ii) No two girls sit together. iii) Boys and girls sit alternately.

Sol:- Given $6B + 6G = 12$ persons.

These can be arranged in a round a circular table is $(n-1)! = (12-1)! = 11!$

i) All the girls sit together :-

6 girls treat as a 1 unit.

$n = 1\text{ unit}(6G) + 6B = 7$.

These can be arranged in a around a circular table is $(n-1)! = (7-1)! = 6!$

The 6 girls are arranging among themselves is $6!$ ways.

The required no. of arrangements $= 6! \times 6!$

[redacted]

ii) No two girls sit together:-

First arranging the 6 boys around a circular table is $(n-1)! = (6-1)! = 5!$
Then we have to find 6 gaps between them.
The 6 gaps can be filled with 6 girls is $^6P_6 = 6!$ ways.
$\therefore$ The required no. of arrangements $= 5! \times 6!$

iii) Boys and girls sit alternately:-

The circular arrangement no importance for the began with boys and girls.

First arranging the 6 boys around a circular table is $(n-1)! = (6-1)! = 5!$
Then we have to find 6 gaps between them.
The 6 gaps can be fill with 6 girls is $^6P_6 = 6!$ ways.
$\therefore$ The required arrangement $= 5! \times 6!$

(13) i) Find the no. of ways of selecting a cricket team of 11 players from 7 batsmen and 6 bowlers such that there will be atleast 5 bowlers in the team.

Sol:- Given no. of bowlers $= 6$.
No. of batsmen $= 7$.

we can select a cricket team of 11 players where atleast 5 bowlers in the team as follows:

| S.NO | 6 bowlers | 7 batsmen | No. of selections |
|---|---|---|---|
| 1 | 5 | 6 | $^6C_5 \times {}^7C_6$ |
| 2 | 6 | 5 | $^6C_6 \times {}^7C_5$ |

$\therefore$ Required no. of selections $= {}^6C_5 \times {}^7C_6 + {}^6C_6 \times {}^7C_5$
$= 6 \times 7 + 1 \times 21$
$= 42 + 21$
$= 63$.

$^7C_2 = \frac{7 \times \cancel{6}\,3}{1 \times \cancel{2}}$

$^6C_1 = 6$.

ii) Find the no. of ways of selecting 11 members cricket team from 7 bats men, 6 bowlers, 2 wickets-keepers, so that team contains 2 wickets-keeps and atleast 4 bowlers.

Sol: Given no.of batsmen = 7.

No.of bowlers = 6.

No.of wicket keepers = 2.

We can select a Cricket team of 11 players the team contains 2 wicket keepers and at least 4 bowlers.

| S.NO | 6 bowlers | 2 W.K | 7 batsmen | No.of selections |
|---|---|---|---|---|
| 1 | 4 | 2 | 5 | $^6C_4 \times {}^2C_2 \times {}^7C_5$ |
| 2 | 5 | 2 | 4 | $^6C_5 \times {}^2C_2 \times {}^7C_4$ |
| 3 | 6 | 2 | 3 | $^6C_6 \times {}^2C_2 \times {}^7C_3$. |

Total no.of selections $= {}^6C_4 \times {}^2C_2 \times {}^7C_5 + {}^6C_5 \times {}^2C_2 \times {}^7C_4 + {}^6C_6 \times {}^2C_2 \times {}^7C_3$.

$= 15 \times 1 \times 21 + 6 \times 1 \times 35 + 1 \times 1 \times 35$

$= 315 + 210 + 35$

$= 560$.

(14) Find the no.of ways of forming a comittee of 5 members out of 6I and 5 Americans. so that always the Indian will be in majority in the comitee.

Sol: Given no. of Indians = 6, No.of Americans = 5.

We can select the comitee of 5 members such that the Indians will be majority in the committee, by using the following table.

| S.NO | 6 Indians | 5 Americans | No.of selections |
|---|---|---|---|
| 1 | 5 | 0 | $^6C_5 \times {}^5C_0$ |
| 2 | 4 | 1 | $^6C_4 \times {}^5C_1$ |
| 3 | 3 | 2 | $^6C_3 \times {}^5C_2$ |

$\therefore$ No. of selections $= {}^6C_5 \times {}^5C_0 + {}^6C_4 \times {}^5C_1 + {}^6C_3 \times {}^5C_2$

$= 6 \times 1 + 15 \times 5 + 20 \times 10$

$= 6 + 75 + 200$

$= 281$.

(15) A candidate is required to answer 6 out of 10 questions which are divided into 2 groups A & B each containing 5 questions. He is not permitted to attempt more than 4 questions from either group. Find the no. of different ways in which the candidate can choose six questions.

Sol:- Total questions = 10.

Candidate has to answer 6 questions.

the required no. of different ways can be done in following ways.

| S.NO | Group A (5) | Group B (5) | No. of selections |
|---|---|---|---|
| 1 | 4 | 2 | $5C_4 \times 5C_2 = 10 \times 5 = 50$ |
| 2 | 3 | 3 | $5C_3 \times 5C_3 = 10 \times 10 = 100$ |
| 3 | 2 | 4 | $5C_2 \times 5C_4 = 10 \times 5 = 50$ |

$\therefore$ Total no. of selections $= 50 + 100 + 50$

$= 200$ ways.

(16) Find the no. of ways of forming a committee of 4 members out of 6 boys and 4 girls. Such that there is atleast one girl in the committee.

Sol:- No. of boys = 6, No. of girls = 4.

we have to form a committee of 4 members containg atleast one girl, this can be done in the following ways.

| S.NO | Boys (6) | Girls (4) | No. of Selections |
|---|---|---|---|
| 1 | 3 | 1 | $6C_3 \times 4C_1 = 20 \times 4 = 80$ |
| 2 | 2 | 2 | $6C_2 \times 4C_2 = 15 \times 6 = 90$ |
| 3 | 1 | 3 | $6C_1 \times 4C_3 = 6 \times 4 = 24$ |
| 4 | 0 | 4 | $6C_0 \times 4C_4 = 1 \times 1 = 1$ |

$\therefore$ Total no. of selections $= 80 + 90 + 24 + 1$

$= 195$ ways.

(17) i) If $1 \le r \le n$, then prove that ${}^{n}C_{r-1} + {}^{n}C_r = {}^{n+1}C_r$.

ii) For $3 \le r \le n$, then prove that ${}^{n-3}C_r + 3\,{}^{n-3}C_{r-1} + 3\,{}^{n-3}C_{r-2} + {}^{n-3}C_{r-3} = {}^{n}C_r$

Sol:- i) $L.H.S = {}^{n}C_{(r-1)} + {}^{n}C_r = \frac{n!}{(n-(r-1))!(r-1)!} + \frac{n!}{(n-r)!r!}$

$= n!\left\{\frac{1}{(n-r+1)!(r-1)!} + \frac{1}{(n-r)!r!}\right\}$ $\quad \boxed{\because {}^{n}C_r = \frac{n!}{(n-r)!r!}}$

$= n!\left\{\frac{r}{(n-r+1)!\,r(r-1)!} + \frac{n-r+1}{(n-r)!r!(n-r+1)}\right\}$

$\boxed{\because n!(n+1) = (n+1)!}$

$= n!\left\{\frac{\cancel{r} + (n-\cancel{r}+1)}{(n-r+1)!r!}\right\} \Rightarrow \frac{n!(n+1)}{(n-r+1)!r!} = \frac{(n+1)!}{((n+1)-r)!r!}$

$= {}^{n+1}C_r = R.H.S$

ii) $L.H.S\ {}^{n-3}C_r + 3.{}^{n-3}C_{r-1} + 3.{}^{n-3}C_{r-2} + {}^{n-3}C_{r-3}$

$= ({}^{n-3}C_r + {}^{n-3}C_{r-1}) + 2.({}^{n-3}C_{r-1} + {}^{n-3}C_{r-2}) + {}^{n-3}C_{r-2} + {}^{n-3}C_{r-3}$

$= {}^{n-2}C_r + 2.{}^{n-2}C_{r-1} + {}^{n-2}C_{r-2}$

$= ({}^{n-2}C_r + {}^{n-2}C_{r-1}) + ({}^{n-2}C_{r-1} + {}^{n-2}C_{r-2})$

$= {}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^{n}C_r = R.H.S$

18) i) Find the sum of all 4 digited numbers that can be formed using the digits 1, 2, 4, 5, 6 without repetation.

ii) Find the sum of all 4 digited numbers that can be formed using the digits 1, 3, 5, 7, 9.

Sol:- i) Given digits 1, 2, 4, 5, 6 Here $r=4, n=5$.

[$\because$ sum of all r digited numbers $= {}^{n-1}P_{r-1}$ (sum of given digits). (111...r times)]

$= {}^{5-1}P_{4-1}.(1+2+4+5+6)(1111)$.

$= {}^{4}P_3\,(18)(1111) = 24 \times 18 \times 1111 = 479952$

ii) Given digits 1, 3, 5, 7, 9 Here $r=4, n=5$.

[$\because$ sum of all r-digited numbers $= {}^{n-1}P_{r-1}$ (sum of given digits)(111...r times)]

(52)

sum of all 4-digited numbers $= {}^{n-1}P_{r-1}$ (sum of n-digits)(111---r times).

$= {}^{5-1}P_{4-1}(1+3+5+7+9)(1111)$

$= {}^{4}P_{3}(25)(1111)$

$= 24 \times 25 \times 1111$

$= 6,66,600$ .

(19) Find the no. of numbers that are greater than 4000 which can be formed using digits 0, 2, 4, 6, 8 without repetation.

Sol:- Given digits are 0, 2, 4, 6, 8 .

Case-i) 4-digit number

The first place cannot be filled with 0, 2

The first place must be filled with ether 4 or 6 or 8 in 3 ways.

Remaining 3 places can be filled with remaining 4-digits in ${}^{4}P_{3}$ ways.

No. of arrangements $= 3 \times {}^{4}P_{3}$

$= 3 \times 4 \times 3 \times 2$

$= 72$ .

Case ii) :- 5-digit number:-

'0' cannot be filled in 1st place must be filled either 2 or 4 or 6 or 8 in 4 ways.

Remaining 4 places can be filled with remaining 4-digits in ${}^{4}P_{4}$ ways.

No. of arrangements $= 4 \times {}^{4}P_{4} = 4 \times 24 = 96$ .

$\therefore$ Total no. of numbers greater than $4000 = 72 + 96$

$= 168$ .

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# PARTIAL FRACTIONS (4M)

1) $\dfrac{x+4}{(x^2-4)(x+1)}$

Sol Given

$$= \frac{x+4}{(x^2-4)(x+1)}$$

$$= \frac{x+4}{(x-2)(x+2)(x+1)}$$

let $\dfrac{x+4}{(x-2)(x+2)(x+1)} = \dfrac{A}{(x-2)} + \dfrac{B}{(x+2)} + \dfrac{C}{(x+1)} \longrightarrow (1)$

$$\frac{x+4}{(x-2)(x+2)(x+1)} = \frac{A(x+2)(x+1)+B(x-2)(x+1)+C(x-2)(x+2)}{(x-2)(x+2)(x+1)}$$

$$x+4 = A(x+2)(x+1)+B(x-2)(x+1)+C(x-2)(x+2) \text{ --- } (2)$$

Put $x-2=0$

$x=2$ in eq (2)

$$2+4 = A(2+2)(2+1)+B(2-2)(2+1)+C(2-2)(2+2)$$

$$6 = A(4)(3)+B(0)+C(0)$$

$$6 = A(12)$$

$$A = \frac{6}{12}$$

$$\boxed{A = \frac{1}{2}}$$

Now

Put $x+2=0$

$x=-2$ in eq (2)

$$-2+4 = A(-2+2)(-2+1)+B(-2-2)(-2+1)+C(-2-2)(-2+2)$$

$$2 = A(0)+B(-4)(-1)+C(-4)(0)$$

$$2 = 4B$$

$$\boxed{B = \frac{1}{2}}$$

Now

Put $x+1=0$

$x=-1$ in eq ②

$$-1+4 = A(-1+2)(-1+1)+B(-1-2)(-1+1)+C(-1-2)(-1+2)$$

$$3 = A(0)+B(0)+C(-3)(1)$$

$$3 = -3C$$

$$\boxed{C=-1}$$

from ①

$$\frac{x+4}{(x+2)(x-2)(x+1)} = \frac{1}{2(x-2)} + \frac{1}{2(x+2)} - \frac{1}{x+1}$$

2) $\dfrac{x-1}{(x+1)(x-2)^2}$

sol given

$$\frac{x-1}{(x+1)(x-2)^2}$$

let $$\frac{x-1}{(x+1)(x-2)^2} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

$$\frac{x-1}{(x+1)(x-2)^2} = \frac{A(x-2)^2+B(x-2)(x+1)+C(x+1)}{(x+1)(x-2)^2}$$

$$x-1 = A(x-2)^2+B(x+1)(x-2)+C(x+1) \quad \text{—— ②}$$

put $x=-1$ in eq ②

$$-2 = A(-1-2)^2+B(-1+1)(-1-2)+C(0)$$

$$-2 = 9A+0+0$$

$$\boxed{A = \frac{-2}{9}}$$

put $x=2$ in eq ②

$$2-1 = A(2-2)^2+B(2+1)(2-2)+C(2+1)$$

$$1 = 0+0+3C$$

$$\boxed{C=\frac{1}{3}}$$

comparing coefficient of $x^2$ on both sides

$$0=A+B$$

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$$0 = -\frac{2}{9} + B$$

$$\boxed{B = \frac{2}{9}}$$

$$\therefore \frac{x-1}{(x+1)(x-2)^2} = \frac{-2}{9(x+1)} + \frac{2}{9(x-2)} + \frac{1}{3(x-2)^2}$$ //

---

3) $\dfrac{x^2-3}{(x+2)(x^2+1)}$

Sol given

$$\frac{x^2-3}{(x+2)(x^2+1)}$$

let $$\frac{x^2-3}{(x+2)(x^2+1)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+1} \quad \text{--- (1)}$$

$$\frac{x^2-3}{(x+2)(x^2+1)} = \frac{A(x^2+1)+(Bx+C)(x+2)}{(x+2)(x^2+1)}$$

$$x^2-3 = A(x^2+1) + Bx(x+2) + C(x+2) \quad \text{--- (2)}$$

Put $x+2=0$

$x=-2$ in eq (2)

$$4-3 = A(4+1)+0+0$$

$$1 = 5A$$

$$\boxed{A = \frac{1}{5}}$$

compare coefficient of $x$ on b.s

$$0 = 2B + C$$

$$0 = 2\left(\frac{4}{5}\right) + C$$

$$\boxed{C = -\frac{8}{5}}$$

comparing coefficient of '$x^2$' on b.s

$$1 = A + B$$

$$1 = \frac{1+5B}{5}$$

$$5 = 1+5B \Rightarrow \boxed{B = \frac{4}{5}}$$

$$\therefore \frac{x^2-3}{(x+2)(x^2+1)} = \frac{1}{5(x+2)} + \frac{4x-8}{5(x^2+1)}$$ //

4) i, $\dfrac{2x^2+3x+4}{(x-1)(x^2+2)}$

Sol Given

$$\frac{2x^2+3x+4}{(x-1)(x^2+2)}$$

Let $\dfrac{2x^2+3x+4}{(x-1)(x^2+2)} = \dfrac{A}{x-1} + \dfrac{Bx+C}{x^2+2}$ —— (1)

$$\frac{2x^2+3x+4}{(x-1)(x^2+2)} = \frac{A(x^2+2)+(Bx+C)(x-1)}{(x-1)(x^2+2)}$$

$2x^2+3x+4 = A(x^2+2) + Bx(x-1) + C(x-1)$ —— (2)

Put $x=1$ in eq (2)

$$2+3+4 = A(1+2)+0+0$$

$$9 = 3A$$

$$\boxed{A=3}$$

coefficient of $x^2$ on b.s

$$2 = A+B$$

$$2 = 3+B$$

$$\boxed{B=-1}$$

coefficient of $x$ on b.s

$$3 = -B+C$$

$$3 = 1+C$$

$$\boxed{C=2}$$

$\therefore \dfrac{2x^2+3x+4}{(x-1)(x^2+2)} = \dfrac{3}{(x-1)} + \dfrac{(-x+2)}{x^2+2}$ ,,

ii) $\dfrac{x^2+5x+7}{(x-3)^3}$

Sol Let $y = x-3$

$x = y+3$

Now, $\dfrac{x^2+5x+7}{(x-3)^3} = \dfrac{(y+3)^2+5(y+3)+7}{y^3}$

$$= \frac{y^2+9+6y+5y+15+7}{y^3}$$

$$= \frac{y^2+11y+31}{y^3}$$

$$= \frac{y^2}{y^3} + \frac{11y}{y^3} + \frac{31}{y^3}$$

$$= \frac{1}{y} + \frac{11}{y^2} + \frac{31}{y^3}$$

$\therefore \dfrac{x^2+5x+7}{(x-3)^3} = \dfrac{1}{x-3} + \dfrac{11}{(x-3)^2} + \dfrac{31}{(x-3)^3}$ ,,

5) i) $\dfrac{x^3}{(2x-1)(x+2)(x+3)}$

Sol Given

fraction is an improper fraction

degree of $f(x)$ = degree of $g(x)$

$$\frac{x^3}{(2x-1)(x+2)(x-3)} = k + \frac{A}{2x-1} + \frac{B}{x+2} + \frac{C}{x-3}$$

$k = \dfrac{\text{coefficient of highest power of } x \text{ in Numerator}}{\text{coefficient of highest power of } x \text{ in Denominator}}$

$$\boxed{k = \frac{1}{2}}$$

$$\frac{x^3}{(2x-1)(x+2)(x-3)} = \frac{1}{2} + \frac{A}{2x-1} + \frac{B}{x+2} + \frac{C}{x-3}$$

$$\frac{x^3}{(2x-1)(x+2)(x-3)} = \frac{(2x-1)(x+2)(x-3)+2A(x+2)(x-3)+2B(2x-1)(x-3)+2C(2x-1)(x+2)}{2(2x-1)(x+2)(x-3)}$$

$$x^3 = (2x-1)(x+2)(x-3)+2A(x+2)(x-3)+2B(2x+1)(x-3)+2C(2x-1)(x+2) \quad \text{——— } (1)$$

Put $x+2=0 \Rightarrow x=-2$ in eq (1)

$$-16 = 2B(-5)(-5)$$

$$-16 = 50B$$

$$B = \frac{-16}{50}$$

$$\boxed{B = \frac{-8}{25}}$$

Put $2x-1=0 \Rightarrow x=\frac{1}{2}$ in eq (1)

$$\frac{2}{8} = 2A\left(\frac{1}{2}+2\right)\left(\frac{1}{2}-3\right)$$

$$\frac{2}{8} = A\left(\frac{5}{2}\right)\left(\frac{-5}{2}\right)$$

$$\boxed{A = \frac{-1}{50}}$$

put $x-3=0$

$\Rightarrow x=3$ in eq (1)

$$2(27) = 0+0+0+2C(5)(5)$$

$$\boxed{C = \frac{27}{25}}$$

$$\therefore \frac{x^3}{(2x-1)(x+2)(x-3)} = \frac{1}{2} - \frac{1}{50(2x-1)} - \frac{8}{25(x+2)} + \frac{27}{25(x-3)}$$

ii) $\dfrac{x^3}{(x-a)(x-b)(x-c)}$

It is an improper fraction

Deg of $f(x)$ = Deg of $g(x)$

$$\frac{f(x)}{g(x)} = \frac{x^3}{(x-a)(x-b)(x-c)} = K + \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

$$K = \frac{\text{coefficient of highest power of } x \text{ in numerator}}{\text{coefficient of highest power of } x \text{ in denominator}}$$

$$K = \frac{1}{1} \quad \boxed{K=1}$$

$$\frac{x^3}{(x-a)(x-b)(x-c)} = 1 + \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

$$\frac{x^3}{(x-a)(x-b)(x-c)} = \frac{(x-a)(x-b)(x-c) + A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b)}{(x-a)(x-b)(x-c)}$$

$$x^3 = (x-a)(x-b)(x-c) + A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b) \quad \text{--- (1)}$$

Put $x-a=0$

$\Rightarrow x=a$ in eq (1)

$$a^3 = 0 + A(a-b)(a-c) + 0 + 0$$

$$a^3 = A(a-b)(a-c)$$

$$\boxed{A = \frac{a^3}{(a-b)(a-c)}}$$

Put $x-b=0$

$x=b$ in eq (1)

$$b^3 = B(b-a)(b-c) + 0$$

$$b^3 = B(b-a)(b-c)$$

$$\boxed{B = \frac{b^3}{(b-a)(b-c)}}$$

Put $x - c = 0$

$x = c$ in eq ①

$c^3 = C(c-a)(c-b)$

$$\boxed{C = \frac{c^3}{(c-a)(c-b)}}$$

$$\therefore \frac{x^3}{(x-a)(x-b)(x-c)} = 1 + \frac{a^3}{(x-a)(a-b)(a-c)} + \frac{b^3}{(b-a)(b-c)(x-b)} + \frac{c^3}{(c-a)(c-b)(x-c)}$$

6) i $\dfrac{x^4}{(x-1)(x-2)}$

Sol It is an improper fraction

deg of $f(x)$ = deg of $g(x)$

$$\frac{f(x)}{g(x)} = Q(x) + \frac{R(x)}{g(x)}$$

$$\frac{x^4}{(x-1)(x-2)} = \frac{x^4}{x^2-2x-x+2}$$

$$\frac{x^4}{(x-1)(x-2)} = \frac{x^4}{x^2-3x+2}$$

$$\begin{array}{r|rrrrr} 3 & 1 & 0 & 0 & 0 & 0 \\ & 0 & 3 & 9 & 21 & 0 \\ & 0 & 0 & -2 & -6 & -14 \\ \hline & 1 & 3 & 7 & 15 & -14 \end{array}$$

$15 \;\; -14 \rightarrow$ Remainder

$Q(x) = x^2+3x+7$

$R(x) = 15x-14$

$$\frac{x^4}{(x-1)(x-2)} = x^2+3x+7 + \frac{15x-14}{(x-1)(x-2)}$$

Now,

$$\frac{15x-14}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$$

$$15x-14 = A(x-2) + B(x-1) \quad \text{——} \quad ①$$

Put $x-1=0$

$\Rightarrow x=1$ in eq ①

$1 = A(-1)$

$\boxed{A=-1}$

Put $x-2=0$

$\Rightarrow x=2$ in eq ①

$16 = 0 + B(1)$

$\boxed{B=16}$

$$\therefore \frac{x^4}{(x-1)(x-2)} = x^2+3x+7+\frac{-1}{x-1}+\frac{16}{x-2}$$

ii) $\dfrac{x^3}{(x-1)(x+2)}$

Sol It is an improper fraction

deg of $f(x) >$ deg of $g(x)$

$$\frac{f(x)}{g(x)} = Q(x) + \frac{R(x)}{g(x)}$$

$$\frac{x^3}{(x-1)(x+2)} = \frac{x^3}{x^2+x-2}$$

By using synthetic division

$$\begin{array}{r|rrrr} & 1 & 0 & 0 & 0 \\ -1 & 0 & -1 & 1 & 0 \\ 2 & 0 & 0 & 2 & -2 \\ \hline & 1 & -1 & \boxed{3} & \boxed{-2} \end{array} \longrightarrow \text{Remainder}$$

$g(x) = x-1 \quad R(x) = 3x-2$

$$\frac{x^3}{(x-1)(x+2)} = x-1+\frac{3x-2}{(x-1)(x+2)}$$

Now,

$$\frac{3x-2}{(x-1)(x+2)} = \frac{A}{x-1}+\frac{B}{x+2}$$

$$3x-2 = A(x+2)+B(x-1) \quad \text{——} \; ①$$

Put $x-1=0$

$\Rightarrow x=1$ in eq (1)

$1=3A$

$$\boxed{A=\frac{1}{3}}$$

Put $x+2=0$

$x=-2$ in eq (1)

$-8=-3B$

$$\boxed{B=\frac{8}{3}}$$

$$\therefore \frac{x^3}{(x-1)(x+2)} = x-1+\frac{1}{3(x-1)}+\frac{8}{3(x+2)}$$

---

7) $\dfrac{3x^3-8x^2+10}{(x-1)^4}$

Sol Given

$$\frac{3x^3-8x^2+10}{(x-1)^4}$$

let $x-1=y$

$y+1=x$

$$\frac{3x^3-8x^2+10}{(x-1)^4} = \frac{3(y+1)^3-8(y+1)^2+10}{y^4}$$

$$= \frac{3(y^3+1+3y^2+3y)-8(y^2+1+2y)+10}{y^4}$$

$$= \frac{3y^3+3+9y^2+9y-8y-8-16y+10}{y^4}$$

$$= \frac{3y^3+y^2-7y+5}{y^4}$$

$$= \frac{3y^3}{y^4}+\frac{y^2}{y^4}-\frac{7y}{y^4}+\frac{5}{y^4}$$

$$\frac{3}{y} + \frac{1}{y^2} - \frac{7}{y^3} + \frac{5}{y^4}$$

$$= \frac{3}{(x-1)} + \frac{1}{(x-1)^2} - \frac{7}{(x-1)^3} + \frac{5}{(x-1)^4}$$

$$\therefore \frac{3x^3-8x^2+10}{(x-1)^4} = \frac{3}{x-1} + \frac{1}{(x-1)^2} + \frac{-7}{(x-1)^3} + \frac{5}{(x-1)^4}$$

# PROBABILITY

$S_{16}$ 1)

A, B, C are three horses in a race. The probability of A to win the race is twice that of B, and the probability of B is twice that of C. what are the probabilities of A, B and C to win the race.

Sol:- A, B, C are 3 horses in a race that events A, B, C respectively.

$$P(A) = 2P(B) = 2P(C) = 4P(C)$$
$$P(A) = 4P(C)$$
$$P(B) = 2P(C)$$

sum of the probabilities = 1

$$P(A) + P(B) + P(C) = 1$$
$$4P(C) + 2P(C) + P(C) = 1$$
$$7P(C) = 1$$
$$\boxed{P(C) = 1/7}$$

$$P(B) = 2P(C)$$
$$P(B) = 2\left(\frac{1}{7}\right)$$
$$\boxed{P(B) = \frac{2}{7}}$$

$$P(A) = 4P(C)$$
$$= 4\left(\frac{1}{7}\right)$$
$$= \frac{4}{7}$$

The Probabilities of A, B, C are

$$P(A) = \frac{4}{7} \qquad P(B) = \frac{2}{7} \qquad P(C) = \frac{1}{7}$$

2) If A and B are independenent events with $P(A) = 0.2$ $P(B) = 0.5$ then find

a) $P(A/B)$ b) $P(B/A)$ c) $P(A \cap B)$ d) $P(A \cup B)$

Sol: Given A, B are independent

$$P(A) = 0.2 \qquad P(B) = 0.5$$

a) $P(A/B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{P(A) \cdot \cancel{P(B)}}{\cancel{P(B)}} = 0.2$

b) $P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{\cancel{P(A)} \cdot P(B)}{\cancel{P(A)}} = 0.5$

c) $P(A \cap B) = P(A) \cdot P(B)$

$= 0.2 \times 0.5$

$= 0.1$

d) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.2 + 0.5 - 0.1$

$= 0.7 - 0.1$

$= 0.6$

ii) If A & B be independent events with $P(A) = 0.6$, $P(B) = 0.7$ then compute.

a) $P(A \cap B)$ b) $P(A \cup B)$ c) $P(B/A)$ d) $P(A^c \cap B^c)$

Sol: $P(A) = 0.6$ $P(B) = 0.7$

a) $P(A \cap B) = P(A) \cdot P(B)$

$= (0.6)(0.7)$

$= 0.42$

b) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.6 + 0.7 - 0.42$

$= 1.3 - 0.42$

$P(A \cup B) = 0.88$

c) $P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{\cancel{P(A)} P(B)}{\cancel{P(A)}} = 0.7$

d) $P(A^c \cap B^c) = P(\bar{A} \cap \bar{B})$

$= P(\overline{A \cup B})$

$= 1 - P(A \cup B)$

$= 1 - 0.88$

$\boxed{P(A^c \cap B^c) = 0.12}$

iii) A and B are events with $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.3$ find the probability that

i) A does not occur

ii) Neither A nor B occurs.

Sol:- Given $P(A) = 0.5$ $P(B) = 0.4$

$P(A \cap B) = 0.3$

i) A does not occur

$P(\bar{A}) = 1 - P(A)$

$= 1 - 0.5$

$P(\bar{A}) = 0.5$

ii) Neither A nor B occurs

By Addition theorem, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.5 + 0.4 - 0.3$

$= 0.6$

$P(\overline{A \cup B}) = 1 - P(A \cup B)$

$= 1 - 0.6$

$= 0.4$

3) If two numbers are selected randomly from 20 consecutive natural numbers. Find the probability.

i) An even number:

let A be the event getting the sum of the 2 numbers is an even number

$n(A) = 10_{C_2} + 10_{C_2}$

$= \frac{10 \times 9}{2} + \frac{10 \times 9}{2}$

$= 45 + 45$

$= 90$

$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{9\cancel{0}}{19\cancel{0}} = \frac{9}{19}$

$\therefore$ The probability that the sum of the 2 numbers is an even number is $\frac{9}{19}$

ii) An odd number:

let B be the event of getting the sum of 2 numbers is an odd number.

$n(B) = 10_{C_1} \times 10_{C_1}$

$n(B) = 10 \times 10 \Rightarrow 100$

$P(B) = \frac{n(B)}{n(S)} = \frac{10\cancel{0}}{19\cancel{0}} = \frac{10}{19}$

$\therefore$ Probability that the sum of two numbers is an odd number is $10/19$.

4) A speaks truth is 75% of the cases and B is 80% cases. what is the probability that their statements about an incident do not match.

Sol: Let A, B be the events that A and B respectively speak truth about an incident.

$$P(A) = \frac{\cancel{75}^{3}}{\cancel{100}_{4}} = \frac{3}{4} \qquad P(B) = \frac{\cancel{80}^{4}}{\cancel{100}_{5}} = \frac{4}{5}$$

$$P(\bar{A}) = 1 - P(A)$$

$$P(\bar{A}) = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(\bar{B}) = 1 - \frac{4}{5} = \frac{1}{5}$$

The Probability that their statements about an incident do not match.

$$= P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= P(A) \cdot P(\bar{B}) + P(\bar{A}) \cdot P(B)$$

$$= \frac{3}{4} \cdot \frac{1}{5} + \frac{1}{4} \cdot \frac{4}{5}$$

$$= \frac{3}{20} + \frac{4}{20}$$

$$= \frac{7}{20}$$

$\therefore$ The probability that their statements about an incident do not match $= \frac{7}{20}$

5) If A, B, C are three independent events of an experiment such that $P(A \cap B^c \cap C^c) = \frac{1}{4}$, $P(A^c \cap B C^c) = \frac{1}{8}$, $P(A^c \cap B^c \cap C^c) = \frac{1}{4}$ then find $P(A)$, $P(B)$ and $P(C)$

Sol: $P(A \cap B^c \cap C^c) = \frac{1}{4}$ —— (1)

$P(A^c \cap B \cap C^c) = \frac{1}{8}$ —— (2)

$P(A^c \cap B^c \cap C^c) = \frac{1}{4}$ —— (3)

$$\frac{eq(1)}{eq(3)} \Rightarrow \frac{P(A \cap \bar{B} \cap \bar{C})}{P(\bar{A} \cap \bar{B} \cap \bar{C})} = \frac{1/\cancel{4}}{1/\cancel{4}}$$

$$\frac{P(A)\cancel{P(\bar{B})}\cancel{P(\bar{C})}}{P(\bar{A})\cancel{P(\bar{B})}\cancel{P(\bar{C})}} = 1$$

$$P(A) = P(\bar{A})$$

$$P(A) = 1 - P(A)$$

$$2P(A) = 1$$

$$\boxed{P(A) = \frac{1}{2}}$$

$$\frac{eq\ 2}{eq\ 3} \Rightarrow \frac{P(\bar{A} \cap B \cap \bar{C})}{P(\bar{A} \cap \bar{B} \cap \bar{C})} = \frac{\frac{1}{\cancel{8}_2}}{\frac{1}{\cancel{4}}}$$

$$\frac{\cancel{P(\bar{A})}\ P(B)\ \cancel{P(\bar{C})}}{\cancel{P(\bar{A})}\ P(\bar{B})\ \cancel{P(\bar{C})}} = \frac{1}{2}$$

$$2P(B) = P(\bar{B})$$

$$2P(B) = 1 - P(B)$$

$$3P(B) = 1$$

$$\boxed{P(B) = \frac{1}{3}}$$

from eq 1

$$P(A \cap B^c \cap C^c) = \frac{1}{4}$$

$$P(A)\ P(\bar{B}) \cdot P(\bar{C}) = \frac{1}{4}$$

$$P(A)(1 - P(B))(1 - P(C)) = \frac{1}{4}$$

$$\frac{1}{\cancel{2}}\left(1 - \frac{1}{3}\right)(1 - P(C)) = \frac{1}{\cancel{4}\,2}$$

$$\frac{2}{3}(1 - P(C)) = \frac{1}{2}$$

$$1 - P(C) = \frac{1}{2} \times \frac{3}{2}$$

$$P(C) = 1 - \frac{3}{4}$$

$$\boxed{P(C) = \frac{1}{4}}$$

$$\therefore \boxed{P(A) = \frac{1}{2}} \quad \boxed{P(B) = \frac{1}{3}} \quad \boxed{P(C) = \frac{1}{4}}$$

6) A problem in calculus is given to two students A and B whose chances of solving it are $\frac{1}{3}$ and $1/4$ respectively. Find the probability of the problem being solved if both of the them try independently

Sol: let A, B be the event that the probability is solved by A and B respectively.

$$P(A) = 1/3 \qquad P(B) = 1/4$$

$$\boxed{P(A) + P(\bar{A}) = 1}$$

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$P(\bar{B}) = 1 - P(B) = 1 - \frac{1}{4} = \frac{3}{4}$$

The probability of the problems being solved if both of them try independently.

$$= 1 - P(\text{not solved problem})$$
$$= 1 - P(\bar{A} \cap \bar{B})$$
$$= 1 - P(\bar{A}) \cdot P(\bar{B})$$
$$= 1 - \frac{\cancel{2}}{\cancel{3}} \times \frac{\cancel{3}^1}{\cancel{4}_2}$$
$$= 1 - \frac{1}{2}$$
$$= \frac{1}{2}$$

---

7) A number x is drawn arbitarily from the set $\{1, 2, 3, \ldots \ldots 100\}$. Find the probability that $\left(x + \frac{100}{x}\right) > 29$

Sol:- let S be the sample space.

$$n(S) = 100$$

let A be the event that x selected randomly from,

$$A = \{1, 2, 3, \ldots \ldots 100\}$$

$$\left(x + \frac{100}{x}\right) > 29$$

$$x^2 - 29x + 100 > 0$$

$$x^2 - 25x - 4x + 100 > 0$$

$$x(x-25) - 4(x-25) > 0$$

$(x-4)(x-25)>0$

$\boxed{(x-\alpha)(x-\beta)>0 \text{ then } x\in(-\infty,\alpha]\cup(\beta,\infty)}$

$x\in(-\infty,4)\cup(25,\infty)$

$A=\{1,2,3,\ldots\ldots 100\}$

$n(A)=78$

$\therefore P(A)=\dfrac{n(A)}{n(S)}=\dfrac{78}{100}=0.78$

8) The probability that Australia wins a match against India in a cricket game is given 1/3. If India and Australia play 3 matches. what is the probability that

i) Australia will loose all 3 matches

ii) Australia will win atleast one match.

Sol: Given $P(A)=1/3$

$\boxed{P(A)+P(\bar{A})=1}$

$P(\bar{A})=1-P(A)$

$P(\bar{A})=1-\dfrac{1}{3}$

$=\dfrac{2}{3}$

i) Australia will loose all the 3 matches:-

$=P(\bar{A})\cdot P(\bar{A})\cdot P(\bar{A})$

$=\dfrac{2}{3}\times\dfrac{2}{3}\times\dfrac{2}{3}$

$=\dfrac{8}{27}$

ii) P(Australia will win atleast one match)

= 1 - P(Australia will loose 3 matches)

$=1-\dfrac{8}{27}$

$=\dfrac{19}{27}$

9) Find the Probability of drawing an ace or a spade from a well shuffled pack of 52 cards.

Sol: let S be the sample space in the experiment of playing cards. $n(S)=52$

let A be the event of getting Aces

$$n(A)=4$$

$$P(A)=\frac{n(A)}{n(S)}=\frac{4}{52}$$

let B be the event of getting spade cards

$$n(B)=13$$

$$P(B)=\frac{n(B)}{n(S)}=\frac{13}{52}$$

$$A\cap B=\{Ace\}$$

$$n(A\cap B)=1$$

$$P(A\cap B)=\frac{n(A\cap B)}{n(S)}=\frac{1}{52}$$

The Probability of getting an ace (or) spade cards

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$

$$=\frac{4}{52}+\frac{13}{52}-\frac{1}{52}$$

$$=\frac{16}{52}$$

$$=\frac{4}{13}$$

$$\boxed{P(A\cup B)=\frac{4}{13}}$$

$\therefore$ The Probability of getting Ace (or) spade cards $=\frac{4}{13}$

10) If A and B are independent events of a random experiment then show that $\bar{A}$ and $\bar{B}$ are also independent events.

Sol: Given A, B are Independent events

$$P(\bar{A}\cap\bar{B})=P(\overline{A\cup B})$$

$$=1-P(A\cup B)$$

$$\boxed{P(\bar{A})=1-P(A)}$$

$= 1 - [P(A) + P(B) - P(A \cap B)]$

$= 1 - P(A) - P(B) + P(A) \cdot P(B)$

$= [1 - P(A)] - P(B)[1 - P(A)]$

$= [1 - P(A)][1 - P(B)]$

$= P(\bar{A}) P(\bar{B})$

$\bar{A}$ and $\bar{B}$ are also independent events.

11) The probability for a contractor to get a road contract is 2/3 and to get a building contract is 5/9. The probability to get atleast one contract is 4/5. Find the probability that he gets both the contracts.

Sol: Let A be the event of road contractor and B be the event of building contractor.

$P(A) = 2/3$

$P(B) = 5/9$

$P(A \cup B) = 4/5$

$P(A \cap B) = ?$

By addition theorem,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$= \frac{2}{3} + \frac{5}{9} - \frac{4}{5}$

$= \frac{30 + 25 - 36}{45}$

$= \frac{19}{45}$

12) In a committee of 25 members, each member is proficient either in mathematics or statistics or in both. If 19 of these are proficient in mathematics, 16 in statistics find the probability that a person selected from the committee is proficient in both.

Sol: Let A be the event of professions in a mathematics and B be the event of profession in statistics

$P(A) = 19/25$

$P(B) = 16/25$

$P(A \cap B) = ?$

From addition theorem,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= \frac{19}{25} + \frac{16}{25} - 1$$

$$= \frac{19+16-25}{25}$$

$$= \frac{10}{25}$$

$$= \frac{2}{5}$$

$\therefore$ The probability that profession in both $= 2/5$

13) A bag contains 12 two rupee coins, 7 one rupee coins and 4 half a rupee coins. If three coins are selected at random, then find the probability that.

i) The sum of 3 coins is maximum

ii) The sum of 3 coins is minimum

iii) Each coin is of different value.

Sol: let 's' be the sample space in the experiment of selecting 3 coins from 23 coins.

$$n(S) = {}^{23}C_3$$

i) The sum of 3 coins is maximum.

let A be the event getting the sum of the 3 coins is maximum.

$$n(A) = {}^{12}C_3$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{{}^{12}C_3}{{}^{23}C_3}$$

$\therefore$ The probability that sum of 3 coins is maximum is $\frac{{}^{12}C_3}{{}^{23}C_3}$

ii) The sum of 3 coins is minimum

let B be the event of getting sum of 3 coins is minimum

$$n(B) = 4_{C_3}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{4_{C_3}}{23_{C_3}}$$

∴ The probability that sum of 3 coins is minimum is

$$\frac{4_{C_3}}{23_{C_3}}.$$

iii) Each coins is different value.

let 'c' be the event getting each coin is different value

$$n(C) = 12_{C_1} \times 7_{C_1} \times 4_{C_1}$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{12_{C_1} \times 7_{C_1} \times 4_{C_1}}{23_{C_3}}$$

∴ The probability that each coin is different value

$$= \frac{12_{C_1} \times 7_{C_1} \times 4_{C_1}}{23_{C_3}}.$$

14) The probability of 3 events A, B, C are such that $P(A)=0.3$, $P(B)=0.4$, $P(C)=0.8$, $P(A\cap B)=0.09$, $P(A\cap C)=0.28$, $P(A\cap B\cap C)=0.08$ and $P(A\cup B\cup C)\geq 0.75$.

Show that $P(B\cap C)$ lies in the interval $[0.21, 0.46]$

Sol: Given

$P(A)=0.3$ $P(B)=0.4$ $P(C)=0.8$

$P(A\cap B)=0.09$ $P(A\cap C)=0.28$ $P(A\cap B\cap C)=0.08$

$$P(A\cup B\cup C) = 0.75$$

We know that,

The probability lies between

$$0 \leq P(A) \leq 1$$

$$0.75 \leq P(A\cup B\cup C) \leq 1$$

$$0.75 \le P(A)+P(B)+P(C)-P(A\cap B)-P(B\cap C)-P(C\cap A)+P(A\cap B\cap C) \le 1$$

$$0.75 \le 0.3+0.4+0.8-0.09-P(B\cap C)-0.28+0.08 \le 1$$

$$0.75 \le 1.21-P(B\cap C) \le 1$$

$$0.75-1.21 \le -P(B\cap C) \le 1-1.21$$

$$-0.46 \le -P(B\cap C) \le -0.21$$

$$\boxed{0.21 \le P(B\cap C) \le 0.46}$$

# Complex Numbers

## Very short answer Question (2 Marks)

① Find the square roots of i) $7+24i$

Ans:- $\boxed{\sqrt{a+ib} = \pm\left[\sqrt{\frac{\sqrt{a^2+b^2}+a}{2}}+i\sqrt{\frac{\sqrt{a^2+b^2}-a}{2}}\right], \text{if } b>0}$

$$\sqrt{7+24i} = \pm\left[\sqrt{\frac{\sqrt{49+576}+7}{2}}+i\sqrt{\frac{\sqrt{49+576}-7}{2}}\right]$$

$$= \pm\left[\sqrt{\frac{25+7}{2}}+i\sqrt{\frac{25-7}{2}}\right] = \pm(4+3i)$$

2. Find the complex conjugate of $(3+4i)(2-3i)$

Sol:- $(3+4i)(2-3i) = 6-9i+8i-12i^2 = 18-i$

complex conjugate of $18-i$ is $18+i$

3. Find the polar form of following complex numbers. $1+i\sqrt{3}$

Sol:- Let $x+iy = 1+i\sqrt{3}$

$P(1,\sqrt{3})$ lies on $Q_1$

$$r = \sqrt{x^2+y^2} = \sqrt{1^2+(\sqrt{3})^2} = \sqrt{4} = 2$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \tan^{-1}\tan\left(\frac{\pi}{3}\right) = \frac{\pi}{3}$$

4. If $z=2-3i$, show that $z^2-4z+13=0$

Sol:- Given $z=2-3i$

Squaring on both sides

$$(z-2)^2 = (-3i)^2 \Longrightarrow z^2-4z+4=-9$$

$$z^2-4z+13=0$$

5. i) If $z_1=-1$ and $z_2=-i$ then find $Arg(z_1z_2)$

Sol:- $z_1=-1 \Rightarrow \cos\pi+i\sin\pi \Rightarrow Arg\ z_1=\pi$

$z_2=-i \Rightarrow \cos(-\pi/2)+i\sin(-\pi/2) \Rightarrow Arg\ z_2=(-\pi/2)$

$Arg(z_1z_2) = Arg\ z_1+Arg\ z_2 = \pi-\frac{\pi}{2} = \frac{\pi}{2}$ $\boxed{Arg(z_1z_2)=Arg\ z_1+Arg\ z_2}$

ii) If $z_1=-1, z_2=i$ then find $Arg\left(\frac{z_1}{z_2}\right)$

$z_1=-1 \Rightarrow \cos\pi+i\sin\pi \Rightarrow Arg\ z_1=\pi$

$Z_2 = i \Rightarrow \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right) \Longrightarrow \operatorname{Arg} Z_2 = \frac{\pi}{2}$

$\operatorname{Arg}\left(\frac{Z_1}{Z_2}\right) = \operatorname{Arg} Z_1 - \operatorname{Arg} Z_2 = \pi - \frac{\pi}{2} = \frac{\pi}{2}$ $\boxed{\because \operatorname{Arg}\left(\frac{Z_1}{Z_2}\right) = \operatorname{Arg} Z_1 - \operatorname{Arg} Z_2}$

6. If $z = (\cos\theta, \sin\theta)$, find $\left(z - \frac{1}{z}\right)$

$z - \frac{1}{z} = (\cos\theta + i\sin\theta) - (\cos\theta - i\sin\theta) = 2i\sin\theta$

$\because z = \cos\theta + i\sin\theta$

$\Rightarrow \frac{1}{z} = \cos\theta - i\sin\theta$

7. Find the additive inverse of $(-6,5)+(10,-4)$

Sol:- $-6+5i+10-4i = -4+i$

$\therefore$ Additive inverse of $4+i$ is $-(4+i)$

8. Find the Multiplicative inverse of $7+24i$

Sol:- Multiplicative inverse of $7+24i = \frac{1}{7+24i} \times \frac{7-24i}{7-24i}$

$= \frac{7-24i}{625}$

9. If $z \neq 0$, find $\operatorname{Arg} z + \operatorname{Arg} \bar{z}$

Sol:- $z = x+iy$, $\operatorname{Arg} z = \theta$ and $\operatorname{Arg}\bar{z} = -\theta$ $\boxed{\begin{array}{l}\operatorname{Arg} z = \theta \\ \operatorname{Arg}\bar{z} = -\theta\end{array}}$

$\therefore \operatorname{Arg} z + \operatorname{Arg}\bar{z} = \theta - \theta = 0$

10. If $(1-i)(2-i)(3-i)\ldots(1-ni) = x-iy$ then prove that $2\cdot5\cdot10\ldots$ $\ldots(1+n^2) = x^2+y^2$

Sol:- $(1-i)(2-i)(3-i)\ldots(1-ni) = x-iy$

Applying 'mod' on both sides.

$|(1-i)(2-i)(3-i)\ldots(1-ni)| = |x-iy| \Longrightarrow |(1-i)(2-i)\ldots||(1-ni)| = |x-iy|$

$\sqrt{1^2+1^2}\,\sqrt{2^2+1^2}\ldots\sqrt{n^2+1^2} = \sqrt{x^2+y^2} \Longrightarrow \sqrt{2}\,\sqrt{5}\ldots\sqrt{n^2+1} = \sqrt{x^2+y^2}$

$\Longrightarrow 2\cdot5\ldots(n^2+1) = x^2+y^2$ $\boxed{\because |x+iy| = \sqrt{x^2+y^2}}$

11. If the $\operatorname{Arg}\bar{Z}_1$ and $\operatorname{Arg} Z_2$ are $\frac{\pi}{5}$ and $\frac{\pi}{3}$ respectively then find $(\operatorname{Arg} Z_1 + \operatorname{Arg} Z_2)$

Sol:- Given $\operatorname{Arg}\bar{Z}_1 = \frac{\pi}{5} \Rightarrow \operatorname{Arg} Z_1 = \frac{-\pi}{5}$

$\operatorname{Arg} Z_2 = \frac{\pi}{3}$ then $\operatorname{Arg} Z_1 + \operatorname{Arg} Z_2 = \frac{-\pi}{5} + \frac{\pi}{3} = \frac{2\pi}{15}$

12. Find the least positive integer $n$, satisfying $\left(\frac{1+i}{1-i}\right)^n = 1$

Sol:- Given that $\left(\frac{1+i}{1-i}\right)^n = 1$

rationalise the denominator.

$$\left[\frac{(1+i)^2}{1+1}\right]^n = 1 \implies \left[\frac{(\cancel{1}+\cancel{i^2})+2i}{2}\right]^n = 1 \implies (i)^n = 1$$

When the least positive integer $n=4$, $(i)^n = 1$

13. If the $(\sqrt{3}+i)^{100} = 2^{99}(a+ib)$ then show that $a^2+b^2=4$

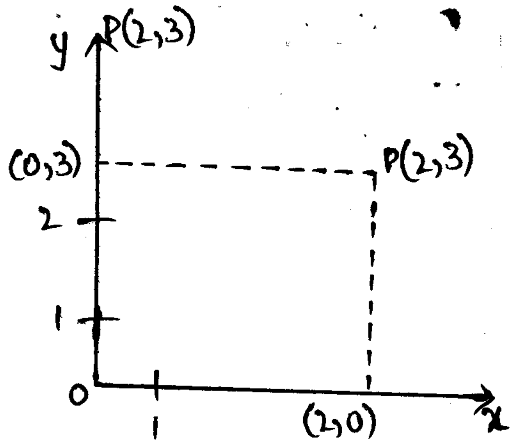
Sol:- Let $|\sqrt{3}+i|^{100} = 2^{99}|a+ib|$

$$\left[\sqrt{(\sqrt{3})^2+1^2}\right]^{100} = 2^{99}\sqrt{a^2+b^2} \qquad \because |x+iy| = \sqrt{x^2+y^2}$$

$$2^{100} = 2^{99}\sqrt{a^2+b^2} \implies a^2+b^2=4$$

14. Represent the complex number $2+3i$ in argand plane.

Sol:- The complex number $2+3i$ denotes the point



# Quadratic Equations & Expressions

## V.S.A.Q's

1. Find the quadratic equation whose roots are $7+2\sqrt{5}$ and $7-2\sqrt{5}$

Sol:- $\alpha = 7+2\sqrt{5}$ $\beta = 7-2\sqrt{5}$

$\alpha+\beta = 7+2\sqrt{5}+7-2\sqrt{5} = 14$ $\alpha\beta = (7+2\sqrt{5})(7-2\sqrt{5}) = 29$.

$\boxed{\text{Q.E is } x^2-(\alpha+\beta)x+\alpha\beta=0}$

$\therefore$ The required quadratic equation is $x^2-14x+29=0$

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2. Find the quadratic equation whose roots are $\frac{p-q}{p+q}, \frac{-(p+q)}{p-q}$ $(p \neq \pm q)$

Sol:- Let $\alpha = \frac{p-q}{p+q}$, $\beta = \frac{-(p+q)}{p-q}$

$$\alpha+\beta = \frac{p-q}{p+q} + \frac{-(p+q)}{p-q} = \frac{(p-q)^2-(p+q)^2}{(p+q)(p-q)} = \frac{-4pq}{p^2-q^2}$$

$\boxed{(a+b)(a-b) = a^2-b^2}$ $\boxed{(a-b)^2-(a+b)^2 = -4ab}$

$\boxed{Q.E = x^2-(\alpha+\beta)x+\alpha\beta}$

$\alpha\beta = \frac{p-q}{p+q}, \frac{-(p+q)}{p-q}$ $\alpha\beta = -1$

The required quadratic equation is $x^2+\frac{4pq}{p^2-q^2}x-1=0$

$(p^2-q^2)x^2+4pqx-(p^2-q^2)=0$

---

3. For what values of 'm' the equation $(m+1)x^2+2(m+3)x+m+8=0$ has equal roots?

Sol:- Given eqn $(m+1)x^2+2(m+3)x+m+8=0$.

$(m+1)x^2+(2m+6)x+m+8=0$. $\boxed{ax^2+bx+c=0}$

$\boxed{\text{Given roots are equal then } \Delta=0 \quad b^2-4ac=0}$

Here

$a=m+1$ $b=2m+6$ $c=m+8$

$b^2-4ac=0 \implies (2m+6)^2-4(m+1)(m+8)=0$.

$4m^2+36+24m-4m^2-36m-32=0$.

$-12m+4=0$

$12m=4$

$m=\frac{1}{3}$

4. If the equation $x^2-15-m(2x-8)=0$ has equal roots find the value of m.

Sol:- Given equation is $x^2-15-m(2x-8)=0$.

$x^2-15-2mx+8m=0$ $\boxed{ax^2+bx+c=0}$

$x^2-2mx+8m-15=0$

$\boxed{\text{Given roots are equal then } \Delta=0 \quad b^2-4ac=0}$

Here $a=1$ $b=-2m$ $c=8m-15$

$(-2m)^2-4(1)(8m-15)=0$.

$4m^2-32m+60=0$

$m^2-8m+15=0$.

$m^2-5m-3m+15=0$.

$m(m-5)-3(m-5)=0$

$(m-5)(m-3)=0$

$m-5=0$ $m-3=0$

$m=5$ $m=3$.

5. If the $\alpha,\beta$ are the roots of the equation $ax^2+bx+c=0$. Find the value of i) $\frac{1}{\alpha}+\frac{1}{\beta}$ ii) $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$

Sol:- i) Since $\alpha,\beta$ aree roots of $ax^2+bx+c=0$

$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\beta+\alpha}{\alpha\beta}=\frac{-b/\cancel{a}}{\frac{c}{\cancel{a}}}=-\frac{b}{c}$

$\boxed{\begin{array}{l}\text{Sum of roots } \alpha+\beta=-\frac{b}{a}\\ \text{Product of roots } \alpha\beta=\frac{c}{a}\end{array}}$

ii) Given equation is $ax^2+bx+c=0$

Since $\alpha,\beta$ are roots of given equation.

$\frac{1}{\alpha^2}+\frac{1}{\beta^2}=\frac{\beta^2+\alpha^2}{\alpha^2\beta^2}=\frac{(\alpha+\beta)^2-2\alpha\beta}{(\alpha\beta)^2}=\frac{\left(\frac{-b}{a}\right)^2-2\left(\frac{c}{a}\right)}{\left(\frac{c}{a}\right)^2}=\frac{\frac{b^2-2ac}{\cancel{a^2}}}{\frac{c^2}{\cancel{a^2}}}$

$=\frac{b^2-2ac}{c^2}$ $\boxed{\alpha+\beta=\frac{-b}{a}}$ $\boxed{\alpha\beta=\frac{c}{a}}$

6. i) Find the maximum (or) minimum of the experession $12x-x^2-32$.

Sol: Let $f(x)=12x-x^2-32 \Rightarrow -x^2+12x-32$

where $\boxed{a=1}$ $\boxed{b=12}$ $\boxed{c=-32}$

coefficient of $x^2$ is $a=-1<0$

Given expression has Maximum value.

$\boxed{\text{Max value} = \frac{4ac-b^2}{4a}}$

Max value $= \frac{4a-b^2}{4a} = \frac{4(-1)(-32)-(12)^2}{4(-1)}$

$= \frac{128-144}{-4} = \frac{-16}{-4} = 4$

ii) Find the maximum (or) minimum of the expression $2x-7-5x^2$

Sol: Let $f(x)=-5x^2+2x-7$

where $\boxed{a=-5}$ $\boxed{b=2}$ $\boxed{c=-7}$

Since coeff of $x^2$ is $a=-5<0$

Given experssion has maximum value.

Max value $= \frac{4ac-b^2}{4a} = \frac{4(-5)(-7)-4}{-20} = \frac{140-4}{-20} = \frac{-136}{20} = \frac{-34}{5}$

7. Prove that roots of $(x-a)(x-b)=h^2$ are always real.

Sol: Given equation is $(x-a)(x-b)=h^2$

$x^2-bx-ax+ab=h^2$

$x^2-x(a+b)+ab-h^2=0$.

$\boxed{a=1}$ $\boxed{b=-(a+b)}$ $\boxed{c=ab-h^2}$

Now Discriminent.

$\Delta = b^2-4ac$

$=(a+b)^2-4(1)(ab-h^2)$

$=(a+b)^2-4ab+4h^2$

$=(a-b)^2+4h^2 \geq 0$

$\therefore \Delta \geq 0$

Roots are always real.

# DEMOIVER'S THEORM (V.S.A.Q)

1.) If $x = cis\theta$, then find the value of $\left(x^6+\frac{1}{x^6}\right)$

Sol:- $cis\theta = x$

$$x^6 = (cis\theta)^6 = (\cos\theta + i\sin\theta)^6$$
$$= \cos 6\theta + i\sin 6\theta$$
$$x^6+\frac{1}{x^6} = \cos 6\theta + i\sin 6\theta + \frac{1}{\cos 6\theta + i\sin 6\theta}$$
$$= \cos 6\theta + \cancel{i\sin 6\theta} + \cos 6\theta - \cancel{i\sin 6\theta}$$
$$= 2\cos 6\theta$$

2.) If A, B, C are angles of $\Delta ABC$ such that $x = cisA$, $y = cisB$, $z = cisC$, find values of $xyz$?

Sol:- Given ABC are angles in $\Delta^{le}$

$$A+B+C = \pi$$

Given $x = cisA$, $y = cisB$, $z = cisC$

$$xyz = cisA \cdot cisB \cdot cisC$$
$$= cis(A+B+C)$$
$$= \cos(A+B+C) + i\sin(A+B+C)$$
$$= \cos\pi + i\sin\pi = -1 + i(0) = -1$$

3.) If the cube root of unity are $1, \omega, \omega^2$ then find the roots of equation $(x-1)^3+8=0$

Sol:- $(x-1)^3 = -8$

$$x-1 = (-8)^{1/3}$$
$$x-1 = (-8)^{1/3}(1)^{1/3}$$
$$x-1 = [(-2)^3]^{1/3}(1)^{1/3}$$

$$x-1 = [(-2)^3]^{1/3}\ (1, \omega, \omega^2)$$

$$x-1 = -2\ (1, \omega, \omega^2)$$

$$x-1 = -2, -2\omega, -2\omega^2$$

$$x = 1-2, 1-2\omega, 1-2\omega^2$$

4.) $(2-\omega)(2-\omega^2)(2-\omega^{10})(2-\omega)^{11} = 49$

Sol:- L.H.S:- $(2-\omega)(2-\omega^2)(2-\omega^{10})$ $\boxed{1+\omega+\omega^2=0}$

$= (2-\omega)(2-\omega^2)(2-(\omega^3)^3\omega)(2-(\omega^3)^3\omega^2)$ $\boxed{\omega^3=1}$

$= (2-\omega)(2-\omega^2)(2-\omega)(2-\omega^2)$

$= [(2-\omega)(2-\omega^2)]^2 = (4-2\omega^2-2\omega+\omega^3)$

$= (4-2(\omega^2+\omega)+1)^2 = (4-2(-1)+1)^2$

$= (4+2+1)^2 = 7^2 = 49$

5) If $1, \omega, \omega^2$ are the cube roots of unity then find the values of the following.

1, $(1-\omega+\omega^2)^6 + (1-\omega^2+\omega)^6$ $[1+\omega+\omega^2=0]$

Sol:- $(1-\omega+\omega^2)^6 + (1-\omega^2+\omega)^6$

$= (1-\omega+\omega^2)^6 + (-\omega-\omega^2)^6$

$= (-2\omega)^6 + (-2\omega)^6$

$= (-2)^6\omega^6 + (-2)^6\omega^{12} = (-2)^6\ [\omega^6 + (\omega^6)^2]$

$= 64\,[1+1] = 64 \times 2 = 128$

6.) If $\omega, \omega^2$ are the cube roots of unity, then prove that $x^2+4x+7=0$ where $x = \omega-\omega^2-2$

Sol:- $x^2+4x+7=0$ where $x = \omega-\omega^2-2$

Here $x = \omega-\omega^2-2 \Rightarrow x+2 = \omega-\omega^2$

S.O.B.S

$$(x+2)^2 = (\omega-\omega^2)^2$$

$$x^2+4x+4 = \omega^2+\omega^4-2\omega^3 \qquad [1+\omega+\omega^2=0]$$

$$x^2+4x+6 = -1$$

$$x^2+4x+7 = 0$$

7) i) $(1-\omega+\omega^2)^5+(1+\omega-\omega^2)^5$

Sol:- $(1-\omega+\omega^2)^5+(1+\omega+\omega^2)^5 = (1+\omega^2-\omega)^5+(1+\omega-\omega^2)^5$

$$= (-\omega-\omega)^5+(-\omega^2-\omega^2)^5 \Rightarrow (-2\omega)^5+(-2\omega^2)^5 \qquad \boxed{1+\omega+\omega^2=0}$$

$$= (-2)^5(\omega^5+\omega^{10}) = -32(\omega^5+\omega^{10}) \qquad \boxed{1+\omega^2=-\omega}$$

$$= -32(\omega^2+\omega) = (-32)(-1) = 32 \qquad \boxed{1+\omega=-\omega^2}$$

ii) $(1-\omega+\omega^2)^6+(1-\omega^2+\omega)^6$

Sol:- $(1-\omega+\omega^2)^6+(1-\omega^2+\omega^6) = (-\omega-\omega)^6+(-\omega^2-\omega^2)$

$$= (-2\omega)^6+(-2\omega^2)^6 = (-2)^6\omega^6+(-2)^6\omega^{12}$$

$$= (-2)^6[\omega^6+(\omega^6)^2] = 64[1+1]$$

$$= 64\times 2 = 128. \qquad \boxed{\omega^6=\omega^3\omega^3=1}$$

# THEORY OF EQUATIONS (V.S.A.Q's)

1.) From the equation whose roots are $2\pm\sqrt{3}$, $1\pm 2i$

Sol:- Given roots are $2\pm\sqrt{3}$, $1\pm 2i$

The roots are $[2+\sqrt{3}, 2-\sqrt{3}, 1+2i, 1-2i]$

$\boxed{\because \text{The } \alpha, \beta, \gamma, \delta \text{ are the roots then } (x-\alpha)(x-\beta)(x-\gamma)(x-\delta)=0}$

$\therefore$ Req equation is $[x-(2+\sqrt{3})][x-(2-\sqrt{3})][x-(1+2i)][x-(1-2i)]=0$

$$[(x-2)-\sqrt{3}][(x-2)+\sqrt{3}][(x-1)-2i][(x-1)+2i]=0$$

$$\boxed{(a+b)(a-b)=a^2-b^2}$$

$$[(x-2)^2-(\sqrt{3})^2][(x-1)^2-(2i)^2]=0$$

$$[(x-2)^2-3][(x-1)^2-4i^2]=0 \qquad \boxed{i^2=-1}$$

$$(x^2-4x+4-3)(x^2-2x+1+4)=0$$

$$(x^2-4x+1)(x^2-2x+5)=0$$

$$x^4-2x^3+5x^2-4x^3+8x^2-20x+x^2-2x+5=0$$

$$\therefore x^4-6x^3+14x^2-22x+5=0$$

---

2.) If $-1, 2$ and $\alpha$ are the roots of $2x^3+x^2-7x-6=0$ then find $\alpha$

Sol:- Given $-1, 2$ and $\alpha$ are the roots of $2x^3+x^2-7x-6=0$ ($S_1$ $S_2$ $S_3$)

Now $S_1 = \frac{-1}{2}$ $\qquad \boxed{\because \text{sum of roots} = \frac{-a_1}{a_0}}$

$$-1+2+\alpha = \frac{-1}{2}$$

$$1+\alpha = \frac{-1}{2}$$

$$\alpha = \frac{-1}{2}-1 \Rightarrow \boxed{\alpha = \frac{-3}{2}}$$

3.) If $1, -2, 3$ are the roots of $x^3 - 2x^2 + ax + 6 = 0$ then find a.

$\quad\quad\quad\quad S_1 \quad S_2 \quad S_3$

Sol:- Now $S_2 = \frac{a}{1}$

$$\boxed{\begin{array}{c} S_2 = \text{Sum of product of roots} \\ S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{a_2}{a_0} \end{array}}$$

$\alpha = 1, \beta = -2, \gamma = 3$

$S_2 = a$

$\alpha\beta + \beta\gamma + \gamma\alpha = a$

$1(-2) + (-2)(3) + 3(1) = a$

$-2 - 6 + 3 = a$

$\boxed{a = -5}$

4.) If the product of the roots of $4x^3 + 16x^2 - 9x - a = 0$ is 9. then find a.

Sol:- let $\alpha, \beta, \gamma$ are the roots of given equation

$\therefore S_3 = \alpha\beta\gamma = 9$

$$\boxed{\because \text{sum of product of roots} = \frac{-a_3}{a_0}}$$

$= \frac{-(-a)}{4} = 9$

$\Rightarrow \frac{a}{4} = 9$

$\boxed{a = 36}$

5.) If $1, 1, \alpha$ are the roots of $x^3 - 6x^2 + 9x - 4 = 0$ then find $\alpha$.

Ans:- Given $1, 1, \alpha$ are the roots of $x^3 - 6x^2 + 9x - 4 = 0$

$\quad\quad\quad\quad S_1 \quad S_2 \quad S_3$

Now $S_1 = 6$

$1 + 1 + \alpha = \frac{-(-6)}{1}$

$2 + \alpha = 6$

$\alpha = 6 - 2$

$\boxed{\alpha = 4}$

6.) If $\alpha, \beta, 1$ are the roots of $x^3 - 2x^2 - 5x + 6 = 0$ then find $\alpha, \beta$

Sol:- Given $\alpha, \beta, 1$ are the roots of $x^3-2x^2-5x+6=0$

Now $S_1 = 2$ $\quad$ $\because$ Sum of roots $= \frac{-a}{a_0}$

$\alpha+\beta+1 = 2$

$\boxed{\alpha+\beta=1}$ —— ①

and $S_3 = -6$ $\quad$ $\because S_3 =$ product of roots $= \frac{-a_3}{a_0}$

$\alpha(\beta)(1) = -6$

$\boxed{\alpha\beta=-6}$

$(\alpha-\beta)^2 = (1)^2 - 4(-6)$ $\quad$ $\because (\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$

$= 1+24$

$(\alpha - \beta)^2 = 25$

$\boxed{\alpha-\beta=5}$ —— ②

solve ① & ②

$\alpha + \cancel{\beta} = 1$

$\alpha - \cancel{\beta} = 5$

$2\alpha = 6 \Rightarrow 2\alpha = 6 \Rightarrow \alpha = \frac{6}{2} \Rightarrow \boxed{\alpha=3}$

put $\alpha=3$ in ①

$3+\beta = 1$

$\beta = 1-3$

$\boxed{\beta=-2}$

---

9.) Find the transformed eq^n whose roots are the negitives of the roots of $x^4+5x^3+11x+3=0$

Sol:- Given eq^n is $f(x) = x^4+5x^3+11x+3=0$

Req eq^n is $f(-x)=0$

$(-x)^4+5(-x)^3+11(-x)+3=0$

$\boxed{x^4-5x^3-11x+3=0}$

$\because$ negitive roots $-\alpha_1, -\alpha_2, -\alpha_3 \ldots \alpha_n$ are the roots of $f(-x)=0$

10) Find the equation whose roots are reciprocals of roots of $x^4-3x^3-7x^2+5x-2=0$

Sol:- Given equation is $f(x)=x^4-3x^3+7x^2+5x-2=0$

Required equation is $f\left(\frac{1}{x}\right)=0$

Reciprocal roots

$\therefore \frac{1}{\alpha_1}, \frac{1}{\alpha_2}, \frac{1}{\alpha_3} \cdots \frac{1}{\alpha_n}$ are the roots of $f\left(\frac{1}{x}\right)=0$

$$\frac{1}{x^4}-3\left[\frac{1}{x^3}\right]+7\left[\frac{1}{x^2}\right]+5\left[\frac{1}{x}\right]-2=0$$

multiply by $x^4$ on both sides

$$1-3x+7x^2+5x^3-2x^4=0$$

$$-2x^4+5x^3+7x^2-3x+1=0$$

$$\boxed{2x^4-5x^3-7x^2+3x-1=0}$$

11) i) If $\alpha, \beta, \gamma$ are the roots of the equation $x^3-2x^2-4x-3=0$, find the equation whose roots are 3times the root of given equation.

Sol:- Given equation is $f(x)=x^3+2x^2-4x-3=0$

Req. equation is $f\left(\frac{x}{3}\right)=0$

$\therefore 3\alpha_1, 3\alpha_2 \cdots\cdots 3\alpha_n$ are the roots of $f\left(\frac{x}{3}\right)=0$

$$\left(\frac{x}{3}\right)^3+2\left(\frac{x}{3}\right)^2-4\left(\frac{x}{3}\right)-3=0$$

$$\frac{x^3}{27}+2\frac{x^2}{9}-4\frac{x}{3}-3=0$$

$$\frac{x^3+6x^2-36x-81}{27}=0$$

$$x^3+6x^2-36x-81=0$$

ii, Find the algebric equation whose roots are two times of the roots of $x^5-2x^4+3x^3-2x^2+4x+3=0$

Sol:- Given equation is $f(x)=x^5-2x^4+3x^3-2x^2+4x+3=0$

Required equation $f\left(\frac{x}{2}\right)=0$

$\therefore 2\alpha_1, 2\alpha_2 ------ 2\alpha_n$ are the roots of $f\left(\frac{x}{2}\right)=0$

$$\left(\frac{x}{2}\right)^5-2\left(\frac{x}{2}\right)^4+3\left(\frac{x}{2}\right)^3-2\left(\frac{x}{2}\right)^2+4\left(\frac{x}{2}\right)+3=0$$

$$\frac{x^5}{32}-2\frac{x^4}{16}+3\frac{x^3}{8}-2\frac{x^2}{4}+4\frac{x}{2}+3=0$$

$$\frac{x^5-4x^4+12x^3-16x^2+64x+96}{32}=0$$

$$\boxed{x^5-4x^4+12x^3-16x^2+64x+96=0}$$

# PERMUTATIONS & COMBINATIONS (VSAQ's)

① If ${}^{n}P_4 = 1680$ then find 'n'.

Sol:- ${}^{n}P_4 = 1680$

$= 168 \times 10$

$= 21 \times 8 \times 10$

$= 7 \times 3 \times 8 \times 2 \times 5$

${}^{n}P_4 = {}^{8}P_4$

$\boxed{n=8}$

ii) If ${}^{n}P_3 = 1320$ then find 'n'?

${}^{n}P_3 = 132 \times 10$

$= 12 \times 11 \times 10$

${}^{n}P_3 = {}^{12}P_3$

$\boxed{n=12}$

② If ${}^{n+1}P_5 : {}^{n}P_6 = 2:7$ then find n.

${}^{n+1}P_5 : {}^{n}P_6 = 2:7.$

$$\frac{{}^{n+1}P_5}{{}^{n}P_6} = \frac{2}{7}.$$

$$\frac{\frac{(n+1)!}{(n+1-5)!}}{\frac{n!}{(n-6)!}} = \frac{2}{7}.$$

$$\frac{\frac{(n+1)n!}{(n-4)!}}{\frac{n!}{(n-6)!}} = \frac{2}{7}.$$

$$\frac{\frac{(n+1)\cancel{n!}}{(n-4)(n-5)\cancel{(n-6)!}}}{\frac{\cancel{n!}}{\cancel{(n-6)!}}} = \frac{2}{7}$$

$$7(n+1) = 2(n-4)(n-5)$$
$$7n+7 = 2(n^2-5n-4n+20)$$
$$7n+7 = 2n^2-10n-8n+40$$
$$2n^2-22n-3n+33=0$$
$$2n^2-22n-3n+33=0$$
$$2n(n-11)-3(n-11)=0$$
$$(n-1)(2n-3)=0$$
$$n=1,\ n=\frac{3}{2}$$

$\therefore$ n is a +ve integer $\boxed{\therefore n=11}$

③ If ${}^{n+1}P_5 : {}^{n}P_5 = 3:2$ then find n.

Sol:- ${}^{n+1}P_5 : {}^{n}P_5 = 3:2$

$$\frac{{}^{n+1}P_5}{{}^{n}P_5} = \frac{3}{2}$$

$$\frac{(n+1)\cancel{n}\cancel{(n-1)}\cancel{(n-2)}\cancel{(n-3)}}{\cancel{n}\cancel{(n-1)}\cancel{(n-2)}\cancel{(n-3)}(n-4)} = \frac{3}{2}$$

$$\frac{n+1}{n-4} = \frac{3}{2}$$

$$2n+2 = 3n-12$$
$$2+12 = 3n-2n$$
$$\boxed{n=14}$$

④ If ${}^{n}C_{21} = {}^{n}C_{27}$ then find ${}^{50}C_n$.

Sol:- ${}^{n}C_{21} = {}^{n}C_{27}$

$\boxed{{}^{n}C_r = {}^{n}C_s \text{ then } n=r+s}$

$$n = 21+27$$
$$n=48.$$

$\boxed{{}^{n}C_r = {}^{n}C_{n-r}}$

Now, ${}^{50}C_n = {}^{50}C_{48} = {}^{50}C_2 = \frac{\overset{25}{\cancel{50}}\times 49}{1\times\cancel{2}}$

$$=1225.$$

5. If ${}^{n}C_5 = {}^{n}C_6$ then find the value of ${}^{13}C_n$

Sol:- ${}^{n}C_5 = {}^{n}C_6$

$\boxed{{}^{n}C_r = {}^{n}C_s \text{ then } n = r+s \text{ (or) } r = s}$

$n = 5+6 = 11$

${}^{13}C_n = {}^{13}C_{11}$ $\boxed{{}^{n}C_r = {}^{n}C_{n-r}}$

$= {}^{13}C_2$

$= \frac{13 \times \cancel{12}^{6}}{1 \times \cancel{2}}$

$= 78$

6. If ${}^{12}P_r = 1320$, find 'r'

Sol:- ${}^{12}P_r = 1320$

$= 132 \times 10$

$= 12 \times 11 \times 10$

${}^{12}P_r = {}^{12}P_3$

$\boxed{\therefore r = 3}$

7. If ${}^{12}C_{r+1} = {}^{12}C_{3r-5}$ find 'r'

Sol:- $\boxed{{}^{n}C_r = {}^{n}C_s \text{ then } n = r+s \text{ (or) } r = s}$

$12 = r+1+3r-5$ | $r+1 = 3r-5$

$12 = 4r-4$ | $1+5 = 3r-r$

$4r = 16$ | $6 = 2r$

$\boxed{r = 4}$ | $\boxed{r = 3}$

$2(n+1) = 3(n-4)$

$2n+2 = 3n-12$

$\boxed{n = 14}$

8. If ${}^{n}P_7 = 42\,{}^{n}P_5$ then find 'n'

${}^{n}P_7 = 42 \cdot {}^{n}P_5$

$\cancel{n}\cancel{(n-1)}\cancel{(n-2)}\cancel{(n-3)}\cancel{(n-4)}(n-5)(n-6) = 42\,\cancel{n}\cancel{(n-1)}\cancel{(n-2)}\cancel{(n-3)}\cancel{(n-4)}$

$(n-5)(n-6)=42$

$(n-5)(n-6)=7\times 6$

| | |
|---|---|
| $n-5=7$ | $n-6=6$ |
| $n=7+5$ | $n=6+6$ |
| $\boxed{n=12}$ | $\boxed{n=12}$ |

(9) If ${}^{12}P_5+5\cdot{}^{12}P_4={}^{13}P_r$ find 'r'.

Sol:- ${}^{12}P_5+5\cdot{}^{12}P_4={}^{13}P_r$

$$\boxed{{}^{n}P_r+r\cdot{}^{n}P_{r-1}={}^{n+1}P_r}$$

Here $r=5$.

(10) If $10\cdot{}^{n}C_2=3\,{}^{n+1}C_3$ then find 'n'

Sol:- $10\cdot\frac{\cancel{n}(n-1)}{1\times\cancel{2}}=\cancel{3}\cdot\frac{(n+1)\cancel{n}\cancel{(n-1)}}{1\times\cancel{2}\times\cancel{3}}$

$10=n+1$

$n=10-1$

$\boxed{n=9}$

| | |
|---|---|
| $16=4r$ | $5+1=3r-r$ |
| $\boxed{r=4}$ | $6=2r$ |
| | $\boxed{r=3}$ |

$\therefore r=3$ or $4$.

(11) If ${}^{12}C_{s+1}={}^{12}C_{2s-5}$ find s.

Sol:- ${}^{12}C_{s+1}={}^{12}C_{2s-5}$

$$\boxed{{}^{n}C_r={}^{n}C_s \text{ then } n=r+s \text{ (or) } r=s.}$$

$s+1=2s-5$

$2s-s=5+1$

$\boxed{s=6}$

(12) If ${}^{15}C_{2r-1}={}^{15}C_{2r+4}$ then find r.

Sol:- ${}^{15}C_{2r-1}={}^{15}C_{2r+4}$

$$\boxed{{}^{n}C_r={}^{n}C_s \Rightarrow n=r+s \text{ (or) } r=s}$$

$n = r + s$

$15 = 2r - 1 + 2r + 4$

$15 = 4r + 3$

$4r = 12$

$\boxed{r = 3}$

(13) Find the value of $^{10}C_5 + 2 \cdot {}^{10}C_4 + {}^{10}C_3$.

Sol:- $^{10}C_5 + 2 \cdot {}^{10}C_4 + {}^{10}C_3$

$^{10}C_5 + {}^{10}C_4 + {}^{10}C_4 + {}^{10}C_3$.

$^{10+1}C_5 + {}^{10+1}C_4$ $\boxed{^{n+1}C_r = {}^nC_r + {}^nC_{r-1}}$

$^{11}C_5 + {}^{11}C_4$.

$^{11+1}C_5 = {}^{12}C_5$

$= \frac{12 \times 11 \times \cancel{10}^{2} \times \cancel{9}^{3} \times \cancel{8}^{2}}{1 \times \cancel{2} \times \cancel{3} \times \cancel{4} \times \cancel{5}}$

$= 792$.

(14) If $^nP_r = 5040$ and $^nC_r = 210$ find 'n' add 'r'.

Sol:- Given $^nP_r = 5040$ and $^nC_r = 210$

$\frac{\cancel{5040}^{24}}{\cancel{210}} = r!$

$r! = 24$ $\boxed{\frac{^nP_r}{^nC_r} = r!}$

$r! = 4!$

$\boxed{r = 4}$

put $r = 4$ in $^nP_r = 5040$

$^nP_4 = 5040$

$n(n-1)(n-2)(n-3) = 504 \times 10$

$= 21 \times 24 \times 10$

$= 7 \times 3 \times 8 \times 3 \times 10$

$= 10 \times 9 \times 8 \times 7$.

$^nP_4 = {}^{10}P_4$

$\boxed{n=10}$

$\therefore$ $\boxed{n=10}$ and $\boxed{r=4}$

(15) find the number of diagnals of a polygon with 12 sides?

Sol:- Here $n=12$

then sides of a polygon is no.of diagonals is $\frac{n(n-3)}{2} = \frac{\overset{6}{\cancel{12}}(12-3)}{\cancel{2}}$

$= 6(9)$

$= 54$.

(16) Find the number of ways of arranging the letters of the word.
1) PERMUTATION 2) INTERMEDIATE 3) INDEPENDENCE 4) MATHMATES.

Sol:- (1) Given word PERMUTATION

Here $\boxed{n=11}$

The given word permutation contain 11 letters in which there are 2I's are alike and remaining different

$\therefore$ The required no. of arrangements is $\frac{11!}{2!}$.

2) Given word INTERMEDIATE

Here $\boxed{n=12}$

The given word intermediate contain 12 letters in which there are 2 I's, 2T's, 3E's are alike and remaining different

$\therefore$ The required no. of arrangements $= \frac{12!}{2!\,2!\,3!}$

3) Given word INDEPENDENCE

Here $\boxed{n=12}$

The given word independence contain 12 letters in which there are 3n's, 2D's, 4E's are alike and remaining are different.

$\therefore$ The required no. of arrangements $= \frac{12!}{3!\,2!\,4!}$

iv) Given word MATHEMATICS

Here $\boxed{n=11}$

The given word mathematics contain 11 letters in which there are 2M's, 2A's, 2T's are alike and remaining are different

$\therefore$ The required no. of arrangements $= \frac{11!}{2!2!2!}$

17) Find the number of 4 letter word that can be formed using the letters of word PISTON in which at least one letter repeated.

Sol Given word PISTON

Here $\boxed{n=6}$ $\boxed{r=4}$ [r=4 letter word] [given]

i) when repetation is allowed is

$$\boxed{n^r = 6^4}$$

ii) when repetation is not allowed is

$$\boxed{{}^nP_r = {}^6P_4}$$

$\therefore$ The no. of 4 letter words in which atleast one letter is repeated is

$$= n^r - {}^nP_r$$
$$= 6^4 - {}^6P_4$$
$$= 1296 - 360$$
$$= 936$$

18) Find the number of positive divisors of 1080.

Sol:- $1080 = 2^3 \times 3^3 \times 5^1$

| | |
|---|---|
| 2 | 1080 |
| 2 | 540 |
| 2 | 270 |
| 3 | 135 |
| 3 | 45 |
| 3 | 15 |
| | 5 |

The no. of positive divisors of n is

$$\boxed{N = p_1^{\alpha_1}, p_2^{\alpha_2} \ldots p_k^{\alpha_k} \text{ is } (\alpha_1+1)(\alpha_2+1)\ldots(\alpha_k+1)}$$

$= (3+1)(3+1)(1+1)$

$= (4)(4)(2) = 32$.

# BINOMIAL THEOREM (VSAQ's)

1. Find the numbers of terms in the expansion of (i) $(2x+3y+z)^n$

Sol:- The no. of terms in the expansion of

$$(2x+3y+z)^7 = \frac{(n+1)(n+2)}{2}$$

$$= \frac{(7+1)(7+2)}{2} = \frac{\cancel{8}^{4}(9)}{\cancel{2}}$$

$$= 4 \times 9 = 36$$

$\boxed{\text{The no. of terms in } (x+y+z)^n \text{ is } \frac{(n+1)(n+2)}{2}.}$

2. Find the no. of terms in the expansion of $\left[\frac{3a}{4}+\frac{b}{2}\right]^9$.

Sol:- No. of terms of expansion of $\left[\frac{3a}{4}+\frac{b}{2}\right]^9$.

$$= n+1 = 9+1 = 10.$$

3. If $^{22}C_r$ is the largest binomial coefficient in the expansion of $(1+x)^{22}$. Find the value of $^{13}C_r$.

Sol:- Given expansion is $(1+x)^{22}$ have $n=22$ is an even integer.

If 'n' is even then largest binomial coefficient is $^nC_{n/2} = {}^{12}C_{11}$

Given that $^{22}C_r$ is the largest binomial coefficient in $(1+x)^{22}$

$$^{22}C_{11} = {}^{22}C_r \Rightarrow r=11.$$

$$\therefore {}^{13}C_r = {}^{13}C_{11} = {}^{13}C_2 = \frac{13 \times 12}{1 \times 2} = 78.$$

4. Find the set of values of 'x' for which $(2+3x)^{-2/3}$ is valid.

Sol:- Given $(2+3x)^{-2/3} = 2^{-2/3}\left[1+\frac{3x}{2}\right]^{-2/3}$

here $X = \frac{3x}{2}$

$\boxed{\text{Binomial expansion is valid for } |x|<1}$

$$\therefore \left|\frac{3x}{2}\right|<1 \Rightarrow |x|<\frac{2}{3} \Rightarrow \frac{-2}{3}<x<\frac{2}{3}.$$

$\therefore$ for $x \in \left[\frac{-2}{3},\frac{2}{3}\right]$ the given expansion is valid.

5. Find the set of values of 'x' for which $(7+3x)^{-5}$ is valid.

Sol:- Given $(7+3x)^{-5}$

$$= 7^{-5}\left[1+\frac{3x}{7}\right]^{-5}$$

here $X = \frac{3x}{7}$

Binomial expansion is valid for $|x|<1$.

$\Rightarrow \left|\frac{3x}{7}\right|<1 \Rightarrow |x|<\frac{7}{3} \Rightarrow \frac{-7}{3}<x<\frac{7}{3} \Rightarrow$ for $x\in\left[\frac{-7}{3},\frac{7}{3}\right]$

The given expansion is valid.

6. Find the set of values of 'x' for which $(3-4x)^{3/4}$ is valid.

Sol:- Given $(3-4x)^{3/4} = 3^{3/4}\left[1-\frac{4x}{3}\right]^{3/4}$ Here $X=\frac{4x}{3}$

Binomial expansion is valid for $|x|<1$.

$\left|\frac{4x}{3}\right|<1 \Rightarrow |x|<\frac{3}{4} \Rightarrow \frac{-3}{4}<x<\frac{3}{4}$.

$\therefore$ for $x\in\left[\frac{-3}{4},\frac{3}{4}\right]$ the given expansion is valid.

7. Find the coefficient of $x^{-7}$ in $\left[\frac{2x^2}{3}-\frac{5}{4x^5}\right]^7$.

Given $\frac{2x^2}{3}-\frac{5}{4x^5}$

The coefficient of $x^m$ in $\left[ax^p+\frac{b}{x^q}\right]$ is $T_{r+1}$ ; where $r=\frac{np-m}{p+q}$

here $n=7, p=2, q=5, m=-7$.

$\therefore r = \frac{7(2)-(-7)}{2+5} = \frac{14+7}{7} = \frac{21}{7} = 3$.

$\therefore T_{r+1} = T_{3+1} = {}^nC_r\, x^{n-r} a^r = {}^7C_3\left[\frac{2x^2}{3}\right]^{7-3}\left[\frac{-5}{4x^5}\right]^3$

$= {}^7C_3\left[\frac{2x^2}{3}\right]^4\left[\frac{-5}{4x^5}\right]^3$.

$= {}^7C_3\left[\frac{2}{3}\right]^4 x^8\left[\frac{-5}{4}\right]^3\cdot\frac{1}{x^{15}}$.

$= {}^7C_3\left[\frac{2}{3}\right]^4\left[\frac{-5}{4}\right]x^{-7}$.

$\therefore$ coefficient of $x^{-7} = {}^7C_3\left[\frac{2}{3}\right]^4\left[\frac{-5}{4}\right]^3$

8. Find the middle terms in the expansion of $\left[\frac{3x}{7}-2y\right]^{10}$.

Sol:- Given $\left[\frac{3x}{7}-2y\right]^{10}$ ; $n=10$.

If n is even, middle term is $T_{\frac{n+2}{2}} = T_{\frac{10+2}{2}} = T_{\frac{12}{2}} = T_6$

$T_{r+1} = {}^nC_r\, x^{n-r} a^r \Rightarrow T_6 \Rightarrow T_{5+1} = {}^{10}C_5\left[\frac{3x}{7}\right]^{10-5}(-2y)^5$.

$= {}^{10}C_5\left[\frac{3x}{7}\right]^5(-2)^5 y^5$.

$= {}^{10}C_5\left[\frac{3}{7}\right]^5(-2)^5 x^5 y^5$.

9. Find the middle term in the expansion of $\left[4a+\frac{3}{2}b\right]^{11}$.

Sol:- Given $\left[4a+\frac{3}{2}b\right]^{11}$ ; $n=11$.

If n is odd, then middle terms are $T_{\frac{n+1}{2}}, T_{\frac{n+3}{2}}$

$$= T_{\frac{11+1}{2}}, T_{\frac{11+3}{2}}$$

$$= T_{\frac{12}{2}}, T_{\frac{14}{2}}$$

$$= T_6, T_7.$$

$$\therefore T_6 = T_{5+1} = {}^{11}C_5 (4a)^{11-5}\left[\frac{3}{2}b\right]^5$$

$$= {}^{11}C_5 (4a)^6 \left[\frac{3}{2}b\right]^5$$

$$= {}^{11}C_5\, 4^6 \left[\frac{3}{2}\right] a^6 b^5.$$

$$\therefore T_7 = T_{6+1} = {}^{11}C_6 (4a)^{11-6}\left[\frac{3}{2}b\right]^6$$

$$= {}^{11}C_6 (4a)^5 \left[\frac{3}{2}b\right]^6$$

$$= {}^{11}C_6 (4)^5 \left[\frac{3}{2}\right]^6 a^5 b^6.$$

10. Find the term independent of 'x' in the expansion of $\left[4x^3+\frac{7}{x^2}\right]^{14}$.

Sol:- General form in the expansion of $(x+a)^n$ is $T_{r+1} = {}^nC_r\, x^{n-r} a^r$.

$$= {}^{14}C_r \left[(4x)^3\right]^{14-r}\left[\frac{7}{x^2}\right]^r$$

$$= {}^{14}C_r\, 4^{42-3r}\, 7^r\, x^{42-3r}\, x^{-2r}.$$

$$= {}^{14}C_r\, 4^{32-3r}\, 7^r\, x^{42-5r}.$$

For the term independent of 'x' we have $42-5r=0$.

$r=\frac{42}{5}$ which is not an integer.

$\therefore$ Hence term independent of 'x' in the given expansion is zero.

11. Find the term independent of 'x' in the expansion of $\left[\frac{2x^2}{5}+\frac{15}{4x}\right]^9$.

Given $\left[\frac{2x^2}{5}+\frac{15}{4x}\right]^9$. For independent of x in $\left[ax^p+\frac{b}{x^q}\right]^n$ is

$T_{r+1}$; where $r=\frac{np}{p+q}$

here $n=9, p=2, q=1 \Rightarrow r = \frac{9(2)}{2+1} = 6$.

$T_{r+1} = T_{6+1} = T_7$ is the term independent of $x$.

$$T_7 = T_{6+1} = {}^9C_6 \left[\frac{2}{5}\right]^3 \left[\frac{15}{4}\right]^6$$

$$\boxed{T_{r+1} = {}^nC_r\, x^{n-r} a^r}$$

12. prove that $C_0+2C_1+4\cdot C_2+8\cdot C_3+----+2^n C_n=3^n$.

we have $(1+x)^n=C_0+C_1x+C_2x^2+----+C_nx^n$.

Put $x=2$.

$(1+2)^n=C_0+C_1\cdot 2+C_2\, 2^2+-----+C_n\cdot 2^n$.

$\therefore C_0+2\cdot C_1+4\cdot C_2+-----+2\cdot{}^nC_n=3^n$.

13. If the coefficients of $(2r+4)^{th}$ term and $(3r+4)^{th}$ term in the expansion of $(1+x)^{21}$ are equal, then find 'r'

Sol:- Given expansion is $(1+x)^{21}$.

coefficient of $T_{r+1}$ is ${}^{21}C_r$

Now the coefficient of $(2r+4)^{th}$ term $={}^{21}C_{2r+3}$.

Now the coefficient of $(3r+4)^{th}$ term $={}^{21}C_{3r+3}$

Given coefficients are equal.

If ${}^nC_r={}^nC_s \Rightarrow r=s$ (or) $r+s=n$.

${}^{21}C_{2r+3}={}^{21}C_{3r+3}$

$2r+3=3r+3$ (or) $(2r+3)+(3r+3)=21$

$r=0$ (or) $5r=15$

$r=3$

$\therefore r=0,3$.

# MEASURES OF DISPERSION (Statistics)

① (i) Find the mean deviation from the mean of the following discrete data 6, 7, 10, 12, 13, 4, 12, 16

Sol:- Given 6, 7, 10, 12, 13, 4, 12, 16

Here $n = 8$

$$\bar{x} = \frac{\text{Sum of observations}}{\text{No. of observations}} = \frac{80}{8} = 10$$

| $x_i$ | $\lvert x_i - x \rvert = \lvert x_i > 10 \rvert$ |
|---|---|
| 6 | 4 |
| 7 | 3 |
| 10 | 0 |
| 12 | 2 |
| 4 | 6 |
| 13 | 3 |
| 12 | 2 |
| 16 | 6 |
| | $\Sigma \lvert x_i - x \rvert = 26$ |

Mean deviation of mean $= \frac{1}{N} \Sigma |x_i - \bar{x}| = \frac{26}{8} = 3.25$

(ii) Find the mean deviation from mean of the discrete data 3, 6, 10, 4, 9, 10.

Sol: Given data 3, 6, 10, 4, 9, 10

Here $n = 6$

$$\bar{x} = \frac{\text{Sum of observation}}{\text{No. of observation}}$$

$$= \frac{3+6+10+4+9+10}{6}$$

$$= \frac{42}{6} = 7$$

| $x_i$ | 3 | 6 | 10 | 4 | 9 | 10 | |
|---|---|---|---|---|---|---|---|
| $\lvert x_i - x \rvert = \lvert x_i - 7 \rvert$ | 4 | 1 | 3 | 3 | 2 | 3 | $\Sigma \lvert x_i - x \rvert = 6$ |

The mean deviation about the mean $= \frac{1}{n} \Sigma |x_i - x|$

$$= \frac{16}{6} = \frac{8}{3}$$

$$= 2.6$$

(2) Find the mean derivation from the median for the following data 13, 17, 18, 11, 13, 10, 16, 11, 18, 12, 17

Sol: Given data is 13, 17, 16, 11, 13, 10, 16, 11, 18, 12, 17

Ascending order: 10, 11, 11, 12, 13, 13, 16, 16, 17, 17, 18

Here $n=11$ (odd)

$\left(\frac{n+1}{2}\right)^{th} = \left(\frac{11+1}{2}\right)^{th} = 6^{th}$ observation

Median (M) = 13

| $x_i$ | 10 | 11 | 11 | 12 | 13 | 13 | 16 | 16 | 17 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\lvert x_i - \text{median}\rvert$ $\lvert x_i - 13\rvert$ | 3 | 2 | 2 | 1 | 0 | 0 | 3 | 3 | 4 | 4 | 5 |

$\Sigma \lvert x_i - \text{Median}\rvert = 27$

Mean deviation about the median $= \frac{1}{n}\Sigma\lvert x_i - \text{Median}\rvert$

$= \frac{27}{11}$

$= 2.45$

(3) Find the variance and standard deviation of the following data 5, 12, 13, 18, 6, 8, 2, 10

Sol: Mean $\bar{x} = \frac{\Sigma x_i}{n} = \frac{5+12+13+18+6+8+2+10}{8} = \frac{64}{8} = 8$

Variance

| $x_i$ | 5 | 12 | 3 | 18 | 6 | 8 | 2 | 10 |
|---|---|---|---|---|---|---|---|---|
| $x_i - \bar{x}$ | -3 | 4 | -5 | 10 | -2 | 0 | -6 | 2 |
| $(x_i-\bar{x})^n$ | 9 | 16 | 25 | 100 | 4 | 0 | 36 | 4 |

$\Sigma(x_i - \bar{x})^n = 194$

Variance $(\sigma^2) = \frac{1}{n}\sum_{n=1}^{8}(x_i-\bar{x})^2 = \frac{1}{8}\times 194 = 24.25$

Standard deviation $(\sigma) = \sqrt{24.25} = 4.95$ (approx)

(4) Find the mean deviation about the median for following data 4, 6, 9, 3, 10, 13, 2

Sol: Given data 4, 6, 9, 3, 10, 13, 2

Here $n=7$ (odd)

Ascending order = 2, 3, 4, 6, 9, 10, 3

$\left(\frac{n+1}{2}\right)^{th} = \left(\frac{7+1}{2}\right)^{th} = 4^{th}$ observation

| $x_i$ | 2 | 3 | 4 | 6 | 9 | 10 | 13 |
|---|---|---|---|---|---|---|---|
| $\lvert x_i - \text{Median}\rvert = \lvert x_i - 6\rvert$ | 4 | 3 | 2 | 0 | 3 | 4 | 7 |

$\Sigma \lvert x_i - \text{medial}\rvert = 23$

~~Median~~ Mean deviation about median $= \frac{1}{n} \Sigma \lvert x_i - \text{median}\rvert$

$= \frac{23}{7}$

$= 3.29$

⑤ Define range for an ungrouped data and also find the range of the given data 38, 70, 48, 40, 42, 55, 63, 46, 54, 44

Sol: Given $n = 10$

$$\bar{x} = \frac{\text{Sum of observations}}{\text{No. of observations}}$$

$$= \frac{38+70+48+40+42+55+63+46+54+44}{10}$$

$$= \frac{500}{10} = 50$$

| $x_i$ | 38 | 70 | 48 | 40 | 42 | 55 | 63 | 46 | 54 | 44 |
|---|---|---|---|---|---|---|---|---|---|---|
| $\lvert x_i - x\rvert$ | 12 | 20 | 2 | 10 | 8 | 5 | 13 | 4 | 4 | 6 |

Mean deviation about mean $= \frac{1}{n} \Sigma \lvert x_i - x\rvert$

$= \frac{84}{10}$

$= 8.4$

# 10. RANDOM VARIABLES & PROBABILITY DISTRIBUTION

## VSAQ's

1.) The mean and varience of a binomial distribution are 4 and 3 respectively. Find $P(x \geq 1)$

Sol:- Given mean=4 Variance = 3

$np = 4$ — ①  $npq = 3$ — ②

$$\frac{eq②}{eq③} \Rightarrow \frac{\cancel{np}q}{\cancel{np}} = \frac{3}{4} \Rightarrow \boxed{q = 3/4}$$

| | |
|---|---|
| $p+q=1$ | From ① |
| $p=1-q$ | $np=4$ |
| $p=1-3/4$ | $n(1/4)=4$ |
| $p=1/4$ | $n=16$ |

$$P(x \geq 1) = 1 - P(X=0)$$
$$= 1 - {}^{16}C_0 \left(\frac{1}{4}\right)^0 (3/4)^{16-0}$$
$$= 1 - (3/4)^{16}$$

② A poisson variable satisfies $P(X=1) = P(X=2)$ find $P(x=5)$

Sol:- $P(X=1) = P(X=2)$

$$\frac{e^{-\lambda}\lambda^1}{1!} = \frac{e^{-\lambda}\lambda^2}{2!}$$

$$1 = \lambda/2$$

$$\boxed{\lambda = 2}$$

$$P(x=5) = \frac{e^{-\lambda}\lambda^5}{5!}$$
$$= \frac{e^{-2}2^5}{2!} = \frac{\cancel{32}^{\,4}}{\cancel{120}_{\,15}} e^{-2}$$
$$P(X=5) = \frac{4}{15} e^{-2}$$

3) If the mean and variance of a bionomial variable X are 2·4 and 1·44 respectively find $P(1 < X \leq 4)$

Sol:- Mean $= 2.4$ Variance $= 1.44$

$np = 2.4$ —— ①  $npq = 1.44$ —— ②

$$\frac{eq②}{eq①} \Rightarrow \frac{npq}{np} = \frac{1.44}{2.4} \Rightarrow q = \frac{\cancel{144}\,\cancel{12}\,3}{\cancel{240}\,\cancel{20}\,5} = 3/5$$

$$\boxed{q = 3/5}$$

$$p+q=1 \Rightarrow p = 1-q \Rightarrow p = 1-3/5 = 2/5$$

$$\boxed{p = 2/5}$$

from eq ①

$$np = 2.4$$

$$n(2/5) = 2.4$$

$$n = \cancel{12}^{6}/\cancel{2}$$

$$\boxed{n=6}$$

$$P(1<x\leq 4) = P(x=2)+P(x=3)+P(x=4)$$

$$= \sum_{r=2}^{4} P(x=r)$$

$$= \sum_{r=2}^{4} {}^6C_r\,[2/5]^r\,[3/5]^{6-r}$$

4) If the difference between the mean and the variance of a bionomial variance variate is 5/9, then find the probability for the event of 2 success, when the experiment is conducted 5 times.

Sol:- Given

Here $n=5 \Rightarrow p+q=1 \Rightarrow p=1-q$

$$np-npq = 5/9$$

$$np(1-q) = 5/9$$

$$np\cdot p = 5/9$$

$$np^2 = 5/9$$

$$\cancel{5}p^2 = \cancel{5}/9$$

$$q = 1-p \Rightarrow 1-1/3 = 2/3$$

$$\boxed{q = 2/3}$$

The probability for the event of 2 success

$p^2 = 1/9$

$p = 1/3$

$$P(X=2) = 5C_2 \left(\tfrac{1}{3}\right)^2 \left(\tfrac{2}{3}\right)^{5-2}$$

$$= \frac{5\times \cancel{4}^2}{1\times \cancel{2}} \times \frac{1}{9} \times \left(\tfrac{2}{3}\right)^2$$

$$= \frac{10}{9} \times \frac{8}{27} = \frac{80}{243}$$

5) If x is a random variable with probability distribution $P(X=k) = \frac{(k+1)c}{2k}$, $k = 0,1,2 - - - - -$ then find c

Sol:- Given $p(X=k) = \frac{(k+1)c}{2k}$ $k = 0,1,2, - - - -$

The sum of the probabilities $=1$

$$\sum_{k=0}^{\infty} P(X=k) = 1$$

$$\sum_{k=0}^{\infty} \frac{(k+1)c}{2k} = 1$$

$$\frac{c}{2^0} + \frac{2c}{2^1} + \frac{3c}{2^2} + - - - - - - = 1$$

$$c\left[1 + 2\left[\tfrac{1}{2}\right]^1 + 3\left[\tfrac{1}{2}\right]^2 + - - - -\right] = 1$$

$$\boxed{1 + 2x + 3x^2 + - - - - - - = (1-x)^{-2}}$$

$$c(1 - 1/2)^{-2} = 1$$

$$c\left(\frac{2-1}{2}\right)^{-2} = 1$$

$$c(1/2)^{-2} = 1$$

$$c(2)^2 = 1$$

$$4c = 1$$

$$\boxed{c = 1/4}$$

6) The range of a random variable X is $\{1,2,3, - - -\}$ and $P(X=k) = \frac{c^k}{k!}$ ; $k = (1,2,3)$. Find the value of c and $P(0 < x < 3)$

Sol:- Given $P(X=k) = \frac{c^k}{k!}$ $k = 1,2,3 - - - -$

The sum of the probabilities $=1$

$$\sum_{i=1}^{k} P(X=K)=1$$

$$\frac{c^1}{1!}+\frac{c^2}{2!}+\frac{c^3}{3!}+\cdots\cdots=1$$

Adding 1 on b.s

$$1+\frac{c^1}{1!}+\frac{c^2}{2!}+\frac{c^3}{3!}+\cdots\cdots=1+1$$

$$\boxed{e^x=1+\frac{x^1}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots\cdots\infty}$$

$$e^c=2$$

$$c=\log^2 e$$

$$P(0<x<3)=P(x=1)+P(x=2)$$

$$=\frac{c^1}{1!}+\frac{c^2}{2!}$$

$$P(0<x<3)=\frac{(\log^2 e)^2}{1!}+\frac{(\log_e 2)^2}{2!}$$

7) The probability that a person chosen at random is left handed (in handwriting) is 0.1. What is the probability that in a group of 10 people, there is one who is left handed?

Sol:- Given

Here $n=10$

$p=0.1$

$p+q=1$

$q=1-p$

$q=1-0.1$

$\boxed{q=0.9}$

The probability that exactly out of $10 = P(x=1)$

$$= {}^{10}C_1 (0.1)^1 (0.9)^{10-1}$$

$$= 10(0.1)(0.9)^9$$

$$=(0.9)^9$$